

國立中央大學

光電科學與工程學系
博士論文

量子與積體鋇酸鋰光路與結構晶疇元件研究
The study of quantum and integrated optical circuits
and domain structured devices in lithium niobate

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中華民國 106 年 6 月



國立中央大學圖書館 碩博士論文電子檔授權書

(104 年 5 月最新修正版)

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論文名稱： 量子受限體能隙與光路與結構晶矽器件研究

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文 量子膜晶體鉬酸鋰光路與結構晶疇元件研究

the study of quantum and integrated optical circuits and domain structured in

係由本人指導撰述，同意提付審查。

lithium niobate

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106 年 6 月 10 日

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論文口試委員審定書

光電科學與工程學系/研究所 鍾宏彬 研究生所

提之論文量子與積體鈮酸鋰光路與結構晶疇元件研究

The study of quantum and integrated optical circuits and
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量子與積體鈮酸鋰光路與結構晶疇元件研究

摘要

本論文之主要貢獻於利用結合高非線性轉換效率之週期性或非週期性極化反轉鈮酸鋰材料、低損耗光學波導、積體化高速調制電極設計之元件，延承傳統非線性光參量轉換系統，拓展至量子光源系統的開發與研究，本論文收錄之各項研究主題條列如下：。

主題一、非週期極化反轉鈮酸鋰電光可調多波長光參量產生器。

主題二、電光波譜可剪裁之內腔式光參量共振器。

主題三、高速鈮酸鋰電光調制器陣列。

主題四之一、超寬頻絕熱光傳輸之鈦擴散式鈮酸鋰波導元件。

主題四之二、超寬頻非對稱絕熱耦合器之雙光子量子偏振態製備元件。

主題五、量子積體光源元件實驗。

論文中分別於第一章至第二章介紹了鈮酸鋰特性與此元件應用範疇、設計與製作原理。第三章節分別介紹了非週期性鈮酸鋰之訊號頻譜電光可調之光參量產生器與共振器之實驗。此外，於第四章節，展示了高頻陣列式電光鈮酸鋰光學強度調制器設計、製作與量化結果。另一方面，本論文第五章節分別收錄了基於絕熱耦合光學傳輸過程所設計製作之高寬頻光學絕熱定向耦合器、可應用於配置量子偏振態之偏振與波長相依之寬頻非對稱性絕熱耦合器，以及基於鈦擴散週期性極化反轉鈮酸鋰光學波導元件之量子光源之量子態定量研究，其中包含：傳統非線性和頻產生過程與同步光參量下轉換之對應實驗、量子光源特性檢驗實驗與配合二維超表面材料之量子偏振態之量子斷層掃描實驗。另一方面，趨向完全積體化電光可調之量子偏振糾纏光源之草擬設計已初步完成，此部分收錄於第六章第二小節之未來工作中。

結語：量子糾纏之背後明確物理機制目前尚未明朗，宇宙萬物於時代更迭中，過去、現在與未來的時間之矢，是否需要重新定義，而四維時空之概念是否為具體之完整解釋，此須待後繼者持續的鑽研與付出，解答或許將會被趨近，但也或許，宇宙從未輕易使人們能得到所謂的解答。由個人趨向群體的集體智慧，使我們持續成長、了解並且領悟。

The study of quantum and integrated optical circuits and domain structured devices in lithium niobate

Abstract

The major contribution of my thesis is proposed a concept of demonstrating the fully integrated quantum optical circuits from the classical optical parametric down conversion process to the novel quantum approach of adiabatic coupling and entangled photon-pairs generation systems based on a domain-engineered lithium niobate crystals with high electro-optically tunable and modulation ability and corresponded diffused-type waveguide devices. With close teamwork of the R&D and fabrication groups of Professor Chen, Yen-Hung laboratory in National Central University, the member of Polaris photonic Co. Ltd. and the resource of Professor Dragomir Neshev laboratory in Nonlinear Physics Centre, Australian National University for quantum light source qualifications. We demonstrated several systems or devices in 2012-2017, which all have the potential market value. The topics of this thesis are divided to topic I to V as the following:

Topic	Research
I	Electro-optical tunable, multi-wavelength optical parametric generators in aperiodically poled Lithium Niobate.
II	Electro-optically spectrum tailorable intracavity optical parametric oscillator.
III	High-speed Lithium Niobate Electro-optical modulators array.
IV-1	Ultra-broadband adiabatic light transfer in titanium diffused Lithium Niobate waveguides.
IV-2	Ultra-broadband asymmetric adiabatic coupler for two-photon quantum-polarization state preparation.
V	Experiments of quantum integrated optical circuits.

In final, we proposed a concept of the fully integrated quantum optical polarization state preparation based on titanium-diffused, domain-engineered lithium niobate waveguide device on the second section of the chapter 6.

Epilogue:

The clear physical mechanism of quantum entanglement is still not yet clear, this universe is in the era of change, for the past, present or future, is the definition of time arrow needed to be redefine? Is the four-dimensional space-time enough to describe the universes? It needs more sustained research and contribution of all successors, the answer may be approached, but maybe, the universe never made it easy for people to get the so-called answers. What we can do is, from personal research to the collective intelligence, so that, we hope we can continue to grow, understand and comprehend “*the universe*”.

To My Dearest

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Journal paper

1. **Hung-Pin Chung**, Kuang-Hsu Huang, Kai Wang, Sung-Lin Yang, Shih-Yuan Yang, Chun-I Sung, Alexander Solntsev, Andrey A. Sukhorukov, Dragomir N. Neshev, Yen-Hung Chen, “Ultra-broadband asymmetric adiabatic coupler for two-photon quantum-polarization state preparation.” (*in preparation*).
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Explanation of Symbols

c	: speed of light in vacuum
λ	: wavelength in vacuum
f	: frequency
ω	: angular frequency
ε	: permittivity
n	: refractive index
n_{opt}	: optical refractive index
n_m	: microwave effective refractive index
n_e	: optical refractive index of extraordinary waves
n_o	: optical refractive index of ordinary waves
n_{eff}	: effective refractive index of waveguide modes
Δn	: refractive index change
β	: propagation constant of waveguide modes
r_{ij}	: electro-optical coefficients
d_{ij}	: nonlinear coefficients
d_{eff}	: effective nonlinear coefficients
η	: impermeability
$E_{x,y,z}$	: applied electric-field along crystal x, y, z axis
D	: duty-cycle of period
Λ	: period
K_m	: reciprocal lattice vector
κ_{ij}	: coupling coefficients
Γ	: overlapping integral
V_π	: half-wave voltage
$EO(f)$	: electro-optical response in microwave band
MFD	: mode field diameter
Υ	: overlapping between fiber modes and waveguide modes
τ	: coating layer thickness
dn/dC	: change rate of metal concentration
ar	: aspect ratio

er	: elliptical ratio
Θ	: the overlapping integral of modes for nonlinear conversion
OF	: Objective function of stimulated annealed algorithm
γ	: complex propagation constant
α	: attenuation
R	: resistance
L	: inductance
C	: capacitance
G	: conductance
Z_c	: characteristic impedance
S_{11}	: return loss, reflection coefficient
S_{21}	: propagation loss, transmission coefficient
γ_{AC}	: The first (major) adiabatic condition
$ \Phi^\pm\rangle$	: quantum pure state with the some polarization/path modes
$ \Psi^\pm\rangle$	: quantum pure state with different polarization/path modes
$ \mathbf{M}^\pm\rangle$	: quantum mixed state

Chapter 1 、 Introduction

In the view of **Industrial Green Technology** is the key developmental goal of our government, the pursuit of many kinds of low carbon emissions, high-efficiency green technologies already become the key indicators for developed or start-up companies. Highly customized components, devices and automatized systems, machines, are the mainstream R & D goals of these companies. In last decade, the use of optical principles for highly efficient, loss-cost, high environmental resistance optical devices and systems have become the mainstream products for optical sensing [1], telecommunication [2], network [3], modulation [4], laser-based industrial [5] or medical applications [6] and automatic identification [7]. Based the developed fiber-optic telecommunication network and the cloud technology, the large data integration, processing and transmission ability is the key index of many products in this decade. For fiber communication, the high-efficiency, high-power, spectrum tunable, wavelength multiplexed laser sources [8] have the key advantage of their diversity for processing the highly information density. With the advantage of large bandwidth, the electro-optical modulators [9] are the very popular components for switching [10], laser shaping [11], pulsed laser modulation [11], ultra-high power laser [12], and the high-speed optical signal modulation of be transmitted electric signals. For the main wide wavelength range tunable laser source are the optical parametric generators/oscillators (**OPGs/OPOs**), which are based on the nonlinear optical conversion, the commercialized OPOs are already used in the communication, sensing, scanning, medical, industrial processing and academic research [13]. On the other hand, in the next generation of telecommunication, the quantum-based optical communication systems are at an alarming rate of development, in a large number of developing quantum source, the high-brightness type-II (o-eo) spontaneous parametric down-

conversion (**SPDC**) photon-pair source generated via the nonlinear integrated optical waveguide devices are particularly noteworthy [14]. With the lithium niobate (**LiNbO₃**) material having excellent electro-optical (**EO**) and nonlinearity characteristics are used as the devices substrate of my studies, me and our group member, we designed, fabricated and demonstrated several LiNbO₃-based devices, including the electro-optical spectrum tunable multi-wavelength OPGs, the electro-optical spectrum tailorable OPOs, the 100GHz electro-optical modulation array, the adiabatic directional couplers, the asymmetric adiabatic couplers as polarization spatial filters, and the quantum light source devices of SPDC. The **Fig.1** shows my thesis architecture, which contains 6 chapters with 5 thesis topics.

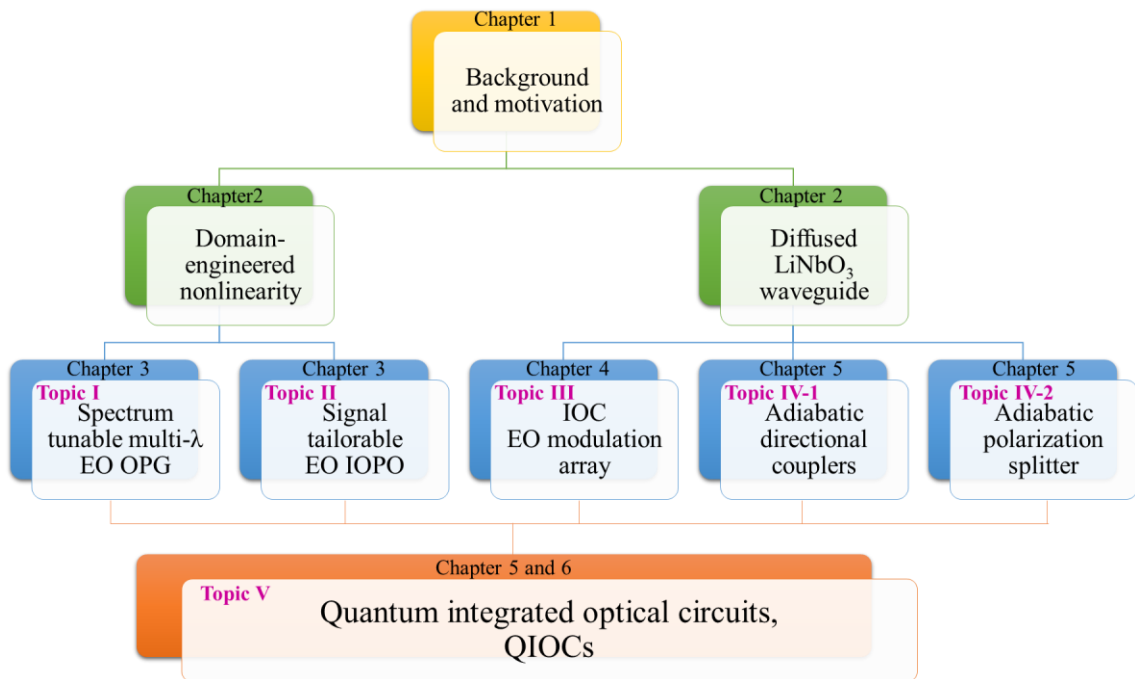


Figure 1. Thesis architecture.

I summarized these studies in the chapter 3, 4 and 5 for topic I, II, III, IV and V, respectively.

Topic I: Electro-Optically Tunable, Multi-wavelength Optical Parametric Generators.

By applying electric-field along crystal z-axis to modulate the refractive index of propagated modes, we can adjust the quasi-phase matched (QPM) condition of multi-wavelength OPGs to shift the signal wavelengths.

Topic II: Electro-Optically Spectrum Tailorable Intracavity Optical Parametric Oscillator.

Through integration of EO polarization mode filtering and QPM optical parametric oscillator in a single aperiodically poled LiNbO₃ (APPLN), we demonstrated the EO spectrum tailorable intracavity optical parametric oscillator (IOPO), the IOPO signal spectrum can be adjusted from single peak, two peaks and multi-peaks by applying electric-field along crystal y-axis under different voltage.

On the other hand, in my PhD period, I lead the development of integrated optical circuits (IOC) and quantum integrated optical circuits (QIOC) projects, the principle of IOC and QIOC are mainly by taking the advantage of μm -level scale, highly integration ability of LiNbO₃ optical waveguides, we can integrate several optical function in a single device, I summarized the study results in chapter 4 and chapter 5 for IOC and QIOC, respectively.

Topic III. High-speed LiNbO₃ Electro-Optical Modulators Array

We demonstrated the high-speed EO modulator array, the results are included in chapter 4. The overall -3dB bandwidth of 12-packaged EO modulator array achieves ~ 100 GHz.

In recent years, our lab have several cooperation projects with Nonlinear Physics Centre, Australian National University (ANU) to develop the quantum source of LiNbO₃ substrate. In my last year of PhD period, I did the experiment in the ANU for seven months via support of doctoral research

abroad program of MOST of Taiwan, study results of QIOCs are concluded as topic IV and topic V.

Topic IV: Light Adiabatic Transfer Devices

Including (**topic IV-1**) the ultra-broadband adiabatic directional coupler and (**topic IV-2**) the high efficiency ultra-broadband polarization and wavelength adiabatic spatial mode splitter in section of 5-1 in chapter 5, we used the adiabatic light transfer features via a special designed waveguide array to demonstrate the ultra-broadband, high tolerance Ti-diffused LiNbO₃ coupler and splitter devices.

Topic V: Quantum Integrated Optical Circuits

The quantum experiments of developed QIOC are summarized in the section 5-2 for quantum source characteristics quantization and quantum polarization state reconstruction via a metasurface. Ti-diffused PPLN waveguide quantum light source QIOC (5-2-1), Type-II sum-frequency generation (SFG) characteristics quantization (5-2-2), SFG-SPDC analogy for Type-II SPDC quantum source characteristics quantization (5-2-3), quantum entanglement and quantum tomography experiments (5-2-4) and quantum polarization state reconstruction via metasurface for single-photon state (5-2-5) are all described, respectively.

Chapter 6 is the conclusion and future outlook, in section 6-2, I design a highly integrated QIOC device of Quantum Entangled source in Multifunction Integrated Ti-diffused PPLN waveguide devices (**QE-MIT**) for demonstrating the EO quantum polarization state tunable pump adiabatic spatially separated entangled photon pair generation in a multifunction integrated Ti-diffused PPLN waveguide device.

Chapter 2 、 Principle and Methods

The substrate material, device applications, working principles, related equations, design tools, simulation methods, fabrication processes flow and precautions of this thesis are all aggregated in the chapter 2. The mainly used substrate material of my thesis is congruent Lithium Niobate (Congruent-LiNbO₃, CLN), introduced in the section (2-1), with domain-engineered technique and/or within metal-diffused or proton-exchanged waveguides. Using the high electro-optical coefficients of LiNbO₃ (2-2), we demonstrate several novel devices or systems, not only bulk systems but also the waveguide approach devices. Because of the fairly close mode sizes, the LiNbO₃ waveguide-based devices are highly suitable for combining the existing telecommunication network with fiber-optic elements, the detail of waveguide design, fabrication, qualification, fiber-to-waveguide modes coupling will be discussed in section (2-3). With the poling technique of domain-engineered (a)periodically poled LiNbO₃, (A)PPLN, we demonstrated several quasi-phase matched (QPM) nonlinear optical processes, such as spontaneous parametric down-conversion (SPDC), optical parameter generation/oscillator (OPG/OPO), or electro-optical polarization-mode converter (EOPMC). The nonlinear optical processes are introduced in the section (2-4).

2-1 Lithium Niobate devices

2-1-1 Lithium Niobate crystals

Lithium Niobate (LiNbO₃) is the first demonstrated in the single crystal form since 1966 at Bell Laboratories, due to the anisotropic properties of its crystal structure, this ferroelectric material has relatively large nonlinear, electro-optic coefficients, and very suitable to be the substrate of nonlinear optical conversion and electro-optical modulation devices [15]. According to different growing

methods, the LiNbO_3 group can be divide into congruent- LiNbO_3 (CLN), stoichiometric- LiNbO_3 (SLN), or MgO-doped congruent- LiNbO_3 (MgO:CLN), or MgO-doped stoichiometric- LiNbO_3 (MgO:SLN). The literature [16] points out the concentration of MgO-doping into CLN or SLN substrates will increase the high optical power tolerance and increase the optical damage threshold by increasing the photoconductivity, but in this thesis, all experiment are operated under damage threshold of CLN crystals, which is around $1\text{KW}/\text{cm}^2$ or $1\text{MW}/\text{cm}^2$ for undoped CLN or 5% mol. MgO:CLN, respectively [17].

2-1-2 Application of Lithium Niobate devices

There are numerous existing application of LiNbO_3 devices, I simply list some common examples in the industry, medical science, defense or academia. In the industry we can find many product made by LiNbO_3 , including Nd:YAG second harmonic generation laser for visible laser generation [18], electro-optical intensity/phase modulator [19], high-peak power laser component [20]. We also can find many applications in the medical science, such optical parametric oscillator scanning [21], high-spectral resolution device [22], and optical coherence tomography (OCT) [23]. In the defense, LiNbO_3 devices are the key component of fiber-optic gyroscope (FOG) [24], which is the heart of many positioning systems, such as attitude and heading reference system (AHRS), inertial measurement unit (IMU), GPS-based navigation or guiding systems [25]. There are more diverse of academia applications, we can find the many nonlinear optical conversions for different laser line generation [26-28], EO Q-switching devices[29], quantum light source generation of quantum information for next generation quantum telecommunication network [30-32], or in the other science applications, such as, remote sensing [33], Photoplethysmography (PPG) in bio-

photo detection [34], airborne multi-wavelength water vapor differential absorption lidar [35], wind profile measurement in atmosphere [36], and laser spectroscopy [37].

2-2 Electro-optical (EO) effect

2-2-1 EO effect in Lithium Niobate crystals

Electro-optical effect is the results of the interaction between optical-field and applied electric-field, which can be understood by the change of impermeability tensor in applied electric-field along three mutually orthogonal axis with respect to the crystal axis. The **Eq. (1)** describes the relation of impermeability tensor, electro-optical coefficients tensor and applied electric-fields.

$$[\Delta\eta_{ij}(E)] = \begin{bmatrix} \Delta\eta_1 \\ \Delta\eta_2 \\ \Delta\eta_3 \\ \Delta\eta_4 \\ \Delta\eta_5 \\ \Delta\eta_6 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \\ r_{41} & r_{42} & r_{43} \\ r_{51} & r_{52} & r_{53} \\ r_{61} & r_{62} & r_{63} \end{bmatrix} \begin{bmatrix} E_x \\ E_y \\ E_z \end{bmatrix} \quad (1)$$

The first-order electro-optical coefficients tensor of LiNbO₃ crystal is :

$$\begin{bmatrix} 0 & -r_{22} & r_{13} \\ 0 & r_{22} & r_{13} \\ 0 & 0 & r_{33} \\ 0 & r_{51} & 0 \\ r_{51} & 0 & 0 \\ -r_{22} & 0 & 0 \end{bmatrix} \quad (2)$$

with $r_{13}^T = 10 \text{ pm/V}$, $r_{13}^S = 8.6 \text{ pm/V}$, $r_{22}^T = 6.8 \text{ pm/V}$, $r_{22}^S = 3.4 \text{ pm/V}$, $r_{33}^T = 32.2 \text{ pm/V}$, $r_{33}^S = 30.8 \text{ pm/V}$, $r_{51}^T = 32 \text{ pm/V}$, $r_{51}^S = 28 \text{ pm/V}$, for bulk congruent LiNbO₃ (CLN) at 632.8 nm wavelength [38].

2-2-2 Electro-Optical Polarization-Mode Converter, EOPMC

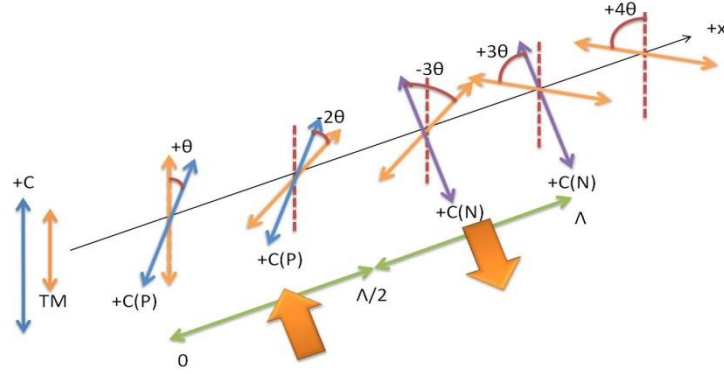


Figure 2. The schematic of polarization evolution in a single positive (P) and negative (N) domains with period Λ . The orange and blue double arrows represent the mode polarization and half-wave plate (HWP) angles respected to the vertical axis. Bold orange arrows are the domain orientation of unpoled/poled domains.

Šolc-type polarization mode convertor is a series of angle staggered half-wave plates (HWPs) with $+\theta$ and $-\theta$ relative to absolute coordinates in the length of half period as the **Fig. 2** shown. The polarization of light (parallel to absolute coordinates) will be tuned by $+2\theta$ after pass through the first $+\theta$ HWP, and the angle of modulated polarization now becomes $+3\theta$ respected to the fast axis of the second HWP, and finally, after passing through a pair of $+\theta$ and $-\theta$ HWPs, the light polarization will be modulated to $+4\theta$ respected to original unmodulated polarization. This process can also be demonstrated by applying the electric-field parallel to the crystal y-axis, E_y of LiNbO_3 . The impedance change of modulated LiNbO_3 crystal can be expressed as the **Eq. (3)**.

$$\begin{bmatrix} \Delta\eta_1 \\ \Delta\eta_2 \\ \Delta\eta_3 \\ \Delta\eta_4 \\ \Delta\eta_5 \\ \Delta\eta_6 \end{bmatrix} = \begin{bmatrix} 0 & -r_{22} & r_{13} \\ 0 & r_{22} & r_{13} \\ 0 & 0 & r_{33} \\ 0 & r_{51} & 0 \\ r_{51} & 0 & 0 \\ -r_{22} & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ E_y \\ 0 \end{bmatrix} \quad (3)$$

Therefore, the modulated crystal impedance tensor is:

$$\eta(E_y) = \begin{bmatrix} \frac{1}{n_o^2} - r_{22}E_y & 0 & 0 \\ 0 & \frac{1}{n_o^2} + r_{22}E_y & r_{51}E_y \\ 0 & r_{51}E_y & \frac{1}{n_e^2} \end{bmatrix} \quad (4)$$

The modulated index ellipsoid becomes:

$$\left(\frac{1}{n_o^2} - r_{22}E_y\right)x^2 + \left(\frac{1}{n_o^2} + r_{22}E_y\right)y^2 + \frac{1}{n_e^2}z^2 + 2yzr_{51}E_y = 1 \quad (5)$$

By rotating the coordinate axes from (x, y, z) to (x, y', z') with the specific chosen θ , the coefficient of $y'z'$ term can be zero, this specific angle θ can be expressed as the function of applied electric-field as **Eq. (6)**:

$$\tan 2\theta = \frac{2r_{51}E_y}{\left(\frac{1}{n_o^2} - \frac{1}{n_e^2} + r_{22}E_y\right)} \approx \frac{2r_{51}E_y}{\left(\frac{1}{n_o^2} - \frac{1}{n_e^2}\right)} \quad (6)$$

Since the electro-optic coefficient of LiNbO_3 is extremely small with ten to the power of minus 12 in unit of m/V respected to index difference term (with ten to the power of minus 3), so with reasonable approximation, $(1/n_o^2 - 1/n_e^2) \gg r_{22}E_y$, the modulated angle with applied electric-field along crystal y-axis can be approximated as: $\theta \approx r_{51}E_y / \left(\frac{1}{n_o^2} - \frac{1}{n_e^2}\right)$

Finally, the index ellipsoid after rotating the coordinate becomes:

$$\begin{aligned} &\left(\frac{1}{n_o^2} - r_{22}E_y\right)x^2 + \left(\frac{1}{n_o^2} + r_{22}E_y + 2r_{51}E_y \tan \theta\right)(y')^2 \\ &+ \left(\frac{1}{n_e^2} - 2r_{51}E_y \tan \theta\right)(z')^2 = 1 \end{aligned} \quad (7)$$

This principle axis rotating angle is the function of applied electric-field, which can be used to form the polarization retardation plate and then to adjust the polarization of the modified modes. To quantify the coupling coefficients with different applied electric-fields, by using the **Eq. (4)** with the approximate of $1/n_o^2 \gg r_{22}E_y$, $n_o^{-2} \pm r_{22}E_y \approx n_o^{-2}$, one can also derive the relationship between

the dielectric tensor after the electric field perturbation, since the EO coefficients of LiNbO₃ crystal of the order in 10^{-12} m/V leads to $n_o^2 n_e^2 r_{51}^2 E_y^2 \ll 1$, therefore the dielectric tensor after the electric field perturbation can be approximated as:

$$\varepsilon(E_y) \approx \varepsilon_0 \begin{bmatrix} n_o^2 & 0 & 0 \\ 0 & n_o^2 & -n_o^2 n_e^2 r_{51} E_y \\ 0 & -n_o^2 n_e^2 r_{51} E_y & n_e^2 \end{bmatrix} \quad (8)$$

The dielectric tensor with perturbation becomes: $\varepsilon(x, y, z) = \varepsilon_0(y, z) + \Delta\varepsilon(x, y, z)$, where, $\varepsilon_0(y, z)$ is unperturbed dielectric tensor distribution in yz plane, and $\Delta\varepsilon(x, y, z)$ is dielectric tensor perturbation, which is

$$\Delta\varepsilon = -\varepsilon_0 \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & n_o^2 n_e^2 r_{51} E_y \\ 0 & n_o^2 n_e^2 r_{51} E_y & 0 \end{bmatrix} g(x) \quad (9)$$

with $g(x) = \pm 1$ depended on the crystal domain axis direction.

The x-propagation polarization-mode conversion process of waveguide systems can be expressed by coupled-mode equations as [39]:

$$\begin{aligned} \frac{dA_o(x)}{dz} &= -j\kappa A_e(x) \exp(+i\Delta\beta x) \\ \frac{dA_e(x)}{dz} &= -j\kappa^* A_o(x) \exp(-i\Delta\beta x) \end{aligned} \quad (10)$$

where, κ , κ^* are the coupling coefficients between two polarized modes, $\Delta\beta$ is the phase-mismatching term of two polarized modes. In the periodic poled domain structure, the phase-mismatching term can be derived as:

$$\Delta\beta = \beta_o - \beta_e - K_m \quad (11)$$

$K_m = 2m\pi/\Lambda$ is the reciprocal lattice vector with order m of the period in Λ . For satisfying the phase matching condition of $\Delta\beta = 0$, the required QPM domain-inverted period must be $\Lambda = m\lambda/(n_o - n_e)$. By solving the **Eq. (11)** and with defining the polarization transmittance after interaction length of L , $T =$

$|A_o(L)/A_e(0)|^2$, under the condition of limited coupling coefficients with $A_e(x) \approx A_e(0)$, $\Delta\beta \gg |\kappa|^2$, the transmittance of polarized mode can be approximated as:

$$T = \left| \frac{A_o(L)}{A_e(0)} \right|^2 \approx |\kappa|^2 L^2 \text{sinc}^2 \left(\frac{\Delta\beta}{2} L \right) \quad (12)$$

With the change sign of EO coefficients after the domain inversion, EOPMC can be formed by a series of periodically poled and unpoled domains (with $-r_{51}$ and $+r_{51}$, respectively) within the applied E_y -field on the LiNbO₃ substrate or waveguides. Thus, according **Eq. (10)** the dielectric tensor perturbation in m -order periodically poled LiNbO₃ structure can be expressed as:

$$\varepsilon_m(y, z) = -\varepsilon_0 n_o^2 n_e^2 r_{51} E_y \frac{2}{m\pi} \sin(m\pi D) \quad (13)$$

with, r_{51}, E_y and D are the EO coefficient, applied electric-field and duty-cycle of periodically poled structure. The coupling coefficients of polarization-mode conversion is the function of mode overlapping integral of two modes [40], the coupling coefficient in LiNbO₃ waveguide is:

$$|\kappa| = \frac{\omega}{c} \frac{\sin(m\pi D)}{m\pi} \frac{n_{o,s}^2 n_{e,s}^2 r_{51} E_y}{\sqrt{n_{o,eff} n_{e,eff}}} \vartheta \quad (14)$$

And the $n_{nr}(y, z) = n(y, z)/n_s$ is the normalized index distribution and n_{eff} are the modes effective index, with the mode overlapping integral, $\vartheta = \int e_o^*(y, z) n_{o,nr}^2(y, z) n_{e,nr}^2(y, z) e_e(y, z) dy dz$, $n_{o,s}, n_{e,s}$ are the maximum index of index distribution for o-wave or e-wave modes,

2-3 Waveguides

In this section, the short introduction of the characteristic and advantage of waveguide devices and the several mainstream type of waveguides in LiNbO₃ crystals will be summarized in the paragraph of (2-3-1) and (2-3-2). To theoretic analysis for characteristics and modeling of Ti-diffused or PE-type LiNbO₃ waveguide, the several methods have been introduced, including Effective Index Method (2-3-3-1), Beam Propagation Method (2-3-3-2), half-wave voltage modeling (2-3-3-3) and high-speed modulator modeling (2-3-3-4). After introducing the design processes of single-mode waveguides on LiNbO₃ crystals (2-3-3-5), the parameters of the fabrication process considerations of two types LiNbO₃ waveguides are shown in the paragraph of (2-3-4-1) and (2-3-4-2). The fabricated waveguide sample test method (2-3-5) of this thesis are summarized in the paragraph of (2-3-5).

2-3-1 Characteristic and advantage of waveguide devices

Waveguide is the structure for restricting the light energy inside specific regions, which is based on the phenomenon of interference results of continuous total reflections, the total reflections are occurred in the boundaries between the high refractive index and low refractive index regions, according to the energy limitation boundaries of different dimensions, for one dimensional limited waveguides are called planar waveguides, and for two dimensional limited waveguides are called channel waveguides, and because of the discrete conditions of standing waves of the interference, the interference patterns of waveguides are called modes, if there is only one mode existed in the waveguide, this mode is called fundamental mode, for more than one modes situation, the rest are all called higher-order modes. For generally common devices, the designer will control the

waveguide parameters to let it only fundamental mode can be existed, the light energy can only concentrated on fundamental mode, and let the optical process can be simplify and efficient, for example, the efficiency of nonlinear conversion or electro-optical modulation are proportional to the intensity of electric-fields, therefore, in comparison of the bulk devices, the needed modulation voltage or applied pump power of the waveguide devices will be effectively reduced significantly, for telecom band, diffused-type LiNbO₃ waveguide devices, the mode size are around $10 \times 10 \mu\text{m}^2$, but bulk system modes, focused by objective lens, are around $100 \times 100 \mu\text{m}^2$, so under the fixed input power, the effective applied intensities in waveguide modes are ~ 100 times than in the bulk system modes. In addition, for the fiber telecommunication network, the LiNbO₃ waveguide mode sizes are very close to the commercial fiber mode sizes in typically, so the coupling loss between waveguide and fiber modes are much less than the coupling loss of bulk system modes to fiber modes, taking the advantage of this efficient and energy saving features of waveguide devices, the integrated optical circuits (IOCs) based on waveguide structures have been extensively studied, examined, and development since 1980, and IOCs have become a practical commodity in the last 20 years, which have an important place for optical modulation, data processing, optical sensing, and optical logic systems of a wide variety of applications.

2-3-2 Type of waveguides in Lithium Niobate crystals

More than 50 years of development history, there are a variety of waveguide fabrication process have been invented since the first production of the single crystal congruent LiNbO₃ in 1965. The main types of process method can be roughly divided into 5 groups: metal-diffused (Ti-diffused, especially), Proton-

Exchanged (PE), etched-ridge, Adhered-Ridge Waveguide (ARW) and the hybrid. In this thesis, I focus on the research and development of the two most mainstream waveguiding techniques of the LiNbO₃ crystals, the Ti-diffused and PE-type waveguides, these techniques will be discussed in depth in the following paragraphs from (2-3-3) to (2-3-5). It is worth mentioning that the biggest difference for Ti-diffused and PE type is the ability to form the high purity single polarization mode. For PE-type, the index change of proton-exchanged region is increased for the index of extraordinary wave, but slightly decreased for the index of ordinary wave in contrast to the un-exchanged region, this feature is resulting in producing the ability of polarization-selective for guiding modes. This is a characteristics that general waveguide does not have, and by using this feature, researchers can develop a lot of super pure polarized waveguide devices. In the **chapter 4**, I will introduce our integrated optical devices made by PE-type LiNbO₃ waveguide. The characteristics of high-speed optical modulator array (made by PE-type LiNbO₃ waveguides) and quantum integrated optical devices (made by Ti-diffused LiNbO₃ waveguides) are summarized in the **chapter 4** and **5**, respectively.

2-3-3 Modeling and design

From the basic Maxwell's wave equations, the x-propagation waveguide transverse mode electric-field under small index perturbation can be expressed as [39]:

$$\left\{ \nabla_t^2 \left[\frac{\omega^2}{c^2} n^2(y, z) - \beta^2 \right] \right\} E(y, z) = 0 \quad (15)$$

where, $\nabla_t^2 = \nabla^2 - \frac{\partial^2}{\partial x^2}$ is the transverse Laplacian operator.

The commonly used numerical analysis of basic and modeling design method for waveguide devices are introduced in this section, it contains several

quantitative methods, which is the Effective Index Method (2-3-3-1), Beam Propagation Method (2-3-3-2), half-wave voltage modeling (2-3-3-3) and high-speed modulation modeling (2-3-3-4), at the end of this section, I summarized our design method for single-mode waveguides (2-3-3-5).

2-3-3-1 Effective index method

To estimate how many waveguide modes can exist in a planar (1D) waveguide, we can start to calculate the effective index of a specific waveguide mode [41] by solving the mode dispersion equation in normalized form [42]:

$$2V \int_0^{y_t} (f(y) - b)^{1/2} dy = \left(2m + \frac{3}{2}\right) \pi \quad (16)$$

where, the V, b is the normalized parameter of thickness or effective mode index of a planar (1D) waveguide, $f(y)$ is the arbitrary index profile of 1D waveguide in depth direction. y_t is the turning point, which $n(y_t) = n_{eff}$.

After mastering the effective mode index of 1D waveguide, we can further to seek the effective mode index of an arbitrary channel (2D) waveguide based on the same fabrication process but lack index difference of one direction, which decreased to the 1D waveguide. The 2D effective mode index modeling procedures are listed below:

1. Choose V_0 to get b_0 in planar (1D) waveguides.
2. Choose the ratio of width (W) and depth (D) of waveguide (W/D) to get channel waveguide normalized parameter V' , where $V' = V_0 b_0^{1/2} \sqrt{W/D}$, where the 1D planar waveguide normalized parameters are :

$$V_0 = \frac{2\pi}{\lambda_0} D \sqrt{(n_{bulk} + \Delta n)^2 - n_{bulk}^2} \quad (17)$$

$$b_0 = \frac{n_{eff,1D}^2 - n_{bulk}^2}{(n_{bulk} + \Delta n_{1D})^2 - n_{bulk}^2} \quad (18)$$

where, n_{bulk} , $n_{eff,1D}$, Δn_{1D} is the index of bulk, 1D waveguide mode and index change of 1D waveguide, respectively.

3. Use Eq. (15) to find $b(x)$ and $h(2x/W)$ for any position of x , where,

$h\left(\frac{2x}{W}\right) = g\left(\frac{2x}{W}\right) \frac{b(x)}{b_0}$, $g(2x/W)$ is the index distribution in horizontal direction, $f(y)$ is the diffusion function in depth direction.

4. Use the variable $u = 2x/W$, and the dispersion equation of

$V' \int_0^{u_t} [h(u) - b']^{1/2} du = (m + 1/2)\pi$ to find b' by numerical calculation, where V' , b' and u_t is normalized guide width, normalized mode effective index and normalized turning point of 2D waveguide, respectively.

5. Finally, we can obtain the mode effective index of the channel (2D) waveguide (N'_{eff}), which is approximately to:

$$N'_{eff} \approx n_{bulk} + \Delta n b_0 b' \quad (19)$$

The mode number and index of each mode in a straight channel waveguide can be obtain from the dispersion curve of an arbitrary index profile.

2-3-3-2 Beam propagation method

After the foundation design of waveguide, if further we want to simulate propagation feature of different waveguide structures, for example, the Y-branch, X-branch, Mach–Zehnder interferometer (MZI) type, directional coupling type (DC), multi-layer waveguide or waveguide array...etc. then the beam propagation method (BPM) [43] can come in handy. The principle of this method is original from the wave equation within the small index change, which can be seen as

perturbation in the wave equation, the wave equation with perturbation is expressed as the following form as **Eq. (20)**.

$$\nabla^2 \phi + k^2 n_0^2(\vec{r}) \phi = -k^2 \Delta n^2(\vec{r}) \equiv p(\vec{r}) \quad (20)$$

If a solution ψ of this differential equation is known at starting point of propagation ($z = z_0$), then the field ψ and its derivative $\partial\psi/\partial z$ can be solved in the any point of propagation after starting point.

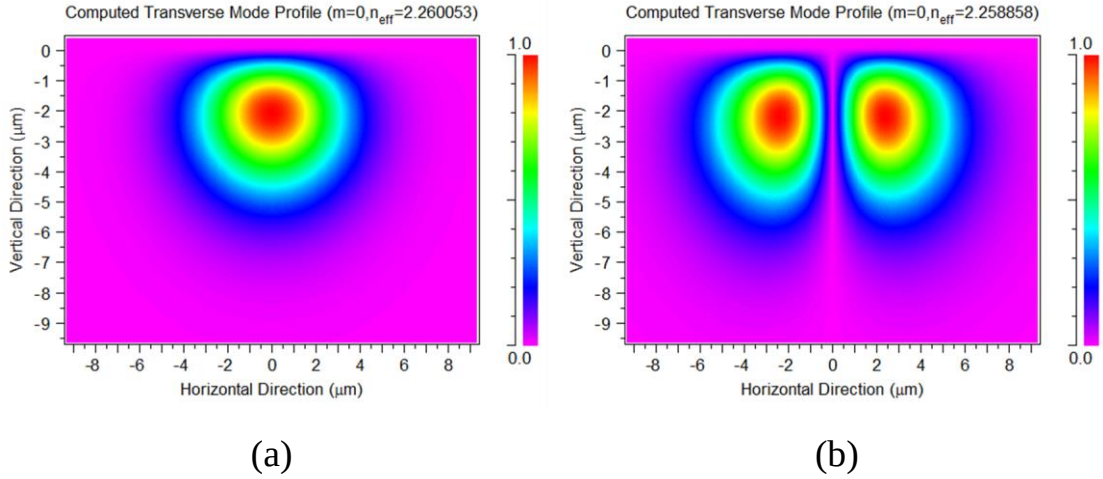


Figure 3. A typical TE-polarized 785-nm mode fields of a diffused LiNbO₃ channel waveguide with fundamental mode (a) and first higher-order mode (b), which are simulated by the commercial BPM simulator.

We can establish the waveguide initial guess parameters, which can be obtained by EIM motioned in the previous section. Use the help of commercial BPM simulator, the mode number, the detail of mode field distributions, the waveguide power transmission characteristics in specific waveguide configuration and the coupling effect of couplers or array all can be simulated for reference. The **Fig. 3** shows the mode fields of a typical diffused-type Lithium Niobate waveguide, which is pumped by a TE-polarized 785 nm laser.

2-3-3-3 Half-wave voltage modeling

In the specification of EO modulator, there is a crucial parameter, which is called half-wave voltage (V_π) with relatively phase change ($\Delta\phi$) of modulated light

field reach π in comparison with unmodulated light field. With different the dimension of electrodes, type of electrodes, type of waveguides, EO coefficients of crystals (r), wavelengths (λ) and working temperature (T), the half-wave voltage can be summarized as the following form,

$$V_{\pi}(\lambda, T) = \frac{G}{L} \frac{\lambda}{r(\lambda, T) n_{eff}^3(\lambda, T) \Gamma P} \quad (21)$$

where, G, L, Γ, P are the electrode-dependent parameters, which is electrode gap, length of electrode, overlapping factor of electro-optical fields and the effective factor of different electrode types, respectively. For common single driving modulation, $P=1$, the single-pass with double driving modulation (push-pull electrodes), $P=2$, double-pass with double driving, $P=4$ and the traveling wave modulation (coplanar waveguide (CPW) electrodes), $P=1.5$.

For seeking the optimal design, to increase the EO overlapping factor (Γ) is extremely important. The transverse overlapping factor is 0~1, in different degree of electric and optical modes overlapping, it can be represented as,

$$\Gamma = \frac{G}{V} \frac{\iint_{-\infty}^{\infty} E_{applied}(x, y) [E_{optical}(x, y)]^2 dx dy}{\iint_{-\infty}^{\infty} [E_{optical}(x, y)]^2 dx dy} \quad (22)$$

where, V is the applied voltage, and the transverse distribution of applied electric-field and modulated optical-field are $E_{applied}(x, y)$ and $E_{optical}(x, y)$, respectively. With setting conditions for an EO LiNbO₃ waveguide under 1550 nm operation, $r = r_{33} \approx 27 \text{ pm/V}$, $n = n_{eff} \approx 2.14$, $\Gamma = 0.65 \sim 0.85$, and CPW push-pull transverse modulation, the estimated half-wave voltage is shown as **Fig. 4** under different overlapping factors, gaps and electrode lengths.

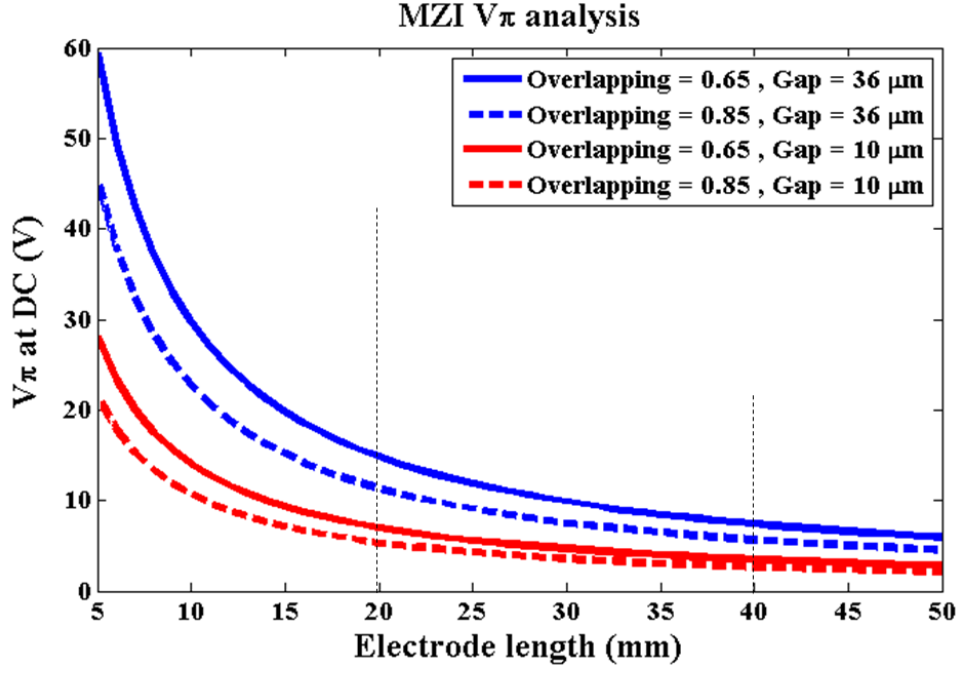


Figure 4. The curves show the electrode length dependence of half-wave voltage of standard transverse EO LiNbO₃ modulator with different electrodes setting. Blue curve are under electrode gaps equal 36μm, and the red ones are under gap widths are 10μm, and solid curve are the case of EO overlapping factor equal 0.65, dash curves are the case of EO overlapping factor equal 0.85.

2-3-3-4 High-speed modulation modeling

For RF modulation from 0.1 ~ 100GHz, the classical uniform electrodes may not applicable due to the velocity mismatching between optical waves and modulation electric waves. The traveling wave modulation comes in handy at this time, with carefully design the configuration of electrodes as the coplanar waveguide (CPW) electrodes, one can choose the effective index of modulation wave (N_m) to close to the effective index of optical waveguides modes (N_{opt}). The effective index of modulation waves are the function of electrodes/buffer layers/waveguides configuration. The crucial design parameters of high-speed EO modulator are the topological parameter of electrodes (W_1, W_2, W_3, G_1, G_2), the buffer layer thickness (T_{buffer}) and waveguide dimension (W_{WG}, D_{WG}) as the following **Fig. 5** shown.

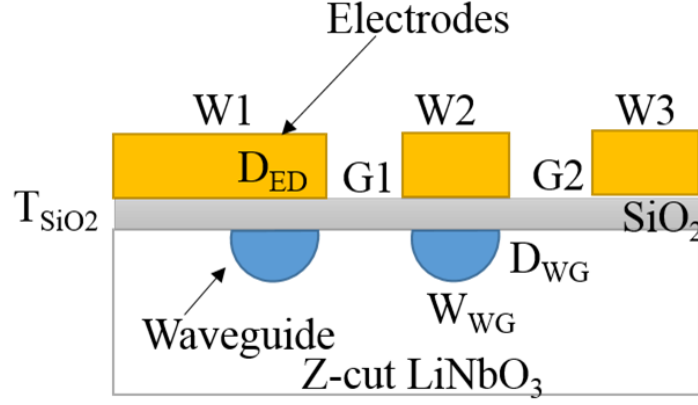


Figure 5. The sectional view of classical high-speed EO LiNbO₃ modulator configuration.

The high modulation frequency electro-optical response in microwave band is given by **Eq. (23)**, which is the function of the effective index of modulation microwave, the effective index of optical mode and microwave characteristic impedance [44]:

$$EO(f) = \left| \frac{1 - S_1 S_2}{(1 + S_2)[\exp(j2u_+) - S_1 S_2 \exp(-j2u_-)]} \times \left[\exp(j2u_+) \frac{\sin(u_+)}{u_+} + S_2 \exp(-j2u_-) \frac{\sin(u_-)}{u_-} \right] \right| \quad (23)$$

where, $u_{\pm} = \frac{1}{c} \pi f L (N_m \mp N_{opt}) - j \frac{1}{2} \alpha L$, $S_1 = \frac{Z_1 - Z_C}{Z_1 + Z_C}$, $S_2 = \frac{Z_2 - Z_C}{Z_2 + Z_C}$, and the N_m , N_{opt} , Z_C , Z_1 , and Z_2 are the effective index of modulation microwave, the optical effective index, the microwave characteristic impedance, the microwave generator's internal impedance, and the shunted loaded impedance, respectively. The α represents the overall loss of microwave and dielectric losses. The 3-dB optical bandwidth Δf is determined approximately by

$$\Delta f = \frac{2c}{\pi |N_{opt} - N_m| L} \quad (24)$$

One can very easy to estimate the optical bandwidth in different designed optical wavelength and modulation speed, the needed modulation half-wave voltage (V_{π}) will decrease if extended the modulation length, but at the same time, the group

velocity mismatch between microwave and optical wave will become more serious if the modulation length becomes longer, there is a trade-off to pursue the relatively maximum bandwidth but in a reasonable modulation voltage. Fortunately, under a certain length, the modulation voltage is likely to be reduced by increasing the overlapping of microwave and optical waves, the overlapping parameter is described by **Eq. (22)**.

2-3-3-5 Design of a single-mode waveguide

In many applications, single-mode waveguide is necessary. The single-mode waveguide usually means that there exists only a fundamental mode of waveguide. To design an optimized single-mode waveguide for minimize loss, there are several issues need to be concerned carefully. First of all, the *cut-off condition*. We can estimate the cut-off condition of specific waveguide in their normalized index dispersion graph calculated by Effective Index Method.

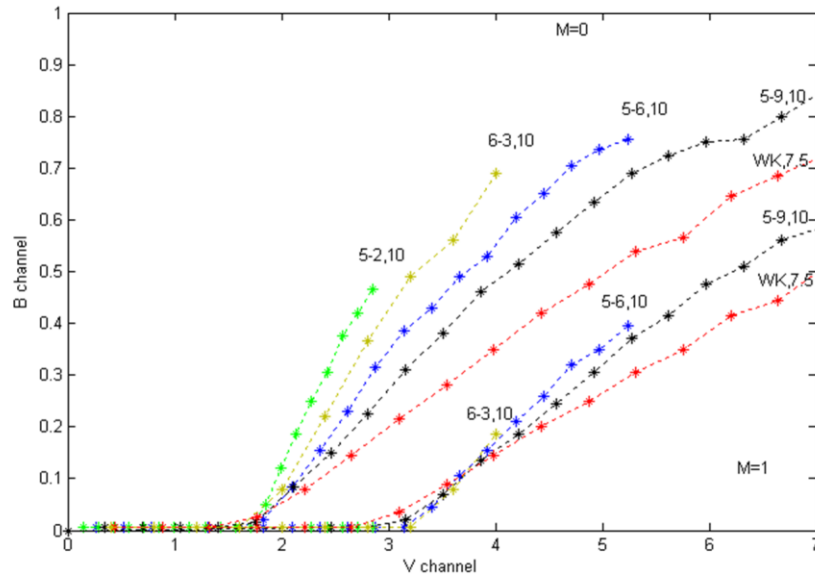


Figure 6. Normalized parameter of diffused channel waveguides for single-mode design. The insert marks are serial number of different waveguide design parameters.

In **Fig. 6**, the cut-off of each waveguide mode are happened at V-value less than 1.5~1.8 in different waveguide, respectively. On the other hand, the first higher-order mode start to appear when V-value is larger than around 2.5, which means the single-mode condition is limited to be V-value from 1.8~ 2.5 in this case. Second, the *confinement*. We all realize that the waveguide mode is the phenomenon of sustainably constructive interference via total reflection of boundaries. With this view, the index different between two boundaries should be as large as possible for pursuing the best confinement, but enlarge the index difference of the cladding and core of waveguide, it will exist more higher-order modes in principle. There is a trade-off between pursuing the higher confinement and seeking the purely single-mode state. Third, last but not least, the *polarized-dependent single-mode condition*. We haven't clarify the single-mode condition in more general case, if there exists two cross-polarized modes in the "single-mode" waveguide, can we still easily think of it as single-mode waveguide? or not? To describe this case, we must need to concern the crosstalk of existed two polarized modes. In some highly single-mode reciprocity applications, the requirement of crosstalk of two cross-polarized modes is extremely strict, which is set for the -60 dB to -90 dB [45]. If there is no special instructions, when we talk about the *single-mode waveguide* in the following sections of thesis, which is referred to the polarized-dependent single-mode case. The procedure for design a single-mode waveguide in diffused-type waveguide via LiNbO₃ crystal is briefly introduce as the following:

1. Choose the wavelength for single-mode operation
2. Establish the index profile of planar waveguide in simulation
3. Establish temperature depended index dispersion of planar waveguide
4. Select the applicable dimension of channel waveguides

5. Testing for single-mode waveguide performance.

The more specifically fabrication and testing methods of diffused-type LiNbO₃ waveguide in this thesis will be introduced in next two sections from (2-3-4) to (2-3-5).

2-3-4 Waveguide fabrication on Lithium Niobate crystals

In method for waveguides fabrication of this thesis, we selected the two of the longest history of development and the most widely used method, which are the Ti-diffused and Proton-Exchanged-type waveguides on the LiNbO₃ substrate. We developed three common diffused-type waveguide on LiNbO₃ substrates for low-loss, high nonlinear coefficients fidelity, which are 1) **Titanium diffused waveguides**, 2) **Annealed proton-exchanged (APE) waveguides**, 3) **Soft proton-exchanged (SPE) waveguides**, the fabrication parameters and expected achievable propagation loss are listed in Table 1.

Method	Type	Fabrication temperature	Fabrication duration	Propagation loss
Titanium diffused	Metal-diffused	> 1000 °C	2-4 days	< 0.4 dB/cm
Annealed proton-exchanged	Proton-exchanged	100-400 °C	1-3 days	< 0.8 dB/cm
Soft proton-exchanged	Proton-exchanged	300-500 °C	3-7 days	< 0.6 dB/cm

Table 1. The fabrication parameters and expected propagation losses comparison of three common diffused-type LiNbO₃ waveguides.

2-3-4-1 Ti-diffused waveguides

In 1974, the first Ti-diffused LiNbO₃ waveguide had been demonstrated by R. V. Schmidt and I. P. Kaminow in Bell Labs [46]. First, the thin-layer of Ti is coated on the –Z surface of LiNbO₃ crystal by standard evaporation or

sputtering coater, then the Ti-diffusion process has been accomplished by a high-temperature furnace tube with 1035 °C operation, which is slightly lower than the Curie temperature of LiNbO₃ of 1130 °C to avoid for disordering the polarization of single crystal domains. In order to analysis of the relation of mode size with index change distribution for Ti-diffusion single-mode LiNbO₃ waveguides [47], we can follow the modeling processes obtained by Steven K.Korotky et. al. in 1982, they used the model of diffused-type channeled two dimensional waveguide index change distribution proposed by G.B.Hocker in 1977. According to their model, the index change distribution of width and depth direction of Ti-diffused channel waveguide can be expressed by error function profile and Gaussian profile, respectively. The entire distribution of two dimensional index change distribution of +X-propagation, -Z Ti-diffused LiNbO₃ waveguide with width of W and depth of D can be expressed as:

$$n(y, z) = n_{bulk} + \Delta n_0 f(y) g(z) \quad (25)$$

where, n_{bulk} and Δn_0 are the index of LiNbO₃ substrate and the constant of index change, Δn_0 is the fabrication-dependent value with:

$$\Delta n_0 = \left(\frac{dn}{dC} \right) erf \left(\frac{W}{2\sqrt{2}D} \right) \sqrt{\frac{2}{\pi}} \left(\frac{\tau}{D} \right) \quad (26)$$

where, dn/dC and are the function of metal concentration with index change and τ is the coated Ti-layer thickness, which can be decided by fitting multi-wavelength index change measurement (described in **2-3-5-1**) with unit of wavelength is μm . In our model,

$$\frac{dn}{dC} = \frac{0.839}{\lambda^2 - 0.0645} \lambda^2 \quad (27)$$

The diffusion depth D is function of diffusion temperature (T) and diffusion time (t), which is:

$$D = \sqrt{2dt} \quad (28)$$

with diffusion constant $d = d_0 \exp(-\frac{T_0}{T})$, and temperature constant $T_0 = 300$ K.

And, the index profile for transverse (width) and longitudinal (depth) direction are

$$f(y) = \left\{ \operatorname{erf} \left[\frac{1}{\sqrt{2D}} \left(y + \frac{W}{2} \right) \right] - \operatorname{erf} \left[\frac{1}{\sqrt{2D}} \left(y - \frac{W}{2} \right) \right] \right\} / \left[2 \operatorname{erf} \left(\frac{W}{2\sqrt{2D}} \right) \right] \quad (29)$$

$$g(z) = \exp \left[-\frac{1}{2} \left(\frac{z}{D} \right)^2 \right] \quad (30)$$

The transverse mode field of waveguide can be expressed as:

$$E_x(y, z) = \Psi(y) \Phi(z) \quad (31)$$

with the horizontal field distribution

$$\Psi(y) = \frac{1}{\sqrt{(w/2)\sqrt{\pi}}} \exp \left[-\frac{1}{2} \left(\frac{y}{w/2} \right)^2 \right] \quad (32)$$

and longitudinal field distribution

$$\Phi(z) = \frac{2}{\sqrt{d\sqrt{\pi}}} \left(\frac{z}{d} \right) \exp \left[-\frac{1}{2} \left(\frac{z}{d} \right)^2 \right] \quad (33)$$

where, w and d are the width and depth of waveguide mode.

So far, we have enough information, the mode field of waveguide will able to be obtained by putting the index profile as Eq. (25) and the mode field as Eq. (29) into the wave equations as Eq. (15). Then, by using the propagation constant, $\beta = k_0 n_{eff} = k_0 (n_{bulk} + \Delta N)$, the wave equation becomes:

$$\begin{aligned} & \left(\frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \Psi(y) \Phi(z) + [n_{bulk} + \Delta n_0 f(y) g(z)]^2 k_0^2 \Psi(y) \Phi(z) \\ & = (n_{bulk} + \Delta n)^2 n_{bulk} k_0^2 \Psi(y) \Phi(z) \end{aligned} \quad (34)$$

As the experiment, index change of Z-cut Ti-diffused LiNbO₃ waveguides is approximately 0.005~0.01 or 0.01~0.02 for TE or TM polarization modes. But e-

wave substrate index of LiNbO₃ in the 1550-nm telecom band is ~ 2.14 , so, with the condition of $\Delta n(y, z) \ll n_{bulk}$, Eq. (34) can approximate to be

$$\begin{aligned} & \left(\frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \Psi(y) \Phi(z) + 2n_{bulk} \Delta n_0 k_0^2 f(y) g(z) \Psi(y) \Phi(z) \\ & = 2\Delta n n_{bulk} k_0^2 \Psi(y) \Phi(z) \end{aligned} \quad (35)$$

Finally, the mode size of width (w) and depth (d) of waveguide modes are able obtained from Eq. (35) by using the dimensionless equation [47] as Eq. (36):

$$\begin{aligned} & 2n_{bulk} \Delta n W^2 k_0^2 \\ & = \frac{-9}{2 \left(\frac{D}{W} \right)^2 \left(\frac{d}{D} \right)^2} - \frac{4}{2 \left(\frac{W}{W} \right)^2} \\ & + \left[2n_{bulk} \left(\frac{dn}{dC} \right) \tau W k_0^2 \right] \frac{W}{D} \sqrt{\frac{2}{\pi}} \left[1 \right. \\ & \left. + \frac{1}{2} \left(\frac{d}{D} \right)^2 \right]^{-3/2} \text{erf} \left[\frac{1}{\sqrt{\left(\frac{W}{W} \right)^2 + 8 \left(\frac{D}{W} \right)^2}} \right] \end{aligned} \quad (36)$$

This relation of mode size and waveguide size are the function of fabrication parameters, including Ti-layer thickness (τ), diffusion temperature (T), diffusion time (t), waveguide width (W), waveguide depth (D) and the effective index of modes $n_{eff} = n_{bulk} + \Delta n$. One can use the partial differential with mode width (w) or mode depth (d) of Eq. (36) to find maximum waveguide width (W) or depth (D) by the dimensionless strength parameter response diagram as the following example of **Fig. 7** shown.

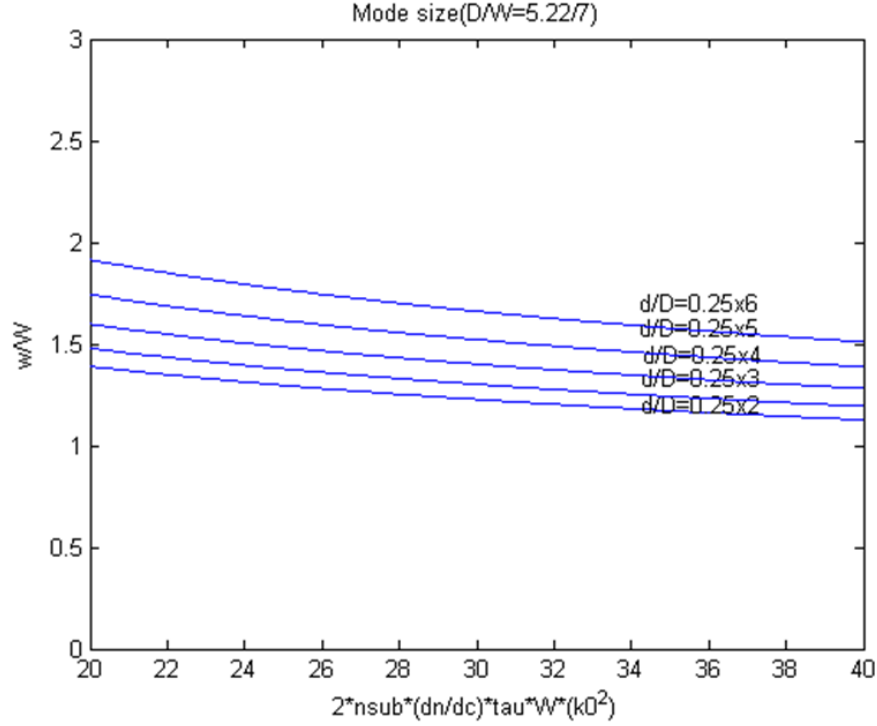


Figure 7. The example of dimensionless strength parameter response of mode size of our Ti-diffused LiNbO₃ waveguides fabrication parameters, which is calculated via Eq. (36).

The blue curves corresponds to the relation of mode size (w, d) and waveguide size (W, D) under different fabrication parameters ($\tau, T, t, dn/dc$) and working conditions (λ, n_{bulk}).

2-3-4-2 Proton-Exchange (PE) type waveguides

The proton-exchange (PE) type waveguides use the exchange the Li⁺ of LiNbO₃ by proton (H⁺) to create relative high index distribution in comparison with unexchange region to form the optical waveguides. The chemical formula of PE process is $\text{LiNbO}_3 + x\text{C}_6\text{H}_5\text{COOH} \longleftrightarrow \text{Li}_{1-x}\text{H}_x\text{NbO}_3 + x\text{C}_6\text{H}_5\text{COOLi}$, which the x is the exchange rate, the different x represent different crystal structure of exchanged LiNbO₃, with the structure of Li_{1-x}H_xNbO₃, the structure can be divided into 3 main types, the low-loss, monocrystalline, α – *phase*, the transition κ_1, κ_2 *phases*, and the high-loss, polycrystalline, $\beta_1, \beta_2, \beta_3, \beta_4$ *phases*, with the $0.01 < x < 0.1$, $0.2 < x < 0.5$ and $0.5 < x < 0.85$, respectively. The

refraction index change of extraordinary (Δn_e) for each phases are approximately $\Delta n_e \leq 0.03$ and $0.05 \leq \Delta n_e \leq 0.12$ for $\alpha - phase$ and $\beta - phase$, respectively. In generally, to form the loss-loss and reserve the nonlinear coefficients, the $\alpha - phase$ is the best choice due to the characteristics of the monocrystalline, but according to the earlier literature, the standard proton-exchange process will accumulate excessively protons on the waveguide surface of LiNbO_3 cause the phase maintains at $\beta - phase$, instead of $\alpha - phase$ [48]. There are exists two main solutions for overcome this issue, the thermal annealing [49] or buffered exchange [50], the thermal annealing method is a two-step processes, the waveguides will be needed to anneal after proton-exchanged process by thermal annually diffusion to decrease the concentration of proton on the exchanged surface and increase the depth of waveguides. After annealing, the index profile of exchanged region will become diffusion-type profiles rather than the step-like index profile of PE process. The crystal phase also becomes simpler monocrystalline, the $\alpha - phase$ after annealing. The another method to form $\alpha - phase$ is the buffered exchange method, called SPE method, by incorporate some buffer medium in exchange solution to increase the concentration of Li^+ in the exchange solution for decelerating the exchange rate, this method is the single step process as PE process, but needs more fabrication duration since the lower exchange rate.

The modeling of APE waveguide can be derived by diffusion theory [49], first, we start from the exchange depth d_{PE} of PE process under exchange time t_{PE} under temperature T_{PE} is

$$d_{PE} = \sqrt{4D_{PE}(T_{PE})t_{PE}} \quad (37)$$

The D_{PE} represents the diffusion temperature-dependent diffusion coefficient,

which is

$$D_{PE}(T_{PE}) = D_0 \exp\left(-\frac{E_a}{kT_{PE}}\right) \quad (38)$$

with Boltzmann coefficient, k the active energy, E_a and the initial diffusion coefficient, D_0 .

The step-like index profile of PE process can be considered as the diffusion source of annealing process. The index profiled $\Delta n_a(x)$ of x-direction APE diffusion waveguides are the function of the concentration of H^+ at the diffused region $C(x)$, with is

$$\Delta n_a(x) = \xi \Delta n_{PE} C(x) / C_{sur,PE} \quad (39)$$

with the coefficient $\xi = 0 \sim 1$ and the surface proton concentration of PE process, $C_{sur,PE}$.

According to the 1D diffusion equation, the index profile of APE for x-cut 1D planar waveguide is

$$\Delta n_a(x) = \frac{\Delta n_{PE}}{2 \operatorname{erf}(d_{PE}/d_a)} \left[\operatorname{erf}\left(\frac{d_{PE} + x}{d_a}\right) + \operatorname{erf}\left(\frac{d_{PE} - x}{d_a}\right) \right] \quad (40)$$

with the waveguide the effective waveguide depth of d_a .

Similarly, we can also model the x-cut channel APE waveguide index distribution by 2D diffusion equations, which is

$$n_a^2(x, z) = n_b^2 + 2\Delta n_a n_b f(x) g(z)$$

where, n_b is the substrate index, and the vertical and horizontal index profile are

$$\begin{aligned} f(x) &= \frac{1}{2 \operatorname{erf}\left(\frac{d_{PE}}{d_a}\right)} \left[\operatorname{erf}\left(\frac{d_{PE} + x}{d_a}\right) + \operatorname{erf}\left(\frac{d_{PE} - x}{d_a}\right) \right] \\ g(z) &= \frac{1}{2 \operatorname{erf}\left(\frac{w}{2d_a}\right)} \left[\operatorname{erf}\left(\frac{w/2 + z}{d_a}\right) + \operatorname{erf}\left(\frac{w/2 - z}{d_a}\right) \right] \end{aligned} \quad (41)$$

From the above analysis, we can select different exchange time, exchange temperature, annealing time and annealing temperature to optimize the index profile for pursuing the low-loss, single-mode or multimode operation with

different working wavelength of all kinds of applications. The **Fig.8** shows the simulation result of standard APE normalized index profile with single-mode operation around 1550-nm band.

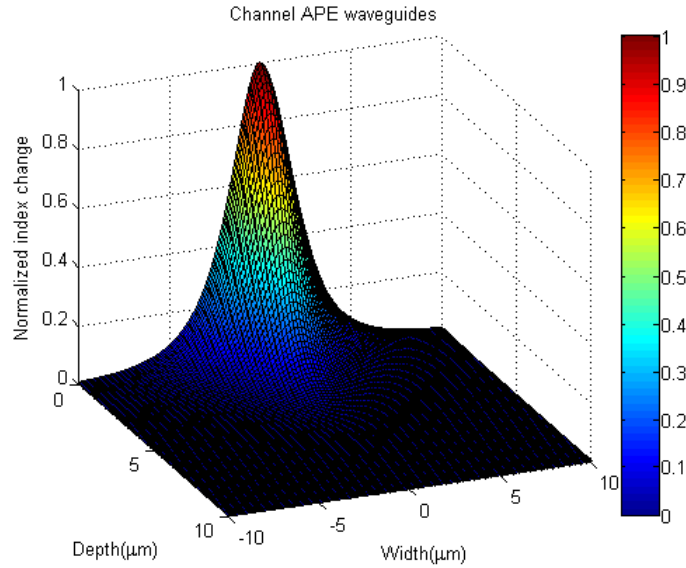


Figure 8. The classical normalized index change profile of channel APE LiNbO₃ waveguides.

2-3-5 Test method

During the qualification of waveguide, we use several methods to determine the waveguide parameters, including the measurement of index profile for diffused waveguides (2-3-5-1), mode size and its aspect ratio (2-3-5-2), mode coupling efficiency between fiber to waveguide and free-space to waveguide (2-3-5-3) and the losses of entire systems from source to end-fiber (2-3-5-4) in this paragraph.

2-3-5-1 Index profile measurement

The main measurement method for index profile of diffused waveguide is using the prism coupler with IWKB method [51]. The working principle of prism coupler is based on the evanescent wave coupling by rotating the incident angle of probe laser beam to testing sample, if there exists some special incident angles can excite the guiding modes of testing sample, the detected power of prism

coupler will be reduced, and according to the prism index and prism angle, we can calculate the effective index of guiding modes and their effective depth by these special angles under different probe laser wavelengths. With the solution of IWKB method, we can estimate the index profile of waveguides in depth direction of 1D planar waveguides by several effective index of guiding modes. The 1D index profile can also be used for estimation the effective index of 2D channel waveguide modes by the Effective Index Method introduced in section (2-3-3-1). We use the commercial prism coupler Metricon 2010/M for testing our samples. The measurement interface shows the rotation steps and received power of the detector as the **Fig. 9** shown, the dip of **Fig. 9** is the relative position of effective index of guiding mode, which can be convert to the effective index.

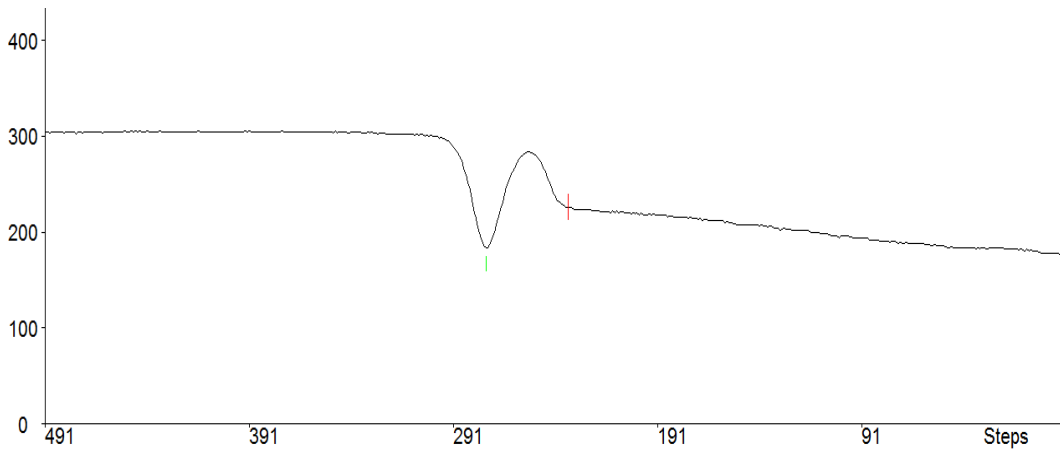


Figure 9. The prism coupler measurement result of standard single-mode waveguide.

2-3-5-2 Mode size and aspect ratio

The mode size of waveguide modes are defined by the mode-field diameter, MFD, by the definition, the MFD is the width of normalized power density reduced to $1/e^2$ of the maximum power position. One can use the MFD to qualification the aspect ratio and coupling efficiency between different interface such as fiber-to-waveguide or free-space-to-waveguide. In experimentally measurement, the MFD can be quantified by the transverse-offset method [52], which is using the

fiber pigtailed scanning at the output end of waveguides to infer the MFD of modes. The measurement results of our standard single-mode waveguide modes are shown as Fig. 9, the transverse mode profile is symmetrical as **Fig. 10 (a)**, but due to the asymmetric characteristics in depth direction of diffused waveguides, the mode is asymmetrical in vertical, so, there are two conditions are used to fit the vertical profile of mode as **Fig. 10 (b)** shown.

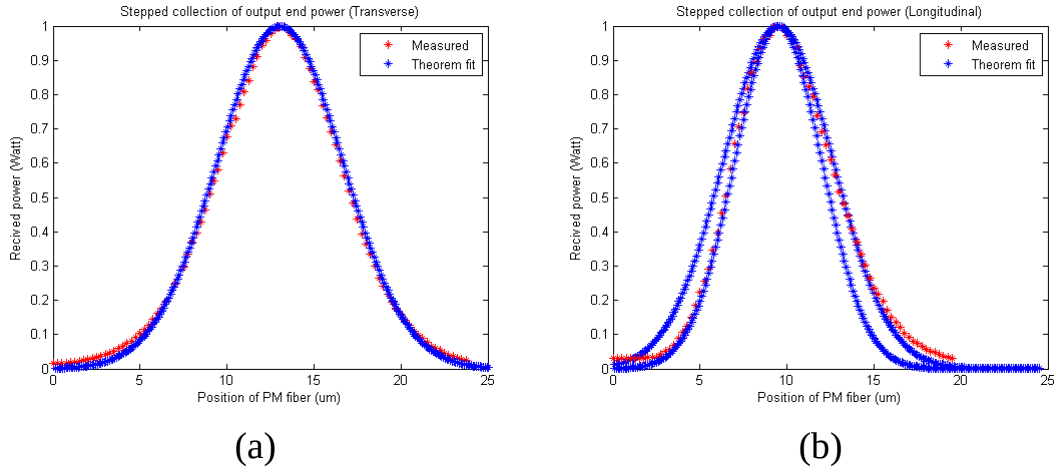


Figure 10. The standard mode fields of diffused LiNbO₃ waveguide in horizontal (a) and in vertical sections (b) with the measured (red) and theorem fitting (blue) results.

The mode aspect ratio (ar) is defined as transversely mode size (MFD_{trans}) divided by longitudinally mode size (MFD_{long}) with $ar = MFD_{trans}/MFD_{long}$ and the elliptical ratio of mode is defined as $er = MFD_{L,up}/MFD_{L,down}$, where $MFD_{L,up}$ and $MFD_{L,down}$ are the mode sizes of up section and down section of asymmetrically modes as the **Fig. 10 (b)** shown.

2-3-5-3 Coupling efficiency

For any integrated optical circuits, the coupling efficiency between interfaces is very critical parameter for developing the low-loss, high-efficient optical devices,

these interfaces are mainly two kinds, the fiber-to-waveguide, or the free-space to waveguide. For fiber telecommunication applications, the fiber-pigtailing of the optical waveguide devices requires highly skilled work for keeping the smallest coupling loss from interfaces. The coupling efficiency can be quantified by calculating the mode size matching parameters, the overlapping between fiber modes and waveguide modes is approximately by [53] :

$$\Upsilon = \frac{4 \left[\left(\frac{4}{w_f^2} + \frac{1}{w_u^2} \right)^{-1/2} + \left(\frac{4}{w_f^2} + \frac{1}{w_d^2} \right)^{-1/2} \right]^2}{\left(\frac{1}{w_f^2} + \frac{1}{w_{trans}^2} \right) w_f^2 w_{trans} w_{long}} \quad (42)$$

where, w_f is the fiber mode size, $w_u = MFD_{L,up}$, $w_d = MFD_{L,down}$ and $w_{trans} = MFD_{trans}$ are the waveguide mode size of up-section and down-section of longitudinally mode profile and the transversely mode profile.

On the other hand, under similar considerations, the free-space coupling efficiency of normal incidence can be modeled by calculating the overlapping integral between the focused beam spot size and the waveguide mode sizes, the microscope objective beam spot size (S.D.), the focused beam spot size is the function of wavelength and Numerical Aperture (N.A.) of the objective as Eq.(43).The suitable objective is 10X with NA=0.1~0.3, which is almost

matching the mode size of LiNbO₃ waveguide single mode in the telecom wavelength range.

$$S. D. = 1.22 \frac{\lambda}{N.A.} \quad (43)$$

2-3-5-4 Losses

There is another key parameter for developing the practical waveguide devices, the losses. The losses of the LiNbO₃ integrated optical devices mainly contain coupling loss, absorption loss, scattering loss, radiation loss and the photorefractive induced loss [54]. The overall loss of an optical component is called insertion loss, which can be calculated in dB scale as Eq. (43)

$$\text{Insertion loss (dB)} = -10 \log \left(\frac{P_{out}}{P_{input}} \right) \quad (44)$$

where P_{input}, P_{out} is the input power and output power with before and after component, respectively.

According to the above description, one can conclude all loss sources to the insertion loss calculation as the following Eq. (45)

$$\text{Insertion loss (dB)} = C. L. + A. L. + S. L. + R. L. + P. L. \quad (45)$$

with C. L. = Coupling loss, A. L. = Absorption loss, S. L. = Scattering loss, R. L. = Radiation loss and P. L. = Photorefractive – induced loss. The overall loss of absorption, scattering and radiation losses can be called propagation loss in unit of (dB/cm), we also can use the loss coefficient α in the unit of (cm^{-1}), the conversion equation is $(dB/cm) = 4.34 \alpha (cm^{-1})$. The main causes of each loss of propagation loss are the electrodes/substrate absorption, the index change discontinuity of waveguides and the bending of waveguides for absorption, scattering and radiation losses, respectively. To reduce

these losses, one can try to study the topological of waveguides, the fabrication parameters, the electrode topology, and the substrates absorption characteristics.

2-4 Nonlinear optical processes

In this section, I further introduce the main physical phenomenon used in the most of devices of this thesis, which is *optical nonlinearity*. In the concept of optical nonlinearity, one can easily regard it as the concomitant higher order dipole oscillation from strong input pump interaction with dipoles of crystal. With point out three nonlinear conversion processes of Sum-Frequency Generation (SFG), Spontaneous Parametric Down-Conversion (SPDC), Optical Parametric Oscillator (OPO) and the necessary phase-matching conditions of Birefringence Phase-Matching (BPM), Quasi-Phase-Matching (QPM) or Multi-Quasi-Phase-Matching (Multi-QPM) in the paragraph of **(2-4-1)** to **(2-4-4)**, respectively, we designed and demonstrated the several nonlinear wavelength conversion devices in this thesis in the bulk or waveguide devices, which are summarized in the **chapter 3** and **5**, These nonlinear wavelength conversion processes can be analyzed by theoretical model shown as nonlinear coupled-mode equations, which is described in the paragraph of **(2-4-5)**, and these devices are fabricated by the standard electric poling method described in the paragraph of **(2-4-6)**.

2-4-1 Nonlinear optical generation

In line with signal wavelength in relative to pump wavelength, the wavelength conversion of nonlinear optical processes are divided to *up-conversion* and *down-conversion* [55]. If signal wavelength is shorter than pump wavelength, which means that signal frequency is higher than pump frequency, the processes are called up-conversion, in the other hand, if pump frequency is

higher than signal frequency, these processes are called down-conversion. In the point of energy conversion, these two processes can be regarded as one of the two-sided reverse process for another process. The up-conversion processes is the results of the combination of two to many relative lower energy pump photons to generate a relative higher energy signal photons, in the category of second-order nonlinearity, the most general process is called Sum-Frequency Generation (SFG), which can be interpreted as lower energy two pump photon to generate one higher energy signal photon. If the two pump photons of SFG process are identical in the wavelength and polarization, the process is called Second Harmonic Generation (SHG), which is the world's first experimental process in wavelength conversion of optical nonlinearity. In the other hand, the down-conversion process is based on one relative higher energy pump photon to split into two relative lower energy signals, in order to distinguish two signals, the shorter wavelength photon is called *Signal*, and higher wavelength photon is called *Idler*. The basis process of down-conversion is Spontaneous Parametric Down-Conversion (SPDC), or Optical Parametric Generation (OPG) in nonlinear optics, the SPDC has been used to generate the non-classical quantum photon source over the past decade, the characteristic of SPDC generated by Ti-diffused periodically poled LiNbO₃ is summarized in the **chapter 5**.

2-4-1-1 Sum-Frequency Generation, SFG

The energy conversion of up-conversion under second-order nonlinearity, including SHG, SFG, is represented by the following formula:

$$h\nu_{p1} + h\nu_{p2} = h\nu_s \quad (46)$$

,which is equivalent to $\frac{1}{\lambda_{p1}} + \frac{1}{\lambda_{p2}} = \frac{1}{\lambda_s}$, in the form of wavelengths.

One can immediately learn that if $\lambda_{p1} = \lambda_{p2} = \lambda_{FW}$, then $\lambda_s = \lambda_{FW}/2 = \lambda_{SH}$, which is called Second Harmonic Generation (SHG) or frequency doubling.

The coupled-mode equation of z-propagation SFG processes can be obtained as Eq. (47).

$$\begin{aligned}\frac{dA_s}{dz} &= -j\kappa_1\Theta^*A_p^*A_{SFG}\exp(-j\Delta\beta z) - \frac{\alpha_s}{2}A_s \\ \frac{dA_p}{dz} &= -j\kappa_2\Theta^*A_s^*A_{SFG}\exp(-j\Delta\beta z) - \frac{\alpha_p}{2}A_p \\ \frac{dA_{SFG}}{dz} &= -j\kappa_3\Theta A_sA_p\exp(+j\Delta\beta z) - \frac{\alpha_{SFG}}{2}A_{SFG}\end{aligned}\tag{47}$$

where, the phase-mismatching $\Delta\beta = \beta_{SFG} - \beta_p - \beta_s$, Θ is the overlapping integral of modes, and the coupling coefficients of processes are $\kappa_i = \left(\frac{8\pi^2 d_{eff}^2}{n_1 n_2 n_3 c \epsilon_0 \lambda_i^2}\right)^{\frac{1}{2}}$, α_s , α_p , α_{SFG} are the loss of signal, pump and SFG waves, respectively. The intensity and power of interaction modes are $I_i = 2n_i \epsilon_0 A_i^* A_i$, $P_i = I_i w_{x,i} w_{y,i}$ with mode size of $w_{x,i}$, $w_{y,i}$. One can simulate the power evolution of the SFG processes during interaction by solving the Eq. (47).

In principle, researchers can generate much higher-order harmonic wave by using cascade circuits to repeat the up-conversion process. For example, these processes can be used to generate the DUV, EUV or X-ray in the industrial use of lithography or imaging. Because the SFG process can be regarded as the reverse process of SPDC under the same phase-matching condition, in this thesis, the SFG process is used to verify the SPDC quantum photon source by measuring the spectrum of SFG signals and the temperature dependence of signal wavelength,

the results of simulation and experiment are summarized in the section (5-2) of chapter 5.

2-4-1-2 Spontaneous Parametric Down-Conversion, SPDC

The optical parametric generation (OPG) in the nonlinear optics is called the spontaneous parametric down-conversion process (SPDC) in the quantum optics field, the conservation of energy of SPDC photons can be by simply reform the Eq. (48).

$$h\nu_p = h\nu_s + h\nu_i \quad (48)$$

,which is equivalent to $\frac{1}{\lambda_p} = \frac{1}{\lambda_s} + \frac{1}{\lambda_i}$, in the form of wavelengths.

In comparison with SFG process, the biggest advantage of SPDC is the signal wavelength tunable ability, in the second-order nonlinearity, the pump of down-conversion can be divided to two any lower energy signals (called signal and idler) if it obeys the energy conservation principle. In the other words, because of the conservation of energy, which means that the SPDC signal and idler photons are able to generate at the same time within the same device but it can be generated in very many different wavelengths (longer than pump wavelength) in different combinations of energy conservation in different working conditions . From the tunability point of view, the SPDC signal and idler is powerful for generating various wavelengths of laser in the same system, but there are one obvious shortcomings of SPDC, that is the efficiency. Because of the SPDC is a single-pass spontaneous process, the efficiency of SPDC is relatively quite low in comparison with up-conversion processes. With an improvement method, one can use the optical parametric amplification (OPA) process by input a small signal to seed the down-conversion process, this process is a stimulated amplification, which can enhance the efficiency of single-pass down-conversion process.

2-4-1-3 Optical Parametric Oscillator, OPO

In addition to use a single-pass optical amplifier process, one can refer to the configuration of laser oscillator by putting mirrors to oscillate SPDC signal or idler. The simplest way is linear bi-directional resonance by using two mirrors in the two end of SPDC generation devices to form a linear oscillator, as the **Fig. 25** shown in the section 3-2 of chapter 3. In the oscillation configuration, one can choose to build into be single oscillation or double oscillation, in case of only signal (or idler) is resonated or signal and idler are all resonated, respectively. In the linear oscillator, there are nodes existed, these nodes is due to the presence of standing waves between mirrors, which are produced more than one longitudinal modes, let the spectrum of OPO to become more broader. The existing solution is using transverse-wave oscillator, which can be formed by the configuration of ring cavity [56]. Our laboratory is building a new approach by combining the EO polarization conversion ability with nonlinear wavelength convertor of special designed domain-engineered LiNbO₃ crystal in only two mirrors linear oscillator scheme. In preliminary results of EO spectrum tailorable intracavity optical parametric oscillator of two mirrors linear oscillator are summarized in the section 3-2 of **chapter 3**.

2-4-2 Birefringence Phase-Matching, BPM

Continuation of the discussion of the previous section, the nonlinearity of optics is original from the behavior of higher order oscillation of dipoles, there is a main factor to limit the efficiency of nonlinear process, which is momentum mismatch of participated different frequency photons in different phase velocity in nonlinear medium. This phenomenon is called *phase-mismatching*. For SFG process, when phase velocities of pump and signal photons are different to each other, the energy from pump photons cannot continually transfer to signal photons

due to the walk-off in the time domain during the propagation in the nonlinear medium, which causes that the efficiency becomes very low. In order to reach the phase-matching, the momentum conservation of nonlinear processes must be satisfied, such as the following general form of SFG process:

$$\vec{k}_{p1} + \vec{k}_{p2} = \vec{k}_s \quad (49)$$

In the collinear SHG processes, the Eq. (49) is easily reform as:

$$n_{SH} = n_{FW} \quad (50)$$

,which the n_{SH} , n_{FW} is the index of harmonic wave and fundamental pump wave. This special and unique condition of phase-matching immediately leads to a problem appeared, which is seems impossible to be satisfied in the dispersion medium. Indeed, according to the dispersion relation of refractive index in different wavelengths in homogeneous isotropic medium, there is no easy way. But fortunately, we can find this special condition in the anisotropic medium. Because of there are existed the extraordinary refractive index in the birefringence medium, one all needs to do is to find the correct angle of incidence to pump the birefringence crystal, thus, in this way, the phase matching is called *Birefringence Phase-Matching (BPM)*. The extraordinary refractive index is the function of incidence angle (θ, ϕ) which is relative to the crystal orientation and crystal temperature (T) , so the birefringence phase-matching condition of SHG process can be rewritten as:

$$n_{SH}(\theta, \phi, T) = n_{FW}(\theta, \phi, T) \quad (51)$$

In the specific incidence angle, one can reach perfect phase velocity matching, but there are accomplished several disadvantage because of the fixed incidence angle under specific working temperature. Due to the nonlinearity coefficients (d) are the function of polarization and crystal orientation, in most cases, there is a

trade-off between to seek the largest nonlinearity coefficients and to be perfect phase matching. Another key issue is energy walk-off in spatial, this phenomenon is original from that the pump and signal energy flows propagation in different directions in the anisotropic medium.

2-4-3 Quasi-Phase-Matching, QPM

To overcome the walk-off and other issues of birefringence phase-matched (BPM) process during the efficient nonlinear conversion, the quasi-phase-matching (QPM) method has been developed for generating high efficiency nonlinear conversion via selected higher nonlinear coefficient with specific collinear conversion during the nonlinear interaction. The QPM process is based on the reciprocal lattice vectors produced by the crystalline domains change in spatially, which is participated in the interaction with the optical waves to compensate the phase difference between waves during the nonlinear conversion processes. The concept of QPM can be understood by the simple model of the reciprocal lattice vectors in k-space, which can be calculated from the crystalline domains distribution in spatially by the Fourier transform. The phase-mismatch ($\Delta\vec{K}$) of interacted optical waves during second-order nonlinear conversion can derived as Eq. (53) with the k-vectors of pump, signal 1 and signal 2, $\vec{k}_p, \vec{k}_{s1}, \vec{k}_{s2}$, respectively.

$$\Delta\vec{K} = \vec{k}_p - \vec{k}_{s1} - \vec{k}_{s2} \quad (53)$$

If we choose the collinear conversion processes, the Eq. (53) becomes

$$\Delta K = 2\pi \left(\frac{n_p(T)}{\lambda_p} - \frac{n_{s1}(T)}{\lambda_{s1}} - \frac{n_{s2}(T)}{\lambda_{s2}} \right) \quad (54)$$

One can immediately find that the Eq. (53) becomes the temperature and wavelengths-dependent only, so, if we deliberately choose the corresponded domain distribution with period of Λ to form the m-order reciprocal lattice vectors of $K = 2\pi m/\Lambda$ in k-space to compensate the phase-mismatching term of ΔK , the government equation of QPM processes becomes Eq. (55).

$$\frac{m}{\Lambda} = \left(\frac{n_p(T)}{\lambda_p} - \frac{n_{s1}(T)}{\lambda_{s1}} - \frac{n_{s2}(T)}{\lambda_{s2}} \right) \quad (55)$$

with $m=1,3,5,\dots$ for first-, third- and fifth- order reciprocal lattice vectors, respectively. The reason of chose the odd-order reciprocal lattice vectors is if using the even-order vectors in QPM processes will let the conversion offset each other to let the overall conversion efficiency equals zero.

2-4-3-1 (A)Periodically Poled Lithium Niobate, (A)PPLN

In the practical implementation of QPM processes, one can deliberately arranged periodically domains with positive and negative nonlinear coefficients of single-axis crystals to form the spatial grating with period of Λ . In these thesis, we use the standard poling technique to arrange the domain-axis of LiNbO_3 crystals, if the periodic is unique, called Periodically Poled Lithium Niobate (PPLN). If the poled LiNbO_3 crystals combined with several domain gratings, which provides more than one reciprocal lattice vectors with $\vec{K}_i, i = 1, 2, \dots$, one can call it as Aperiodically Poled Lithium Niobate (APPLN).

2-4-3-2 QPM in domain-engineered LN waveguides

In the waveguide system, the QPM process is used to deal with the phase-mismatch of the propagation constants of optical modes during the interactions. The Eq. (56) shows the QPM condition of second-order nonlinear conversion

within waveguide modes of pump, signal 1 and signal 2 with propagation constants of $\vec{\beta}_p, \vec{\beta}_{s1}, \vec{\beta}_{s2}$, respectively.

$$\vec{K} = \vec{\beta}_p - \vec{\beta}_{s1} - \vec{\beta}_{s2} = 0 \quad (56)$$

Due to dimensions of the optical modes of waveguides between pump and signals will possible be very different, the nonlinear conversion efficiency strong depends the mode overlapping between interacted modes as the role Θ played in Eq. (47).

2-4-4 Multi-QPM design

The multi-QPM structure provides the possibility to satisfy several QPM processes in a specific superlattice, the main method used in the past literature are the Cascaded periodic structure, CPS [57], Fibonacci optical superlattice, FOS [58], Continuous phase modulation, CPM [59], Non-periodic optical superlattice, NOS [60] and Aperiodic optical superlattice, AOS [61]. CPS usually suffers from low normalized conversion efficiency; while FOS and CPM methods have low freedom of spectral design; though the NOS method has been shown higher optimizing conversion efficiency and spectral fidelity in comparison with the AOS method, but it costs much longer computation time in typically. In this thesis, we chose the AOS technique to construct the devices because of its high freedom and working efficiency, **Fig. 12** shows the flow chart of the simulated-annealing algorithm, which has been used to optimize the AOS structure in this thesis.

2-4-4-1 Aperiodic Optical Structure, AOS

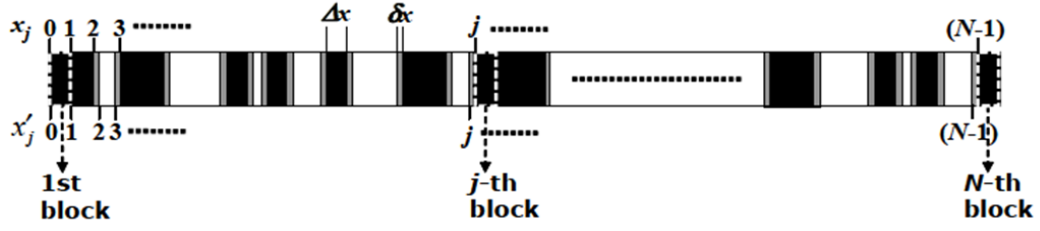


Figure 11. The schematic diagram of AOS structure for N blocks with unit domain length Δx and domain errors δx .

The AOS structure is consisted by a series of lattices with positive or negative domains in aperiodically to form multi-reciprocal lattice vectors as the **Fig. 11** shown, the black regions are the inverted-domains, called “negative domains” and the white regions are the original domains, called “positive domains” conversely. The many kinds of multi-reciprocal lattice vectors are formed by different permutations and combinations of positive and negative domains. One can use these selected reciprocal lattice vectors to satisfy the phase-matching conditions of several QPM processes through a single AOS structure. In the recent studies, the AOS has been successfully expended to 2D aperiodic structure to simultaneously provide orthogonal and diagonal reciprocal lattice vectors for several applications [62].

2-4-4-2 Simulated-Annealed method, SA

To optimize the aperiodic domain structure, we use the simulated-annealed method for searching the maximum efficient aperiodic structure of selected multi-QPM processes. For example, if we want to design a multi-wavelength QPM processes within an AOS structure, we can calculate the AOS structure by simulated-annealed method through the efficiencies of selected QPM processes of calculated AOS structure integrates into an *Objective Function*, OF as the Eq. (57) with the target efficiencies $\eta_{target}(\lambda_q)$, the efficiencies of recent calculated

structure $\eta(\lambda_q)$, the weighting of each QPM process efficiencies $w(\lambda_q)$ and the balance factor β for controlling the relatively efficiencies between different QPM processes [49].

$$\begin{aligned}
 OF = & \sum_{q=1}^M (|\eta_{target}(\lambda_q) - \eta(\lambda_q)|w(\lambda_q)) \\
 & + \beta [Max(\eta(\lambda_1), \eta(\lambda_2), \dots, \eta(\lambda_M)) \\
 & - min(\eta(\lambda_1), \eta(\lambda_2), \dots, \eta(\lambda_M))]
 \end{aligned}
 \tag{57}$$

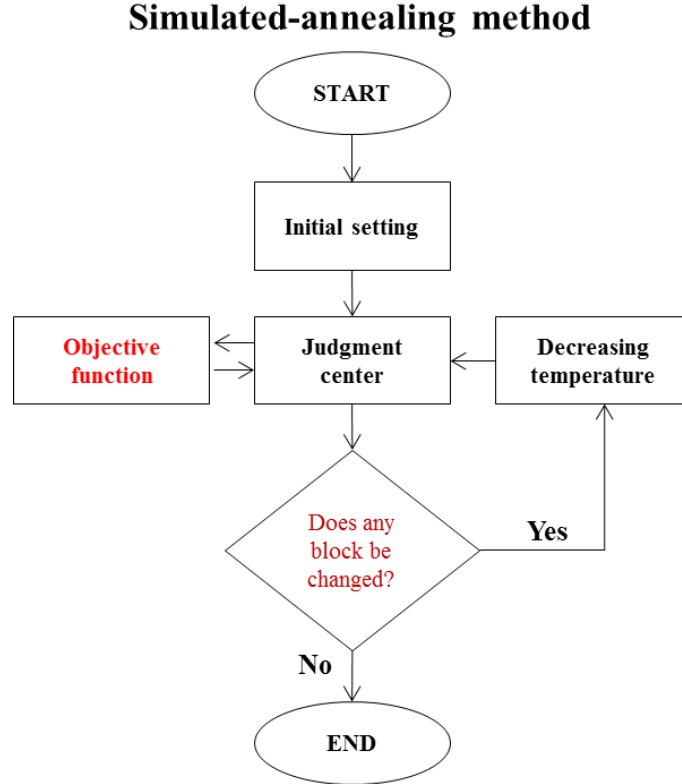


Figure 12. The design flow-chart of simulated-annealing method for AOS optimization.

The flow-chart of simulated-annealing algorithm is shown as **Fig. 12**, the initial setting of algorithm includes the working wavelengths, working temperature, target efficiencies, the relatively efficiencies between several QPM processes, the simulated-annealing temperature and the cooling rate of algorithm. It is worth

mentioning that we insert an escape function into the judgment center of algorithm for avoiding the final solution been trapped at local optimal solution instead of possible global optimal solution.

2-4-5 Fabrication of domain-engineered LiNbO₃ devices

We use standard μm -level photolithography technique in clear room to transfer the designed pattern from mask to the LiNbO₃ 3-inch wafer or chips, and the periodic or aperiodic structures are poled by standard electric-field poling technique [63]. The photolithography is based on the processes of photoresist-coating, soft-bake, mask alignment, exposure, hard-bake and development, after lithography, the designed patterns can be transfer to the photoresist structures on the substrate surface, and after baking, the photoresist becomes the insulating layer during the poling process to define the poled or unpoled regions. It must be noted, the poling procedure of LiNbO₃ waveguide devices must be in accordance with the type of waveguide to avoid the reverse poling phenomenon, metal-diffused types especially. Our standard production flow-chart of Ti-diffused PPLN waveguide devices is shown as **Fig. 13**. We start the fabrication processes from lithography I of channel waveguide geometric distribution definition, then after titanium thin-film coating and lift-off, we use the thermal diffusion process to form the Ti-diffused channel waveguide on the surface of LiNbO₃. The poling process is followed by the lithography II, the lithography II is used for defining poling domains, after the waveguide and poling fabrication processes, and we can add the optional processes for electrode fabrication on some devices for specific EO modulation applications, which are completed by lithography III to V and the coating / lift-off processes. The final step of chip fabrication is optical-grade polishing and sample cutting to the design size. In addition, for packaged devices,

one can use the end-fiber pigtailling, housing and package techniques to integrate the LiNbO₃ sample with the fiber systems.

Production flow-chart of Ti-WG PPLN devices

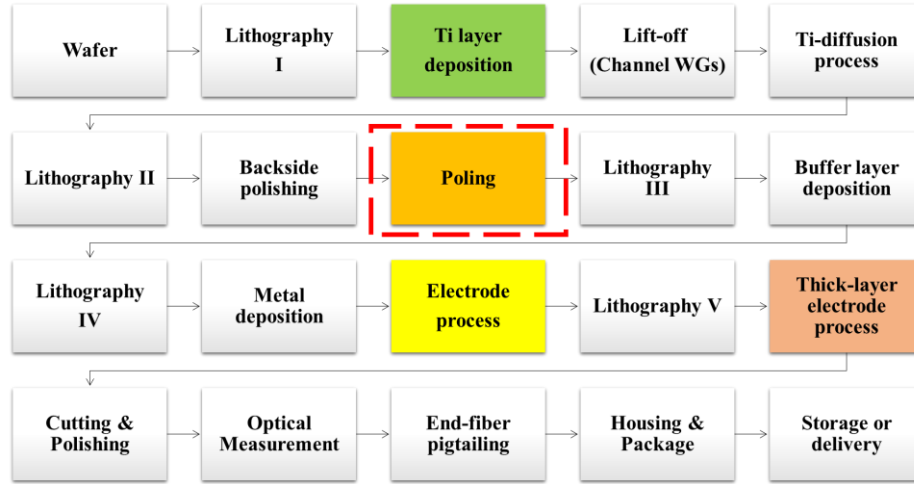


Figure 13. The standard fabrication processes of Ti-diffused PPLN waveguide devices.

2-5 Summary

In this chapter, I organized and introduced the main background, principles of optical nonlinearity and electro-optical effect, quasi-phase matching conditions, device applications, design methodology, coupled-mode equations, simulation and optimization algorithm for multi-QPM aperiodic structures, the fabrication parameters analysis for diffused-type LiNbO₃ waveguides, and the poling process of LiNbO₃ bulk or waveguide samples. These concepts and methods are used in the following chapters and throughout this thesis.

Chapter 3 、Aperiodic Optical Structure in EO APPLN

OPG/OPO

In chapter 3, I summarized two of my major study of electro-optical spectrum adjustable optical parametric generator and oscillators (EO OPG/OPO) with **thesis topic I:** Electro-optically tunable, multi-wavelength optical parametric generators in aperiodically poled Lithium Niobates in the section 3-1, and the **thesis topic II:** Electro-optically spectrum tailorable intracavity optical parametric oscillator in the section 3-2.

The results of these study were published in [63] *Optics express* 20.27 (2012): 28989-29001 and [64] *Optics letters* 40.22 (2015): 5132-5135, respectively.

Thesis topic I:

Y. H. Chen, H. P. Chung, W. K. Chang, H. T. Lyu, J. W. Chang, and C. H. Tseng. "Electro-optically tunable, multi-wavelength optical parametric generators in aperiodically poled lithium niobates." *Optics express* 20.27 (2012): 28989-29001.

Thesis topic II:

H. P. Chung, W. K. Chang, C. H. Tseng, R. Geiss, T. Pertsch, and Y. H. Chen. "Electro-optically spectrum tailorable intracavity optical parametric oscillator." *Optics letters* 40.22 (2015): 5132-5135.

3-1 Topic I: EO tunable, multi-wavelength OPGs in APPLN

3-1-1 Multi-wavelength EO phase-matching tuning

The wavelength-tuning methods in OPG/OPO systems can be classified into these main groups, including the thermal [65], mechanical [66-68] and electro-optical controls. Since a fast, precise, and stable spectral tuning technique has been wide used in many applications, electro-optic (EO) tuning of a multi-wavelength light source has been developed in this work. There are two main types of the EO control methods, the “*Refractive index variation method*” [69] and the “*Asymmetric domain structure method*” [70-71], and second one has been used in a periodic QPM OPG/OPO device. Because of the use of aperiodic structure in this work, we will extend the concept of “asymmetric domain structure” to a multiple wavelength converter, a single asymmetric domain is shown as the **Fig. 14**.

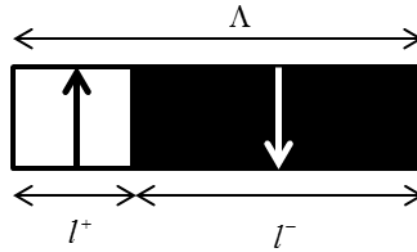


Figure 14. The principle of asymmetric domain structure method.

The period of the structure is Λ , the length of “*Positive domain* (which is unpoled)” is l^+ , and the length of “*Negative domain* (which is poled)” is l^- . The QPM condition for this structure can be derived as Eq. (58), where $\Delta k(0)$ is the phase mismatch under $E_z = 0$. If l^+ is equal to l^- , the QPM condition is independent of the applying E_z -field, so one can design the asymmetric domain structure in deliberately to modify the QPM condition by varying the

applying E_z -field. Further, we can use the Aperiodic Optical Structure, AOS to generate the multi-wavelength optical parametric generation signals.

In the study, we derives the EO wavelength-tuning mechanism into an AOS structure to demonstrate the electro-optically tunable, multi-wavelength converter. In AOS technique, structure is constructed by a sequence of domain blocks with each a thickness of Δx and a domain polarity is the $+z$ or $-z$ crystal orientation. The EO wavelength-tuning mechanism in an aperiodic structure can be understood by the calculating the phase error accumulated over the device length of L as Eq. (58) and the phase-matching conditions for two modulation techniques as Eq. (59) with technique-I (ΔL) or technique-II (ΔN) methods.

$$\Delta\phi(E_z) = (\Delta k_i(0) - K_i)L + \delta k(E_z) \sum_{j=0}^{N-1} s(x_j)\Delta x \quad (58)$$

where, $\sum_{j=0}^{N-1} s(x_j)\Delta x = (L^+ - L^-) = (N^+ - N^-)\Delta x$, $s(x) = \pm 1$, and $\Delta k_i(0)$, $\delta k(E_z)$ and K_i are the phase-mismatching at $E_z = 0$, induced phase-mismatching when $E_z \neq 0$, and reciprocal vectors of AOS, respectively.

$$\begin{aligned} \Delta k_{AOS}(E_z) &= (\Delta k_i(0) - K_i) + \delta k(E_z) \frac{L^+ - L^-}{L} \\ &= (\Delta k_i(0) - K_i) + \delta k(E_z) \frac{(N^+ - N^-)\Delta x}{L} = 0 \end{aligned} \quad (59)$$

If the total lengths of positive and negative domains are the same, we find that the QPM condition will be independent of the E_z -field, which means we cannot tune the phase-matched wavelength by varying the E_z -field. The same idea can be applied to the difference of the block numbers of the positive and negative domains. Therefore, in this work, we

proposed two methods to enhance (ΔL) or (ΔN) value of domain structure.

3-1-2 Design, simulation and experiments

We use the simulated-annealing method (SA algorithm, mentioned as section 2-4-4-2) for optimizing the AOS structure with the objective function as Eq. (60).

$$OF = \sum_{q=1}^M \left(|\eta_{target}(\lambda_q) - \eta(\lambda_q)| w(\lambda_q) \right) + \beta \{ \max[\eta(\lambda_1), \eta(\lambda_2), \dots, \eta(\lambda_M)] - \min[\eta(\lambda_1), \eta(\lambda_2), \dots, \eta(\lambda_M)] \} \quad (60)$$

To investigate the EO tunability in a multi-wavelength OPG, we first calculated the domain structures of conventional design without ΔL or ΔN enhanced AOS. The **Fig. 15** shows the calculated output signal spectra of three generators optimized for generating 2, 3, and 4 signal wavelengths.

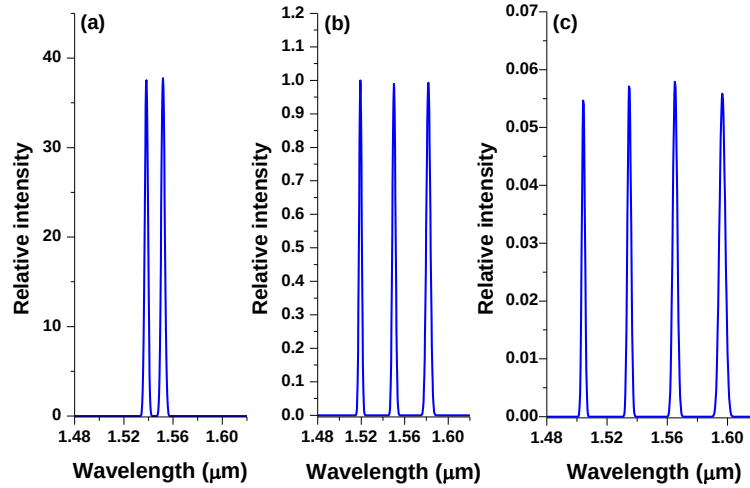


Figure 15. The simulated multi-wavelength optical parametric generation signals in three different designs.

The Table 2 summarizes the design parameters and the domain structure information of these generators. Noticeably for all the three designs the block

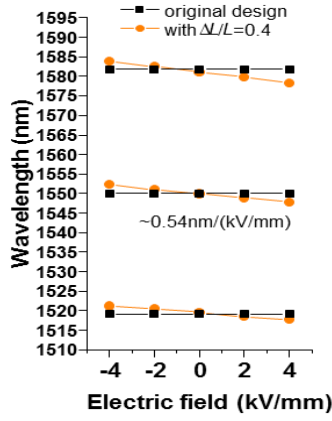
numbers of unit positive and negative domains are about the same. Because of pursuing the maximum conversion efficiency in the SA algorithm, the value of N^+ and N^- will be very close to each other in the structure, which means the EO wavelength-tuning will be very difficult.

	Design parameters			Domain structure analysis			
Number of signal generation	Δx (μm)	N	$w; \beta$	$N^+; N^-$	$\Delta N/N$	Average Fourier coefficient	Spectral tuning rate (nm/(kV/mm))
2	5	8000	0.1; 1.0	3992; 4008	0.0020	~ 0.39	0.00375 @1550nm
3	5	8000	0.6; 1.0	4001; 3999	0.00025	~ 0.32	0.00125 @1550nm
4	5	8000	1.0; 1.0	3988; 4012	0.0030	~ 0.28	0.007125 @1535nm

Table 2. The design parameters and spectral tuning rates of three designs for 2, 3, and 4 signals optical parametric generation.

Technique I, ΔL enhancement method

To enhance the EO tuning rate of the devices, we performed domain overpoling during the fabrication process for an aperiodically-poled structure designed by the standard SA method. We called this method as the ΔL enhancement method. The EO tuning rate is thus proportional to the overpoled quantity δx multiplied by number of total domains. The **Fig. 16 (a)** shows the calculated signal spectra of a generator, which plots the calculated EO spectral tuning curves of the signals from the domain over-poled and standard-design one. After ΔL enhancement, the spectral tuning rate is ~ 0.54 nm/(kV/mm), which is 432 times enhancement in comparison with the standard-design one.



(a)

$N^+ \approx N^-$	standard poling	domain over-poled
L^+	$N^+ \Delta x$	$N^+ (\Delta x - \tilde{\Delta} x)$
L^-	$N^- \Delta x$	$N^- (\Delta x + \tilde{\Delta} x)$
$\Delta L = L^+ - L^-$	$(N^+ - N^-) \Delta x$ ≈ 0	$(N^+ - N^-) \Delta x - (N^+ + N^-) \tilde{\Delta} x$ $\approx -N_{total} \tilde{\Delta} x$
Tuning rate (nm/(kV/mm))	~ 0.00125	~ 0.54

(b)

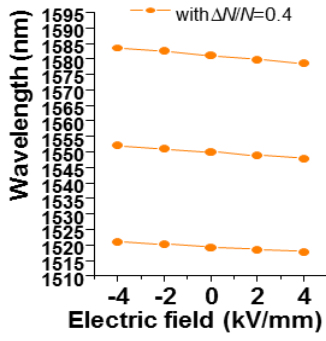
Figure 16. (a) the comparison of the simulated spectrum tuning and the tuning rate (b) for standard poling and domain over-poled (ΔL -enhanced method).

Technique II, ΔN enhancement method

The second method is direct to modify the design algorithm to optimize both the ΔN value and the conversion efficiency, we called it as the ΔN enhancement method. The modified objective function as Eq. (61) is different from the standard one by adding ΔN enhancement weighting ($\Gamma_{Ez-tuning}$) into the calculation. So, the ratio of N^+ to N^- is controllable by varying this weighting.

$$\begin{aligned}
 OF = & \sum_{q=1}^M \left(|\eta_{target}(\lambda_q) - \eta(\lambda_q)| w(\lambda_q) \right) + \\
 & \beta \{ \max[\eta(\lambda_1), \eta(\lambda_2), \dots, \eta(\lambda_M)] - \min[\eta(\lambda_1), \eta(\lambda_2), \dots, \eta(\lambda_M)] \} + \\
 & \Gamma_{Ez-tuning} |\Delta N_{target} - \Delta N_{chip}|
 \end{aligned} \tag{61}$$

The ΔL in this method is equal to ΔN multiplied by Δx . The EO tuning rate is thus proportional to this value. The **Fig. 17** shows the calculated signal spectra of the generator designed by the ΔN enhancement method. The obtained EO tuning rate was ~ 0.54 nm/(kV/mm) a, which is the same as that with the ΔL enhanced one.



(a)

	ΔL enhancement	ΔN enhancement
N^+ / N^-	~ 1	Controllable
$\Delta x^+ / \Delta x^-$	$(\Delta x - \tilde{\Delta}x) / (\Delta x + \tilde{\Delta}x)$	1
$\Delta L = L^+ - L^-$	$(N^+ - N^-)\Delta x - (N^+ + N^-)\tilde{\Delta}x$ $\approx -N_{total} \tilde{\Delta}x$	$(N^+ - N^-)\Delta x$ $= \Delta N \Delta x$

(b)

Figure 17. (a) the comparison of the simulated spectrum tuning and (b) the positive/negative domain length relation for ΔL-enhanced or ΔN-enhanced methods.

Because the ΔL is almost equal to 0 in the stranded design, the wavelengths of OPG signals are hard to be tuned with E_z . It is interesting to note that the ΔN enhanced device has better signal conversion efficiency but lower spectral-shape fidelity in the EO tuning. This can be explained with the Fourier spectra of the two EO-tuning enhancement methods structures at $E_z = 0$ and ± 4 kV/mm at 40°C . The **Fig. 18** shows the calculated Fourier coefficients of the ΔN enhanced OPGs are ~ 1.1 times higher than ΔL enhanced ones. A small discrepancy in the effective nonlinear coefficient can already result in a large difference in the output signal intensity because of the exponentially nonlinear parametric gain. This advantage of the ΔN enhancement method can be attributable to specific optimization process, which the efficiency and tuning rate are simultaneously pursued. The peak value of the Fourier spectrum of the ΔN -enhanced device is a more sensitive function of the EO effect than that of the ΔL -enhanced one, leading to a lower spectral-shape fidelity with the ΔN -enhanced device.

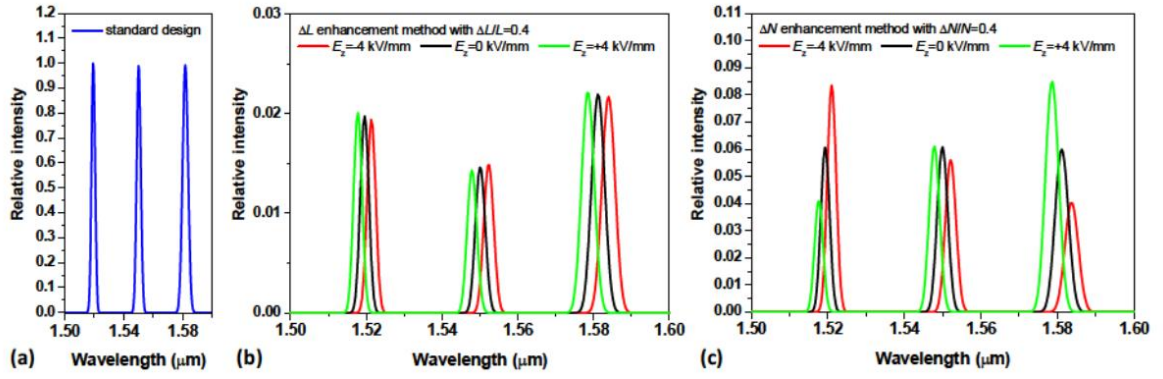


Figure 18. The simulated OPG spectrums for standard design (a) and the spectrum tunability of ΔL -enhanced (b), and ΔN -enhanced (c) designs.

3-1-3 Results

After chips design and performance simulation, we use standard electric-field poling process to fabricate the aperiodic poled Lithium Niobate crystals. Those crystals are 40-mm long, 5-mm wide, and 0.5-mm thick. In this case, the minimum domain length (which is the δx) is 5 μm . In experiment, the crystal was pumped by a 10-Hz, bandwidth < 0.2 nm, 9-ns pulse duration, Q-switched Nd:YAG 1064 nm laser with pump intensity $\sim 92 \text{ MW/cm}^2$, the sample was mounted on a voltage and temperature-controllable stage. The crystal temperature was stabilized at 140 $^{\circ}\text{C}$ to avoid any possible effect from the photorefraction. The output spectrum of the OPG was measured by an optical spectrum analyzer.

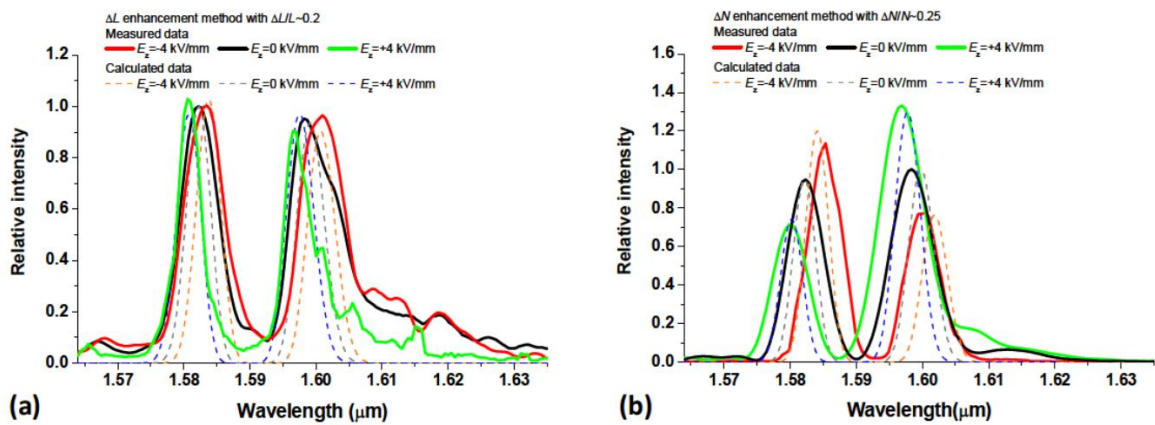


Figure 19. The experiment measured and theory fitting OPG spectrums for ΔL -enhanced (b), and ΔN -enhanced (c) designs.

Two OPGs designed for simultaneous two signal-wavelength OPGs had been demonstrated.

In short summary of this study, we applied and demonstrated two new methods for EO spectrum tunable optical parametric generators by asymmetric cumulative lengths of positive /negative domains in EO APPLN within ΔL or ΔN enhancement features. **Fig. 19** shows the output signal spectra from the two samples with $E_z=0$ and ± 4 kV/mm. It's apparent that the spectral-shape fidelity of the ΔL enhancement method is better than the ΔN enhancement method, which is consistent to our calculation. The tuning rate of these two methods are 0.38 and 0.49 nm/kV/mm due to their different ΔL or ΔN ratio, the ratios are different from the design values, which can be attributable to the non-uniformity of structure after poling. The signal conversion efficiencies of $\sim 8.6\%$ and $\sim 13\%$ were obtained. These measurement results show that the ΔN -enhanced device has better signal conversion efficiency but lower spectral-shape fidelity with EO tuning in comparison with the ΔL -enhanced one.

3-2 Topic II: Electro-optically spectrum tailorable IOPO

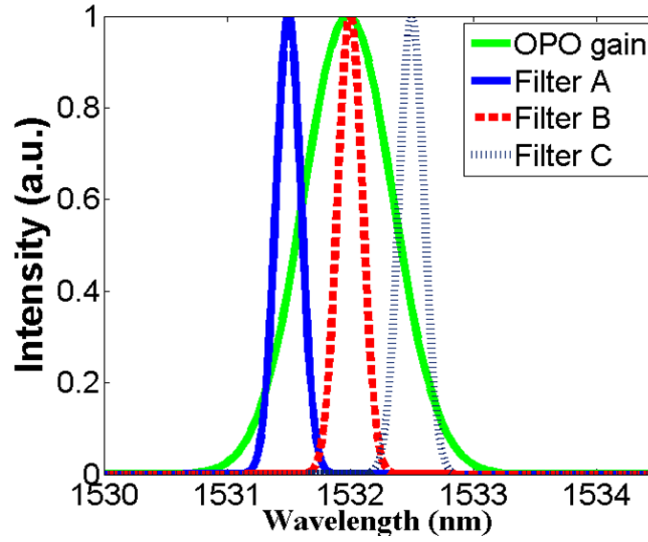


Figure 20. The mechanism of spectrum filtering of OPO signals in wavelength domain.

The mode selection of OPO spectrum is a method to narrow the OPO signal spectrum in wavelength domain, as the **Fig. 20** shown, the mode selection can be achieved by inserting the filter into the OPO systems, this mode selectable filter can be tune by mechanical controlled, such as Etalon [72], grating [73], Volume Bragg Grating, VBG [74], or it's can be a faster electro-optical controlled system by using multi-section EOPPLN as the Solc-type polarization filters [78]. Our prior proof-of-principle demonstration [78] of “Electro-Optical Narrow line, spectral tailorable OPO systems” is used three periodic-poled sections with two periods as a PPLN Sandwich, two PPLN as EO filters at both ends of central PPLN, the central PPLN is used to be an optical parametric gain medium, OPGM, to provide the nonlinear gain in the systems, we can generate 1, 2 or 3 signals peaks under appropriate conditions with different temperature and applied electric-fields. The e-wave transmittance of EO filter are Ey-field depend. In this study, we further use the concept of aperiodic structure to build an aperiodically poled LiNbO₃ EO polarization mode filters in the IOPO systems to simplify the system.

3-2-1 Electro-optically spectrum tailorability, EOST

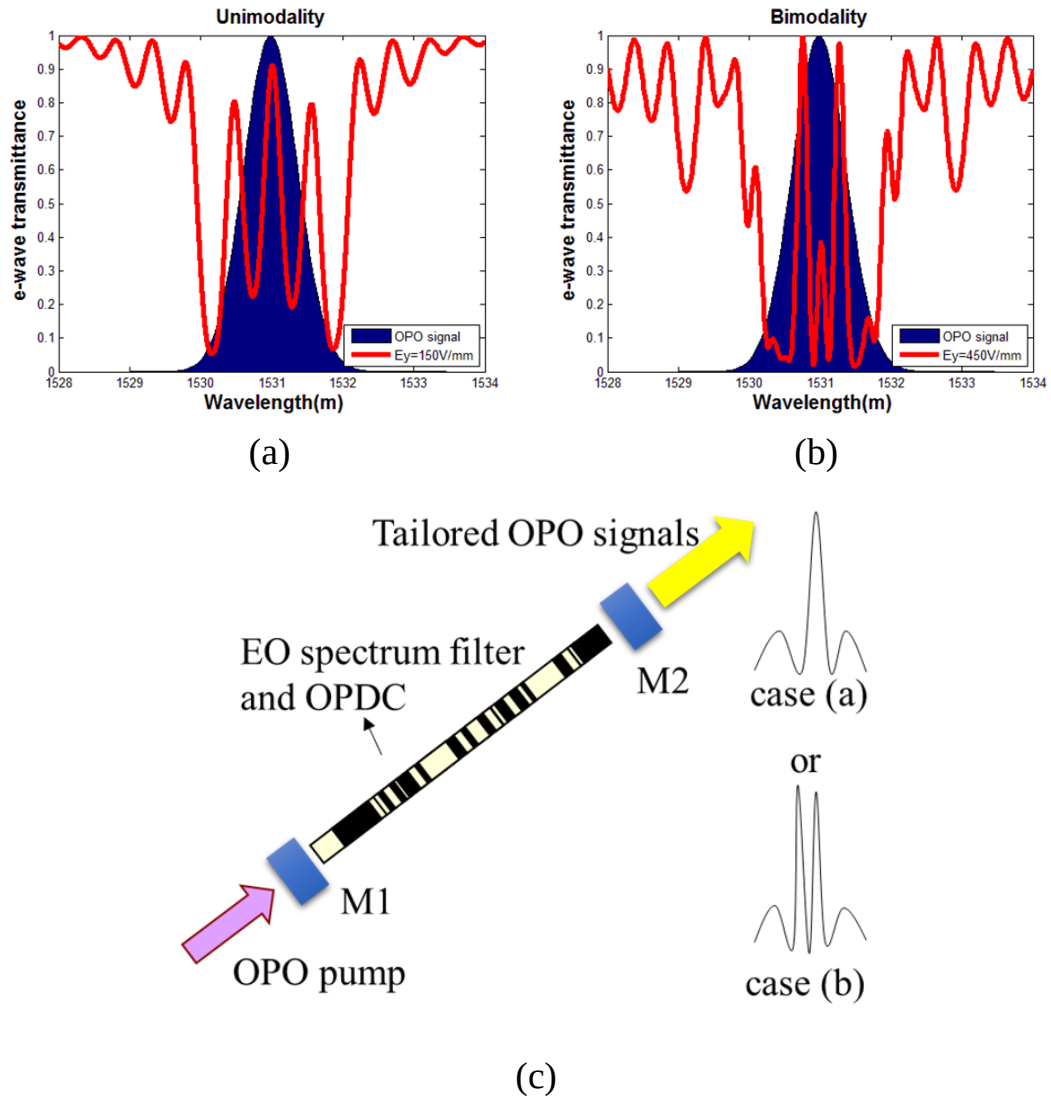


Figure 21. The single-peak (a) or two-peak spectrum filtering (b) of tailored OPO signals, and the description schematic of the EOST modulation principle (c), the insert spectrum in (c) are the schematic diagram of tailored OPO signals spectrum.

In this study, there are two important processes, the multiple QPM electro-optical polarization-Mode converter as filter (Multi-QPM EO PMF) and optical parametric down converter (OPDC). We want to combine the concept of “*Multi-QPM EO PMF*” and “*OPDC*” in a special designed single APPLN device. Because of the diversity of aperiodic structure, we find we can only use one AOS structure to be a single peak, twin peak or multi-peak polarization mode filter in the OPO gain region in different electric-field

operation conditions, the transmittance of filter are shown by the red curves in the **Fig. 21 (a) and (b)**. But the simulation results here, are not the true OPO signals, because of the nonlinear gain has not to be considered yet.

3-2-2 Design, simulation and experiments

Coupled-mode equations for deriving EO gain tailored OPDC signal evolution with x-propagation is Eq. (62).

$$\begin{aligned}
\frac{dE_{s,o}(x)}{dx} &= -i \frac{\omega_s}{c_0 n_{s,o}} [\kappa_{eo1} s(x) E_{s,e}(x) e^{i\Delta k_{eo} x} - \kappa_{eo2} s(x) E_{s,o}(x)] \\
\frac{dE_{s,e}(x)}{dx} &= -i \frac{\omega_s}{c_0 n_{s,e}} [\kappa_{eo1} s(x) E_{s,o}(x) e^{-i\Delta k_{eo} x} + d_{33} s(x) E_{p,e}(x) E_{i,e}^*(x) e^{-i\Delta k_{op} x}] \\
\frac{dE_{i,e}(x)}{dx} &= -i \frac{\omega_i}{c_0 n_{i,e}} d_{33} s(x) E_{p,e}(x) E_{s,e}^*(x) e^{-i\Delta k_{op} x} \\
\frac{dE_{p,e}(x)}{dx} &= -i \frac{\omega_p}{c_0 n_{p,e}} d_{33} s(x) E_{s,e}(x) E_{i,e}(x) e^{i\Delta k_{op} x}
\end{aligned} \tag{62}$$

where $E_{s,o}(x)$, $E_{s,e}(x)$, $E_{i,e}(x)$ and $E_{p,e}(x)$ are the electric-fields of o-wave signal, e-wave signal, e-wave idler and e-wave pump. ω_s , ω_i , ω_p are the signal, idler and pump angular frequencies, the domain axis function is $s(x) = \pm 1$, $n_{s,o}$, $n_{s,e}$, $n_{i,e}$, $n_{p,o}$ are the refractive index of o-wave signal, e-wave signal, e-wave idler and e-wave pump, the nonlinear coefficient is d_{33} , the coupling coefficient of EO tuning is κ_{eo1} and the phase-mismatching terms of EO tuning and OPDC are Eq. (63) and Eq. (64).

$$\Delta k_{eo} = \frac{2\pi}{\lambda_s} (n_{s,e} - n_{s,o}) \tag{63}$$

$$\Delta k_{op} = \frac{2\pi}{\lambda_{p,e}} n_{p,e} - \frac{2\pi}{\lambda_{s,e}} n_{s,e} - \frac{2\pi}{\lambda_{i,e}} n_{i,e} \tag{64}$$

In order to investigate the simultaneously OPDC and EOPM filter of the structure, we start from the coupled-mode equations, show as Eq. (62), these equations are consider the EO polarization mode conversion and the optical parametric gain at the same time, we design a special simulation algorithm,

the EO OPO Spectrum Tailoring Simulation Algorithm (EOSTSA), the EOSTSA is an OPO signals spectrum simulator under EO polarization mode filtering, we can get the output OPO signal spectra by solving the coupled-mode equations block-by-block, round-trip-by-round-trip. For example, for round-trip filtering, the solving process is shown as **Fig. 22 (a)**. This process is solving coupled-mode equation of Eq. (62) from crystal to mirror 2 and back to the crystal and mirror 1 and propagate crystal again, and after several round-trip, in final, we can get the output tailored OPO signals after Mirror 2 with applied appropriate electric-fields along crystal y-axis.

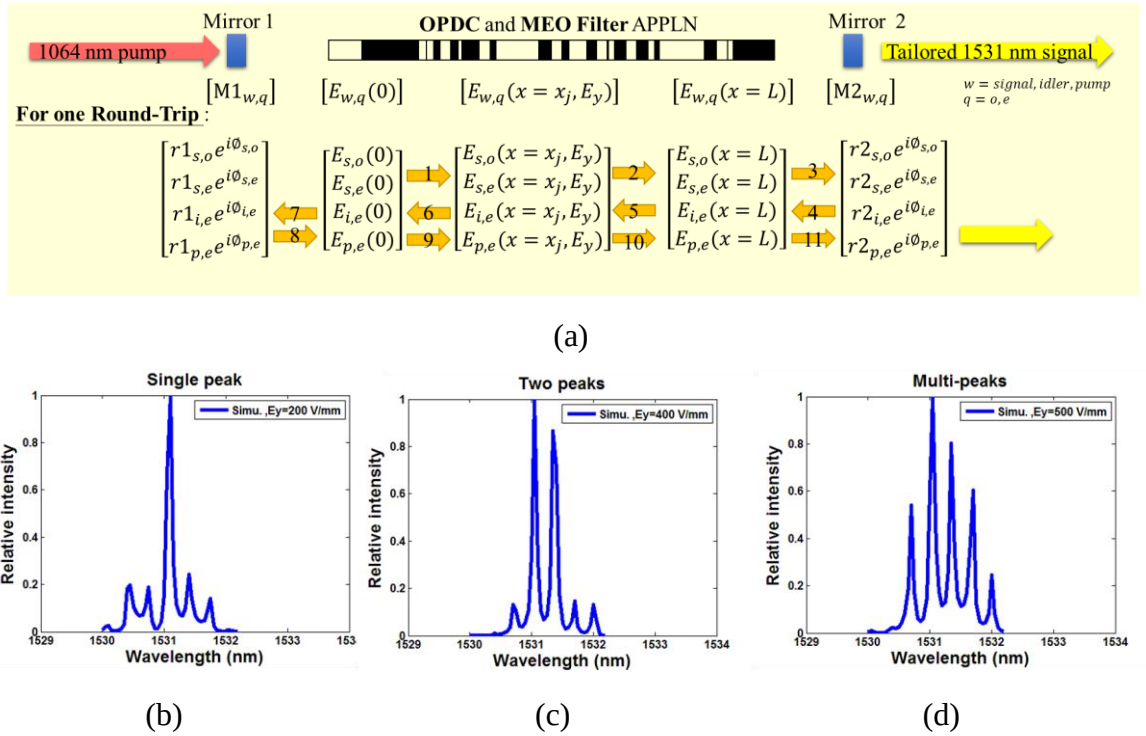


Figure 22. (a) The EO OPO Spectrum Tailoring Simulation Algorithm (EOSTSA). (b-d) Simulation results of EOST IOPO signals for a specific AOS structure under different applied electric-fields for single, two and multi-peaks, respectively

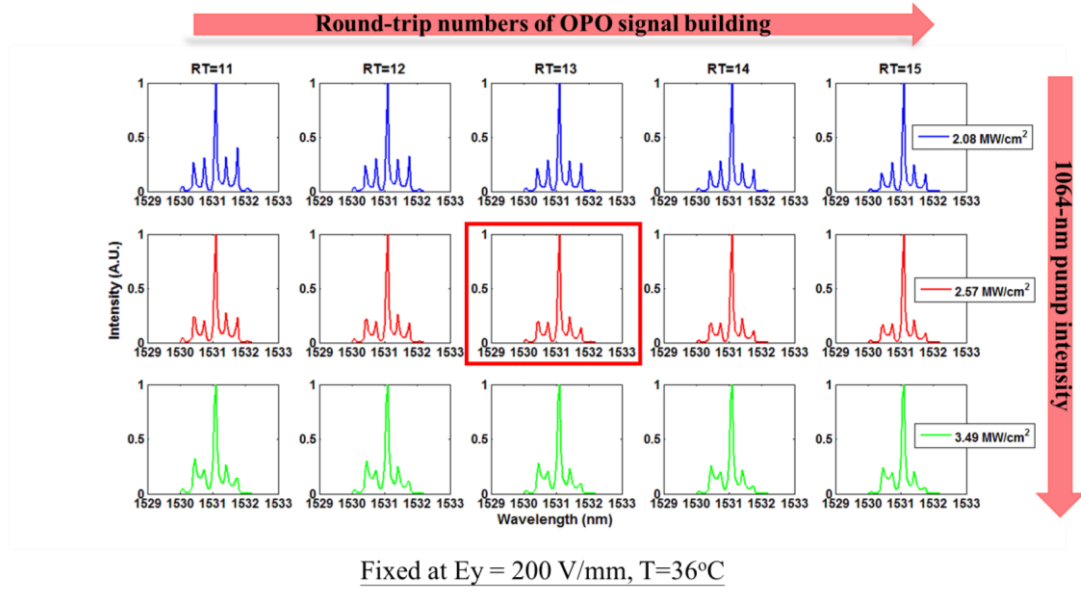


Figure 23. Simulation results of EOST IOPO tolerance of tailored signals with the round-trip from 11 to 15 with the three 1064-nm pump intensity which is from 2.08, 2.57 and 3.49 MW/cm².

Before the measurement, we use the EOSTSA simulator again, the spectrum tolerance of single-peak tailored signals are shown as **Fig. 23**. The tailored signals are simulated in the different conditions, including number of round-trip from 11 to 15, and the different 1064 nm pump intensity from 2.08 to ~3.5 MW/cm², we notice that there are not much different between each other.

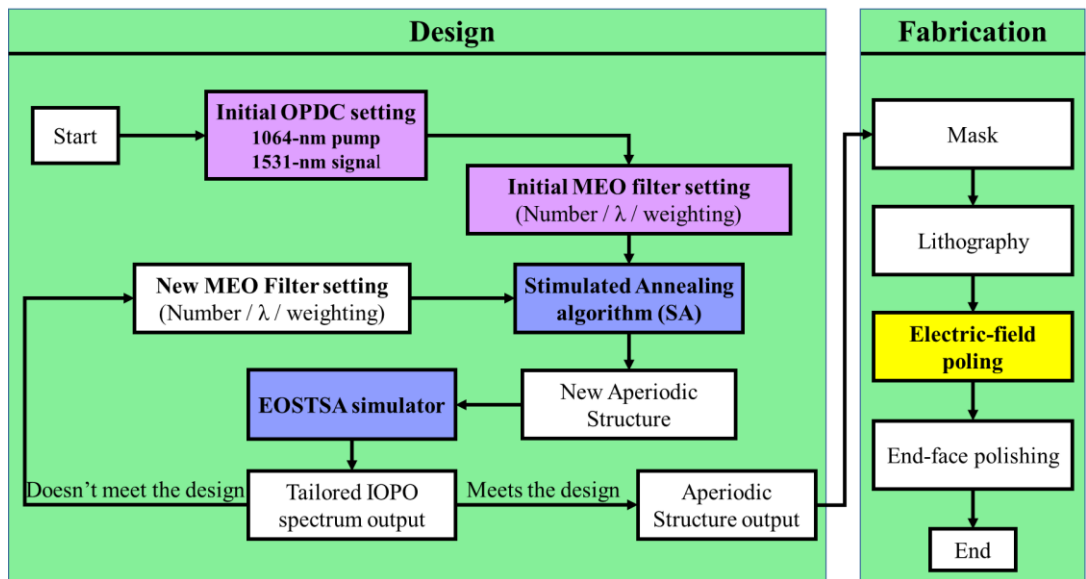


Figure 24. The flow chart of our APPLN design and poling fabrication process of EOST OPO samples.

Fig. 24 shows the flow chart of our design and fabrication process. The design is start from setting OPDC conditions, 1064-nm pump and 1531 nm signals, and initial aperiodic EO filter setting, number of EO filters, central wavelengths, and weighting of filters, and we use the stimulated annealing algorithm (SA) to modify the aperiodic structure, and the SA will give a modified aperiodic structure to the EOSTSA simulator, and we get tailored OPO spectra after EOSTSA simulation, if the output meets the design, the system will output a final aperiodic structure. After lithography, we use our standard electric-field poling to define the structure and after end-face optical-grade polishing process, the chip of design is ready.

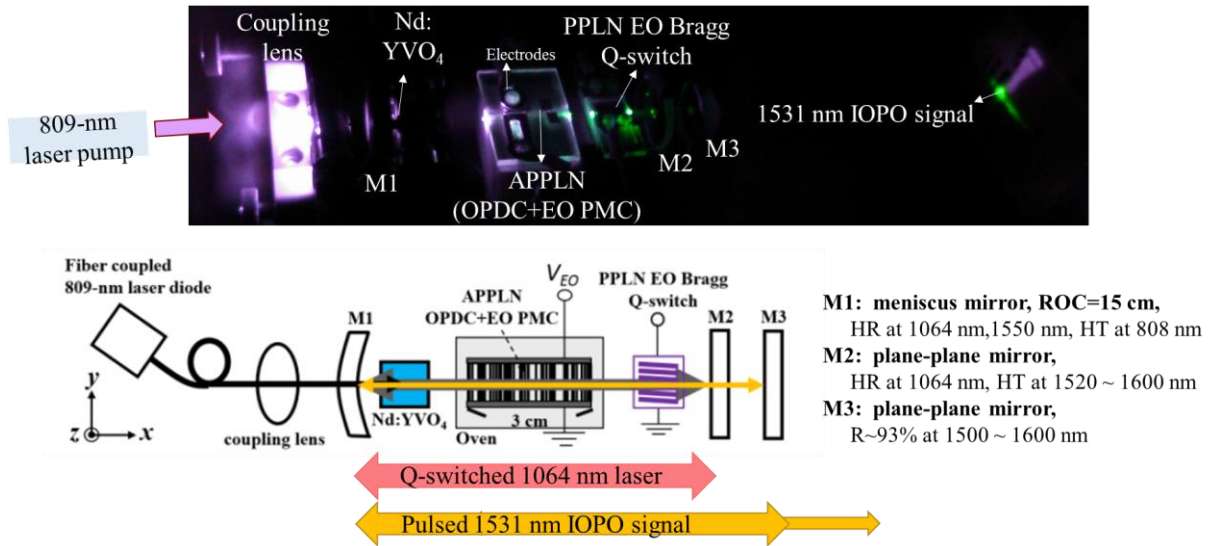
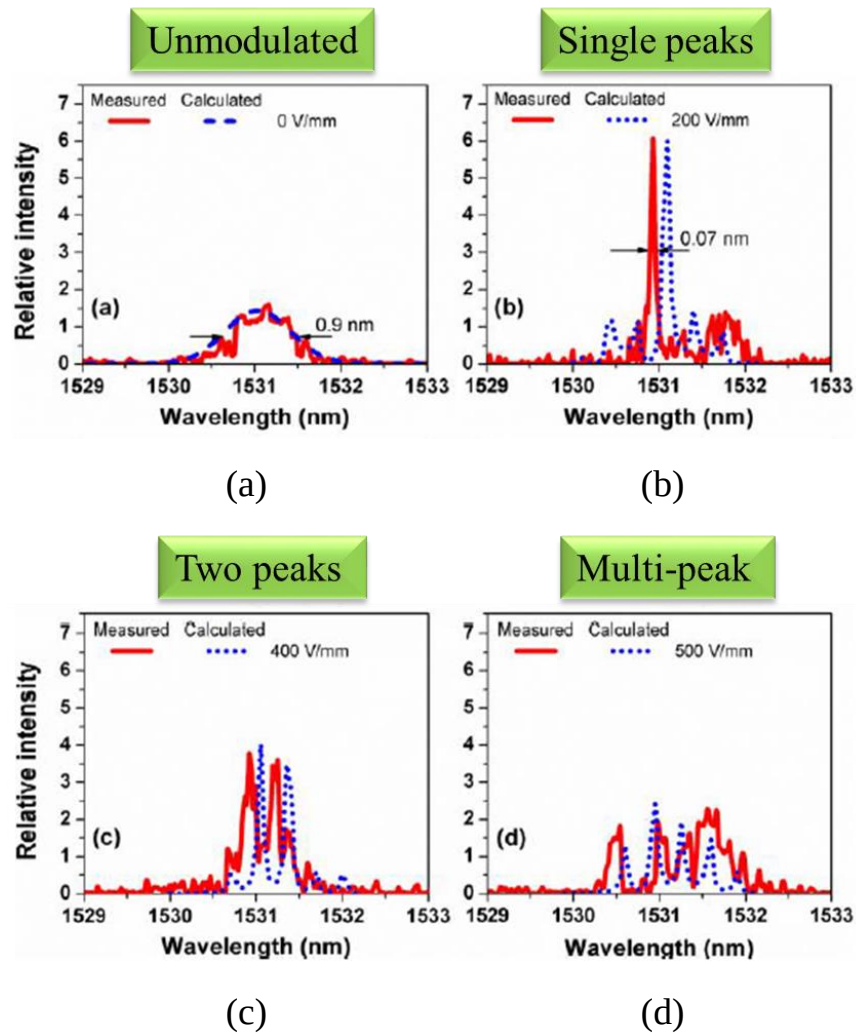


Figure 25. The experimental setup of EOST IOPO system.

Our experiment is using a 809nm fiber coupled laser diode as the continues-wave (CW) pump, and a PPLN EO Bragg device as a Q-switching device to generate the 1064nm with 1-kHz, Q-switched as the intracavity optical parametric oscillator (IOPO) pump, the Q-switched 1531nm IOPO signals is oscillating in the cavity constructed by mirror 1 (M1) and mirror 3 (M3), the coating conditions of all mirror are also marked on the **Fig. 25.**

3-2-3 Results and applications

The experimental and simulation results are shown as **Fig. 26 (a)-(d)**, according the design and measurement, we get single, twin, multi=peak IOPO signals in the same APPLN but in the electric-field equals 200, 400 and 500 V/mm. The spectral width of single-peak signal is ~ 0.067 nm, which is narrowed by the factor of ~ 13.7 in comparison with original untailored-signals and the peak intensity of single-peak signals is ~ 6.1 times than non-tailored signal.



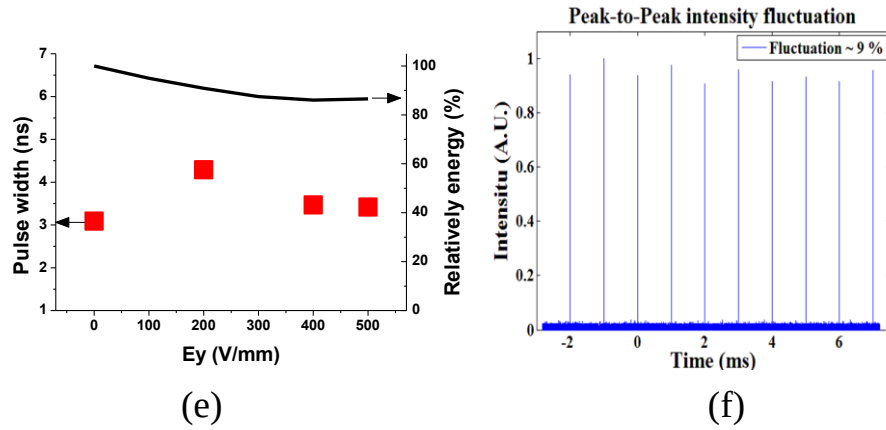


Figure 26. Experimental and simulated fitting results, (a)-(d) the IOPO signals spectrum under non-tailored, single-peak, two peak and multi-peak operations, respectively, (e) the pulse width and relatively energy of output IOPO signals under different operations and (f) the peak-to-peak intensity fluctuation.

The energy and pulse width of different operation conditions are shown on the **Fig. 26 (e)**, we find the pulse width of single-peak tailored signal is slightly larger than untailored signal, we find its original from we narrowed the signals during the single-peak operation and the peak-to-peak intensity fluctuation of single peak signals is $\sim 9\%$ as **Fig. 26 (f)** shown. The specification of different measured IOPO signals are compared in the table 3.

IOPO Signals	E_y (V/mm)	Spectrum FWHM (nm)	Spectral width narrowing factor	Peak intensity enhancing factor
Non-tailored	0	~ 0.92	1.0	1.0
Single peak	200	~ 0.067	~ 13.7	~ 6.1
Twin peaks	400	$\sim 0.16, \sim 0.13$	$\sim 5.9, \sim 7.1$	$\sim 4.1, \sim 3.7$
Multi-peaks	500	$\sim 0.15, \sim 0.11, \sim 0.22$	$\sim 6.1, \sim 8.4, \sim 4.2$	$\sim 2.6, \sim 2.9, \sim 2.4$

Table 3. The measured EOST IOPO signals specification of different operations.

3-3 Summary

In the first study of chapter 3, EO spectrum tunable multi-wavelength OPGs, we successfully realized a unique electro-optically tunable, multi-wavelength light source based on APPLN OPGs constructed by AOS technique. The domain structures of the APPLN OPGs have been engineered to increase the

difference of either the lengths or block numbers of the two opposite-polarity domains constituting the structure (the ΔL or ΔN enhancement method) to effectively enhance the EO tuning rate of the devices. We have also successfully demonstrated the EO spectral tunings with APPLN OPG/OPOs fabricated for simultaneous two signal-wavelength generation, where tuning rates of ~ 0.38 and ~ 0.49 nm/(kV/mm) were obtained from the devices with the ΔL and ΔN enhancement methods, respectively.

Further, at the second study in chapter 3, the EO spectrum tailorable IOPO, we design a special simulator for EOPMF and/or OPG/OPO simulator, EOSTSA, this simulator is suitable for periodic or aperiodic structure simulation. And by using the special designed aperiodic structure, we demonstrate an “electro-optically spectrum tailorable intracavity optical parametric oscillators”, which can generate single- twin or multi-peaks by only switching the applied voltage of the system with fixed temperature. The line width of peak intensity enhanced single-peak signal is ~ 0.07 nm, which is narrowed by a factor of ~ 13.7 in comparison with original untailored signals, and we can only use a single APPLN as a dual-function integrated optical circuits. Due to the simple dual-function integrated optical circuit, our IOPO system only need one temperature controller and one power supplier to control the output signals. By using waveguide involved, we hope this device can operate at relatively low voltage in the future work.

Chapter 4 、 Integrated Optical Circuit, IOC

In chapter 4, I brief our integrated optical circuit designs with **thesis topic III: High-speed Lithium Niobate EO Modulators Array**, including the device topology, performance modeling method and the measured device characteristics of high-speed LiNbO₃ electro-optical modulator array.

4-1 Topic III: High-speed Lithium Niobate EO modulators array

In section 4-1, I summarized our design method and device performance for integrated array-type high-speed Mach–Zehnder interference (MZI) optical intensity modulators. The modulation 3-dB EO response of each modulator on the array is > 8 GHz, for 12-modulator integrated array, the overall modulation bandwidth is ~ 100 GHz.

4-1-1 Principle and design

As the discussion on the section (2-3-3-4), with essentially phase mismatch between modulation microwaves and optical waves due to the large effective index difference, the topology of electrodes, buffer-layer and substrate are playing important roles for device performance. For the appropriate design, one can decrease the effective index of microwave (N_m) to try to approach the optical mode effective index (N_{opt}) to compensate the phase mismatch between microwave modes and optical modes.

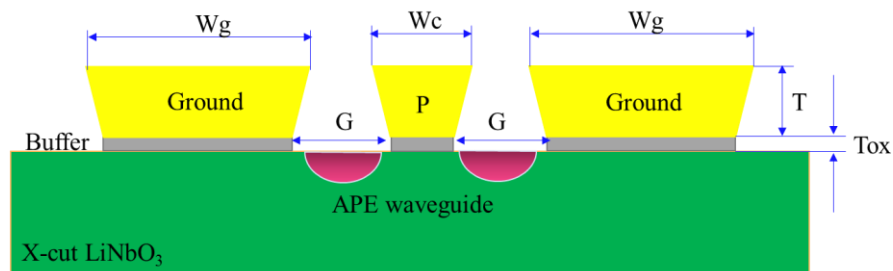


Figure 27. The sectional view of critical topological parameters of electrodes and waveguides for designing an APE-based y-propagation x-cut LiNbO₃ high-speed intensity modulator.

On the other side, the modulation microwave will suffer reflection loss due to the impedance mismatch between device and external circuit instrument. To overcome these issues, one can start from designing the optimized transverse modulation electrodes as the coplanar waveguide (CPW) electrodes of the modulator, the sectional view of topology for high-speed modulator is plotted as **Fig. 27** with the critical parameters of electrode topological parameters (W_g, W_c, G, T) and the buffer-layer thickness (T_{ox}). The method we use to model the N_m and characteristic impedance (Z_c) with different electrode structures under 0-40 GHz modulation is called RLCG-method [76] as discussed in paragraph (4-1-2). The performance of our intensity modulator array is summarized in paragraph (4-1-3).

4-1-2 Modulation performance simulation (RLCG method)

The cross section of coplanar waveguide electrodes for transverse modulation is shown as **Fig. 27**, we simulate the effective refraction index and attenuation of microwave by solving the complex propagation constant γ [77], which is the function of attenuation α and the microwave effective propagation constant β , $\gamma = \alpha + j\beta$, the $\beta = \omega n_{eff}/c$, and the Resistance (R), Inductance (L), Capacitance (C) and Conductance (G) can be solved by the **Eq. (65)** [76] and **Eq. (66)** [78].

$$\gamma Z_0 = j\omega L + R \quad (65)$$

$$\gamma/Z_0 = j\omega C + G \quad (66)$$

We can change the geometric relationship between electrodes, buffer-layer and substrate to design the microwave effective refraction index and the attenuation bandwidth. Under thin-line approximation [76], the structure parameters is

function of hot-electrode width (W_c), electrode gaps (G), electrode thickness (T), as the **Eq. (67)**.

$$\begin{bmatrix} A \\ B \\ C \\ D \end{bmatrix} = \begin{bmatrix} \frac{W_c}{2} \\ \frac{W_c}{2} + \frac{T}{\pi} \\ \frac{W_c}{2} + G - \frac{T}{\pi} \\ \frac{W_c}{2} + G \end{bmatrix} \quad (67)$$

The relation of R, L, C, G with characteristic impedance $Z_0(f)$ and complex propagation constant $\gamma(f)$ can be derived as **Eq. (67-68)**,

$$\gamma(f)Z_0(f) = j\omega L + R \quad (67)$$

$$\gamma(f)/Z_0(f) = j\omega C + G \quad (68)$$

therefore, the characteristic impedance $Z_0(f)$ can be simply expressed as function of R, L, C, G as **Eq. (69)**.

$$Z_0(f) = \sqrt{\frac{j\omega L + R}{j\omega C + G}} \quad (69)$$

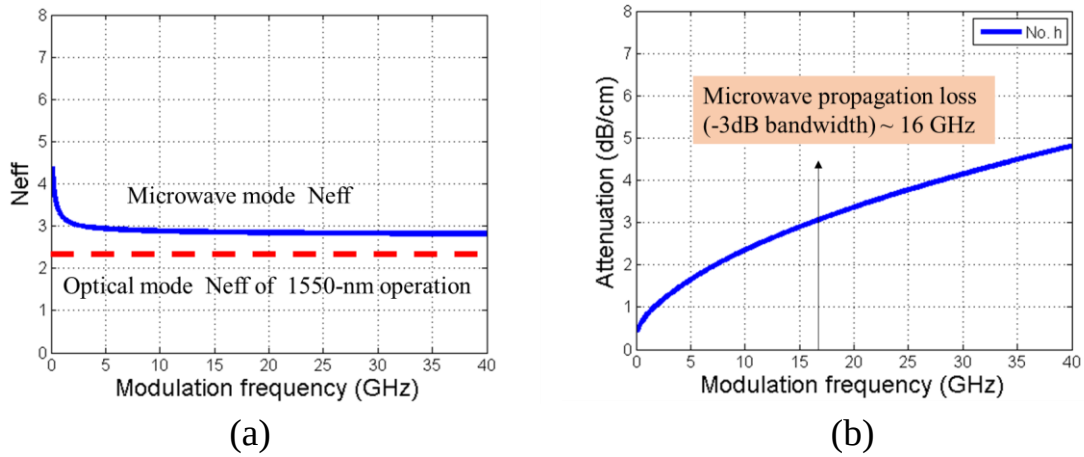


Figure 28. The simulated result of RLCG-method is plotted for microwave mode effective refractive index and the microwave attenuation under different modulation frequency as the blue curve in the (a) and (b), and the optical mode effective refractive index of 1550-nm band are also plot in the (a) as the dash-red line.

There are two methods can use to search the closed form solutions of $Z_0(f)$ and corresponding $\gamma(f)$, the *Direct-search method* by analytical approximation for exact solutions or *numerical approach method* to search the approximated

solution of structure parameters (A,B,C,D). The simulated results of one of our designs at **Fig. 28** shows that we can let the microwave n_{eff} close to the optical mode n_{eff} at optical wavelength around 1550 nm with modulation wave from DC to 40 GHz, and the microwave attenuation -3dB bandwidth is $\sim$ 16GHz.

4-1-3 Array-type intensity modulators

The devices are fabricated by our standard Ti-diffused LiNbO₃ waveguide with coplanar waveguide electrode processes, the traveling wave modulation features are tested via the network analyzer system, the testing picture of 3-inch LiNbO₃ substrate modulator array is Fig. 28 (a). The representative results of measured electrode response parameters S₂₁, S₁₁ are plotted in the Fig. 28 (b), which are corresponding to the electrode loss and the electrode reflection coefficients, respectively. One can set a standard of reflection coefficient, in typically, the S₁₁ less than -10 dB is acceptable for practical modulation applications, which is > 10 GHz in our measured results. On the other hand, the electrode coefficients can be used to calculate the attenuation and the effective refractive index of modulation waves under different applied modulation frequency [79] by calculating the complex propagation constant, γ as **Eq. (70)**.

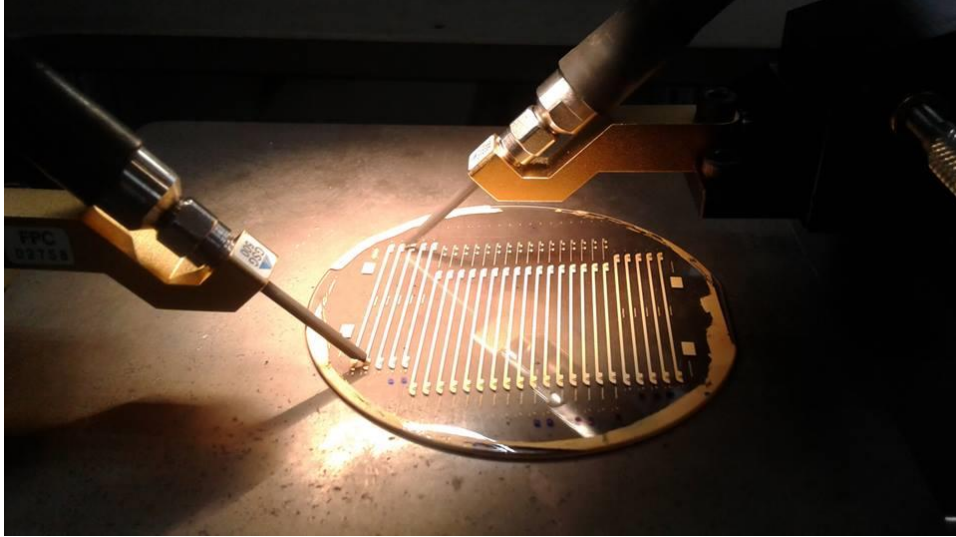
$$\gamma = \alpha + j\beta = \frac{1}{L} \cosh^{-1} \left(\frac{1 - S_{11}^2 + S_{21}^2}{2S_{21}} \right) \quad (70)$$

One can also directly extract the transmission-line parameters ($\alpha, \beta, n_{eff}, Z_c$) of coplanar waveguides by measured S₁₁, S₂₁ data [80] as **Eq. (71)-(72)** with transmission-line length, L .

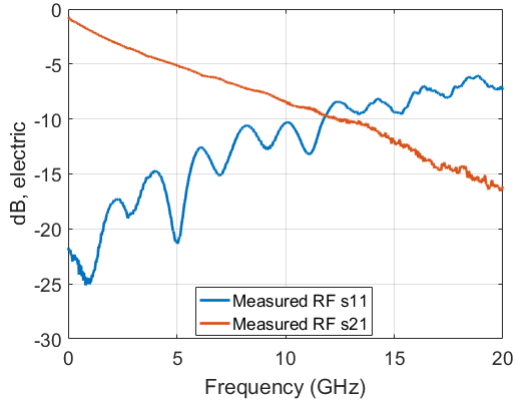
$$e^{-\gamma L} = \frac{2S_{21}}{1 - S_{11}^2 + S_{21}^2 \pm \sqrt{(1 + S_{11}^2 - S_{21}^2)^2 - 4S_{11}^2}} \quad (71)$$

$$Z_c = \pm Z_0 \sqrt{\frac{(1+S_{11})^2 - S_{21}^2}{(1-S_{11})^2 - S_{21}^2}} \quad (72)$$

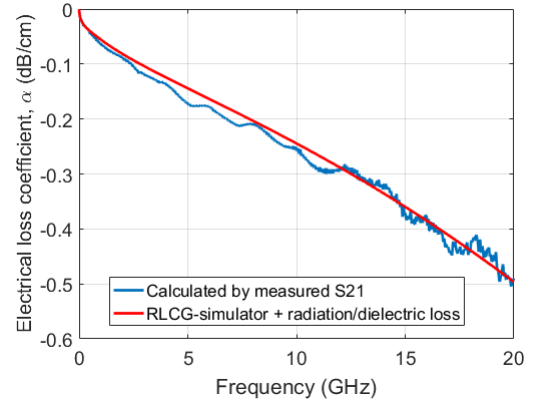
The extracted results of measured data and RLCG-method simulated fitting results for attenuation (α), effective refractive index (n_{eff}), characteristic impedance (Z_c) are shown as **Fig. 29 (c)-(e)**, respectively. The electro-optical response and corresponding half-wave voltage can also be extracted by measured electrode response parameters S21, S11, the extraction equation is **Eq. (22)** shown in chapter 2, the results of EO response and V_π are shown as **Fig. 29 (f)-(g)**, which shows the -3dB bandwidth of EO response is > 10 GHz, and the $V_\pi < 10$ V , from DC-10GHz.



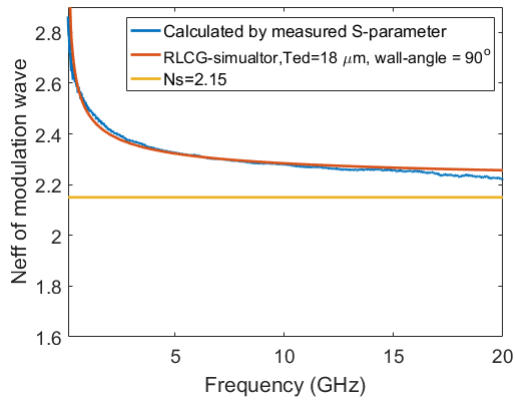
(a)



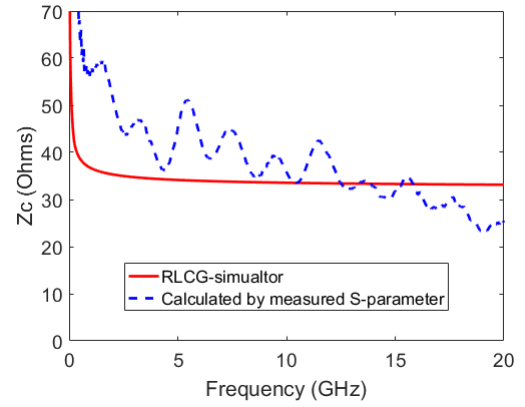
(b)



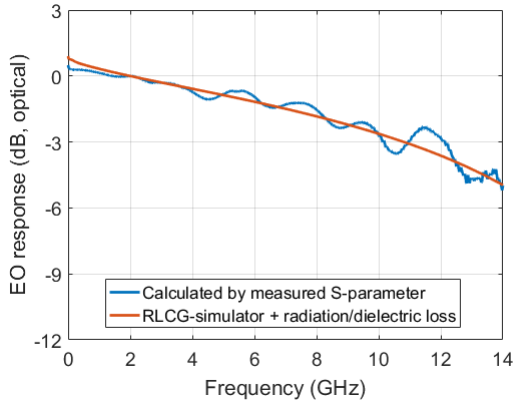
(c)



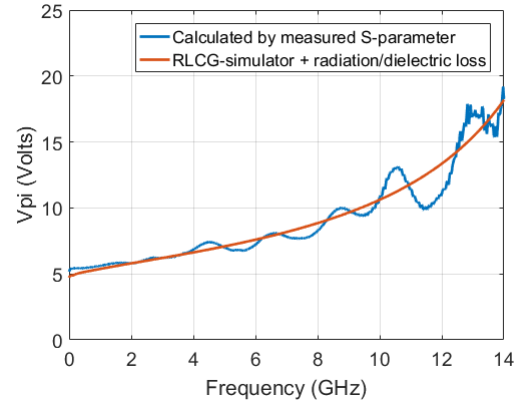
(d)



(e)



(f)



(g)

Figure 29. (a) the testing picture of 3-inch LiNbO₃ modulator array, and the measured and theory fitting data of our LiNbO₃ intensity modulator for several specifications under 0-20 GHz includes the (b) electrode parameters (s11, s21), (c) electrical loss coefficient, (d) microwave effective refractive index, (e) characteristic impedance, (f) EO response and (g) the half-wave voltage, respectively.

4-2 Summary

In this study, we review the RLCG-method for transverse modulation of coplanar waveguide electrodes and demonstrate the 12-modulator-integrated EO LiNbO₃ intensity modulator array, and achieve effective 100 GHz bandwidth with separately control ability. Measurement result shows the each modulator -3dB bandwidth of EO response is > 10 GHz, and the $V_{\pi} < 10$ V, from DC-10GHz, all specification meets the commercial standards of 10 GHz EO LiNbO₃ intensity modulator. The real photo of packaged EO LiNbO₃ intensity modulator array is shown as **Fig. 30 (a)**, the packaged modulator array has 12 fiber input, 12 fiber output and 12 independent electric modulation input port as **Fig. 30 (b)** shown.

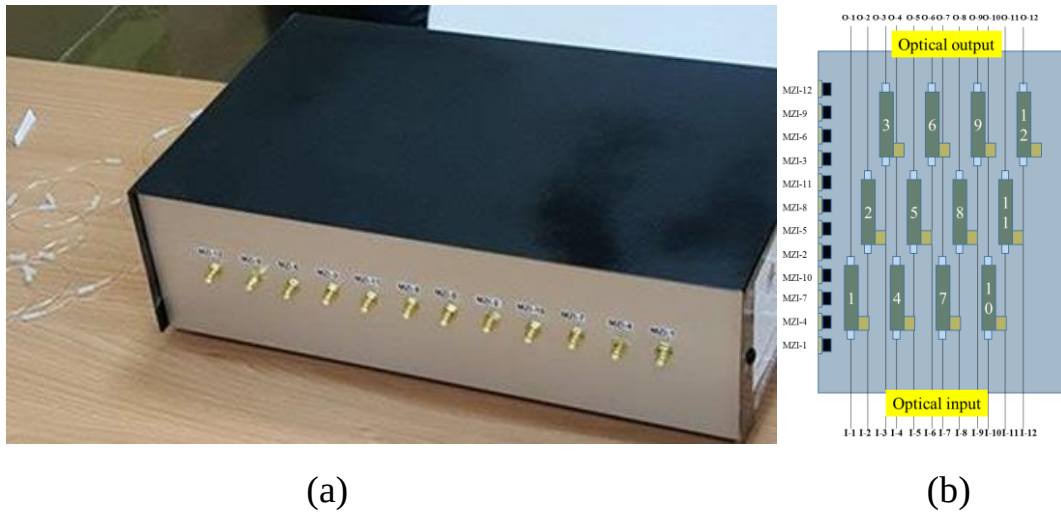


Figure 30. (a) Photo of packaged EO LiNbO₃ modulator array, (b) the content and operation ports schematic.

Chapter 5 、 Quantum Integrated Optical Circuits, QIOCs

In Chapter 5, I summarized several study results of quantum characteristics of Ti-diffused LiNbO₃ waveguide integrated optical devices with **thesis topic IV: Light adiabatic transfer devices** in the section 5-1, including **topic IV-1: Ultra-broadband adiabatic light transfer in titanium diffused Lithium Niobate waveguides** and **topic IV-2: Ultra-broadband asymmetric adiabatic coupler for two-photon quantum-polarization state preparation**, and the **thesis topic V: Experiments of quantum integrated optical circuits** in the section 5-2. The main collaborators of these studies are Kai Wang, Professor Dragomir N. Neshev and A/Professor Andrey A. Sukhorukov of Nonlinear Physics Centre, Research School of Physics and Engineering, Australian National University.

The results of these study were published or posted in ***Optics Express***, 23(24), 30641-30650. (2015), ***CLEO: 2016***, San Jose, CA, USA, AAC paper (*in preparation*) and ***CUDOS: 2017 quantum workshop***, ***CUDOS:2017 annual workshop*** and SPDC with MS paper (*in preparation*), respectively.

Thesis topic IV:

H. P. Chung, K. H. Huang, S. L. Yang, W. K. Chang, C. W. Wu, F. Setzpfandt, T. Pertsch, D. N. Neshev, and Y. H. Chen “Adiabatic light transfer in titanium diffused lithium niobate waveguides.” ***Optics Express***, 23(24), 30641-30650. (2015).

H. P. Chung, K. H. Huang, S. L. Yang, W. K. Chang, C. W. Wu, F. Setzpfandt, T. Pertsch, D. N. Neshev, and Y. H. Chen “Ultra-Broadband Adiabatic Light Transfer in Titanium Diffused Lithium Niobate

Waveguides.” **CLEO: QELS_Fundamental Science 2016**, San Jose, CA, USA, (2016.)

Hung-Pin Chung, Kuang-Hsu Huang, Kai Wang, Sung-Lin Yang, Shih-Yuan Yang, Chun-I Sung, Alexander Solntsev, Andrey A. Sukhorukov, Dragomir N. Neshev, Yen-Hung Chen, “Ultra-broadband asymmetric adiabatic coupler for two-photon quantum-polarization state preparation.” (*in preparation*)

Thesis topic V:

Hung-Pin Chung, Shih-Yuan Yang, Sung-Lin Yang, Tsai-Yi Chien, Kuang-Hsu Huang, Yen-Yu Chou, Kai Wang, Dragomir N. Neshev, and Yen-Hung Chen. “Electro-optic aperiodically poled Lithium Niobate directional couplers” **CUDOS 2017 Annual workshop**, Australia, **Best Poster award of nonlinear quantum photonics**, (2017).

Kai Wang, Sergey. S. Kruk, Lei Xu, Matthew Parry, **Hung-Pin Chung**, Alexander S. Solntsev, James Titchener, Yuri S. Kivshar, Dragomir N. Neshev and Andrey A. Sukhorukov, “Quantum photon state reconstruction with nanostructured metasurfaces.” **CUDOS 2017 quantum workshop**, Australia, (2017).

Kai Wang, Sergey S. Kruk, Lei Xu, Matthew Parry, **Hung-Pin Chung**, James Titchener, Ivan Kravchenko, Yen-Hung Chen, Alexander S. Solntsev, Yuri S. Kivshar, Dragomir N. Neshev, Andrey A. Sukhorukov, “Quantum imaging with all-dielectric metasurfaces for polarization state tomography.” **CUDOS 2017 Annual workshop**, Australia, (2017).

5-1 Topic IV: Light adiabatic transfer devices

In contrast to conventional waveguide directional couplers (DCs), adiabatic DCs are in an arrayed waveguide configuration performing a counterintuitive light transfer between the two outer waveguides of the system without the excitation of the intermediate waveguide(s) [81 -83] shown as **Fig. 31 (a)**. Such an adiabatic coupling is characterized by several unique features including the capability of almost complete power transfer in a (quasi-) monotonic (non-back conversion) fashion, a high tolerance to waveguide fabrication errors, and a low sensitivity to the wavelength permitting broadband operation, simulated light transfer during propagation are shown as **Fig. 31 (b)-(c)**. Previous studies on adiabatic DCs discussed several applications based on this peculiar adiabatic tunneling effect. Most interesting, it has been proposed to spatially filter the pump from modally entangled photon pairs generated via the spontaneous parametric down-conversion (SPDC) process in a nonlinear-optical waveguide array system [84].

We implemented the adiabatic DCs in a simple three-waveguide coupling configuration [85] on a z-cut LiNbO₃ substrate. The adiabatic coupling occurs between the two parallel outer waveguides (**a** and **c**) built along the crystallographic x axis, which are apart by a sufficiently long distance to suppress direct evanescent coupling. The energy exchange between waveguides **a** and **c** is mediated via waveguide **b**, which is slightly oblique with respect to the outer waveguides.

5-1-1 Adiabatic coupling

In the design, we have defined several parameters to configure the waveguide system shown in **Fig. 31 (a)**: L for the length of the coupler, S_{ac} for the distance between the inner-edges of the waveguides **a** and **c**, $S_{ab}(x)$ and $S_{bc}(x)$ for the

distances between the inner-edges of the waveguides **a** and **b** and the waveguides **b** and **c**, respectively, $S_0 = S_{ab}(0) = S_{bc}(0)$ for the (inner-)edge-to-edge distance from the waveguide **b** to the waveguide **a** as well as to the waveguide **c** in the middle of the device (i.e., at $x=0$), and $\Delta S = S_{ab}(-L/2) - S_{bc}(-L/2) = S_{bc}(L/2) - S_{ab}(L/2)$ (the structure is centrally symmetric with respect to point $(x,y)=(0,0)$ in this design). These structure parameters determine the coupling behavior of the waveguides and therefore the adiabaticity of light transfer in such a system. In general, two adiabatic coupling conditions are used as criteria in design for such a three-waveguide coupling scheme [82], one is associated with the coupling strengths (coupling coefficients) among the waveguides, while the other is related to the slope of the inclined waveguide **b**. The coupling coefficients between the waveguides **a** and **b**, **b** and **c**, and **a** and **c** can be expressed as [86]:

$$\begin{aligned}\kappa_{ab}(x) &= \frac{\pi}{2l_{c,ab}(x)} = \frac{\pi}{2l_{c_0}e^{\frac{S_{ab}(x)}{r}}} = \kappa_0 e^{\alpha x} \\ \kappa_{bc}(x) &= \frac{\pi}{2l_{c,bc}(x)} = \frac{\pi}{2l_{c_0}e^{\frac{S_{bc}(x)}{r}}} = \kappa_0 e^{-\alpha x} \\ \kappa_{ac} &= \frac{\pi}{2l_{c,ac}} = \frac{\pi}{2l_{c_0}e^{\frac{S_{ac}}{r}}}\end{aligned}\tag{73}$$

In **Eq. (73)**, $l_{c,ab}$ is the coupling length for directional coupling between waveguides **a** and **b** (the same definition applies to $l_{c,bc}$ and $l_{c,ac}$), l_{c0} and r are parameters depending on the waveguide characteristics (the three waveguides are identical in channel width and fabricated with the same fabrication process) and can be experimentally determined by characterizing separately fabricated conventional DCs, $\kappa_0 = \kappa_{ab}(0) = \kappa_{bc}(0) = \pi/(2l_{c0}\exp(S_0/r))$ (i.e., κ_{ab} is

equal to κ_{bc} in the middle of the device), and $\alpha \equiv \Delta S/rL$ is related to the slope of the inclined waveguide **b** and therefore the adiabatic rate of the coupling process. The first (major) adiabatic condition γ_{AC} thus applies to α for achieving a slow adiabatic light transfer in the system, given by **Eq. (74)**.

$$\gamma_{AC} = \alpha/\kappa_0 = \Delta S/\kappa_0 rL \ll 1 \quad (74)$$

The second adiabatic condition is set to ensure a negligible direct coupling between waveguides **a** and **c** and a pure excitation of waveguide **a** at the input end of the coupler, given by **Eq. (75)**.

$$\kappa_{ac} \sim 0 \text{ and } \kappa_{ab} \left(-\frac{L}{2}\right) < \kappa_{bc} \left(-\frac{L}{2}\right) \left(\text{or } \xi = \left(\kappa_{ab} \left(-\frac{L}{2}\right) / \kappa_{bc} \left(-\frac{L}{2}\right)\right) < 1\right) \quad (75)$$

5-1-2 Topic IV-1, Ultra-broadband adiabatic light transfer in titanium diffused Lithium Niobate waveguides

Ultra-broadband adiabatic light transfer was demonstrated for the first time in lithium niobate. Coupling bandwidths and efficiencies of ~ 540 nm and >0.96 , respectively, were obtained with 5-cm long Ti:LiNbO₃ adiabatic couplers for both polarization modes. In this work, we report to the best of our knowledge the first experimental demonstration of an ultra-broadband (~ 540 nm) adiabatic DC in the iconic quadric nonlinear material, LiNbO₃. We designed and fabricated the adiabatic DCs with the structure schemes #1 (with $S_{ac}=22$ μm and $\Delta S=11$ μm) and #2 (with $S_{ac}=22$ μm and $\Delta S=7$ μm) designed for TE and TM guided modes, respectively, in the same LiNbO₃ chip of 50-mm in length, 5-mm in width, and 0.5-mm in thickness using the titanium thermal diffusion method. All the Ti:LiNbO₃ adiabatic DCs were based on channel waveguides formed by high-temperature diffusion of Ti strips coated on the -z surface of a LiNbO₃ crystal.

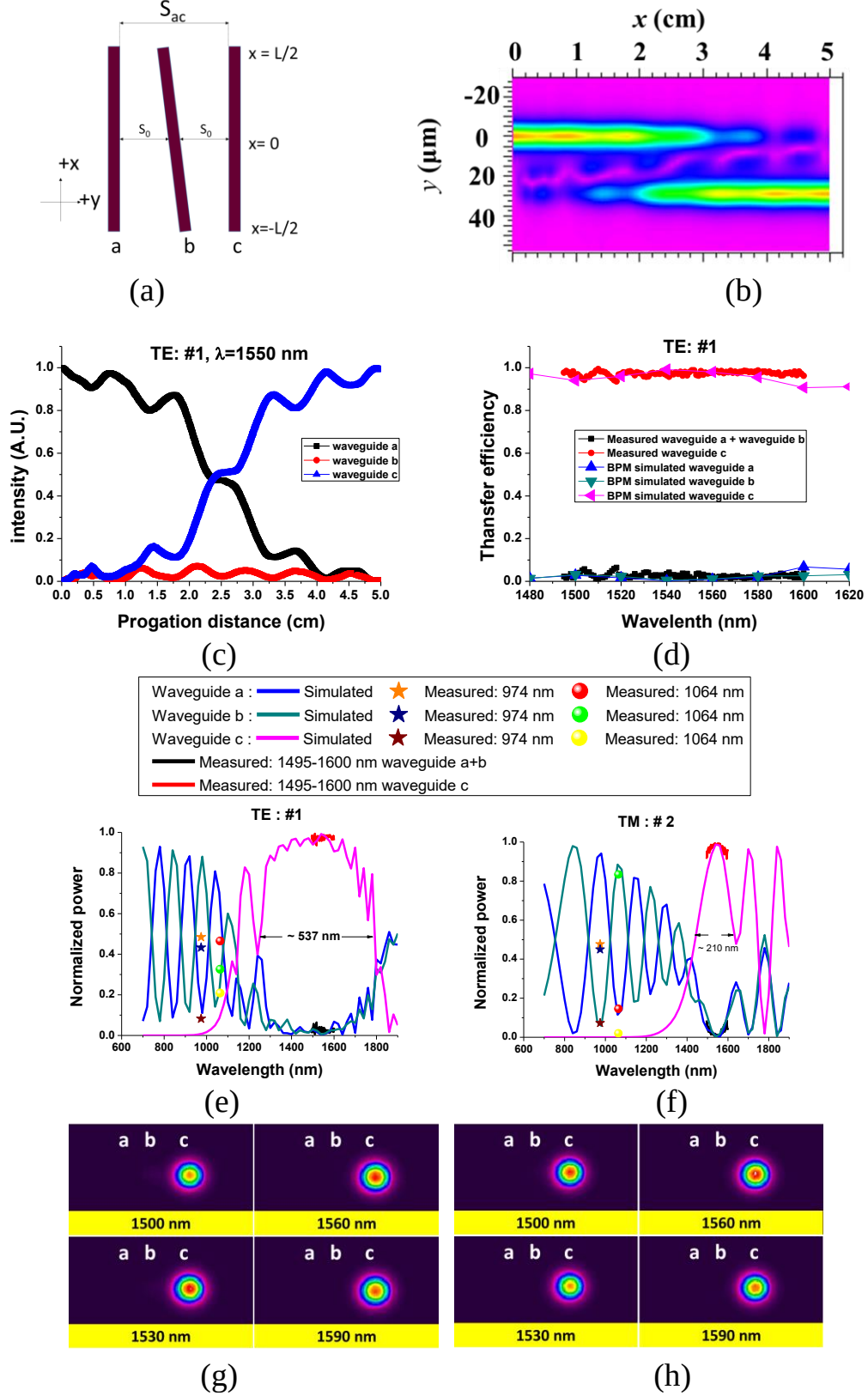


Figure 31. The 50-mm ultra-broadband Ti:LiNbO₃ adiabatic directional coupler (a) structure design, (b) simulated mode field transfer during propagation, (c) adiabatic power transfer during propagation, (d) simulated and measured coupling efficiencies of telecom band (1495 nm -1600 nm) of TE-polarized modes, (e-f) simulated (700 nm -1900 nm) and measured coupling efficiencies of 974 nm, 1064 nm and 1495 nm -1600 nm and (g-h) output mode fields with 1500 nm, 1530 nm, 1560 nm and 1590 nm modes of TE-polarized and TM-polarized modes, respectively.

The waveguides support single-mode guiding of waves in the spectral range 1400-1700 nm for both TE and TM polarizations. The characteristics of the fabricated LiNbO₃ adiabatic DCs were then investigated by using an external cavity laser. The laser excited only the waveguide **a** of each adiabatic DC with an input power of $< \sim 1$ mW. The output powers and mode profiles from the three-waveguide system were measured using a photodetector and an optical mode profiler system, respectively. **Fig. 31 (e) and (f)** show the calculated normalized output powers (normalized to total output power) from the three output ports of the LiNbO₃ adiabatic DCs as a function of wavelength over 700-1900 nm. The measured data at 974, 1064, and 1495-1600 nm are also plotted for comparison (which agrees well with the calculated results). The results show ultra-broad coupling bandwidths (-3 dB) of ~ 537 and 210 nm and peak coupling efficiencies of >0.96 for TE and TM polarization modes, respectively, were obtained. **Fig. 31 (g) and (h)** show the captured output mode intensity profiles from the adiabatic DCs at several different excitation wavelengths. It clearly shows that a great majority of the energy has been transferred from input waveguide a to waveguide c so that only the output mode from waveguide c was sensed by the beam profiler over the scanning wavelength range for both the TE and TM mode excitations.

In short summary of 5-1-2, we have designed and fabricated adiabatic DCs in Ti-diffused LiNbO₃ waveguides and successfully demonstrated the adiabatic light tunneling process in such devices consisting of three waveguides. The optimal structure parameters for 5-cm long LiNbO₃ adiabatic DCs were found to be $(S_{ac}, \Delta S) = (22, 11)$ and $(22, 7)$ μm for TE and TM modes, respectively, yielding coupling bandwidths of ~ 540 nm or ~ 210 nm and efficiencies of >0.96 for both polarization modes.

5-1-3 Topic IV-2, Ultra-broadband asymmetric adiabatic coupler for two-photon quantum-polarization state preparation

In a large number of developing quantum source, the high-brightness type-II (o-eo) spontaneous parametric down-conversion (SPDC) photon-pair source generated via nonlinear waveguides is particularly noteworthy [87]. However, due to unused pump will remain in the same waveguide of photon-pairs, which may influences the generated Bell's states of photon-pairs [84]. In this study, we clever used of the different coupling feature of two cross-polarization mode of Ti-diffused LiNbO₃ waveguides [88] and proposed an integrated high spatial splitting efficiency ultra-broadband asymmetric adiabatic coupler (AAC) based on a specially designed waveguide array. In general, for the same diffusion process of the Ti-diffused LiNbO₃ waveguide, Δn_{TM} is about two times than Δn_{TE} in the wavelength range of visible to near-IR, which leads to the coupling coefficients of two cross-polarized modes will be obviously different. Using this feature, we integrated two adiabatic coupling structures of slash and taper type to demonstrate the wavelength and polarization adiabatic spatial mode splitter in the 1550-nm band. The **Fig. 32** (a) shows the design configuration of AAC composed by No. 1~7 waveguides with devices length of L_{AAC} with the slashes and taper section in the length of L_{slash} and L_{taper} , respectively. There are one input port and with three output exits, the exits are marked as L, C and R , which are corresponding to output the mode of TM-polarized 1480-nm band, TE-polarized 775-nm band and TE-polarized 1627-nm band of type-II SPDC modes, respectively. Using the different coupling coefficient of two cross-polarized modes, we designed the first section of AAC to maintain TM-polarized 1480 nm signals and TE-polarized 775 nm pump in the input waveguide 1 and synchronous

to maximum power transfer of TE-polarized 1627 nm idlers to the waveguide 5 and then output from exit R . In ideal case, we can assume the propagation constant of slash waveguide 2~4 are the same ($\beta_2 = \beta_3 = \beta_4$), and the propagation constant of straight waveguide 1 and waveguide 5 are the same ($\beta_1 = \beta_5 \equiv \beta$). The modes with phase difference during propagation is formed as $a_n(z) = b_n(z)\exp(i\beta z)$, the phase difference between slashes and straight waveguide modes is $\Delta\beta = \beta_2 - \beta$, and the coupling coefficient of neighboring waveguides are $K_{i,i+1}(x) = K_{i+1,i}(x)$, the transfer equations and matrix of slashes adiabatic coupling can be expresses as: $-i d\mathbf{b}(x)/dx = \mathbf{M}(x)\mathbf{b}(x)$, with the matrix of **Eq. (76)**.

$$\mathbf{M}(x) = \begin{bmatrix} 0 & K_{12}(x) & 0 & 0 & 0 \\ K_{21}(x) & \Delta\beta & K_{23} & 0 & 0 \\ 0 & K_{32} & \Delta\beta & K_{34} & 0 \\ 0 & 0 & K_{43} & \Delta\beta & K_{45}(x) \\ 0 & 0 & 0 & K_{54}(x) & 0 \end{bmatrix} \quad (76)$$

The second section of AAC is joined with the waveguide taper to shorten adiabatic coupling length as a shortcut design [89] to transfer the power of TM-polarized signals to the waveguide 7 and remain the TE-polarized pump in the waveguide 1 with taper waveguide 1 and 7. Similar to the above analysis, we can rewrite the propagation constant of waveguide 1 and 7, which are the function of propagation x with $\beta \equiv \beta_7(z = L_{taper}) = \beta_1(z = 0_{taper})$, $\beta_1(x) = \beta - x\delta\beta$ and $\beta_7(x) = \beta + (x + L_{taper})\delta\beta$, respectively, and the transfer matrix of taper adiabatic coupling is written as **Eq. (77)**:

$$\mathbf{M}(x) = \begin{bmatrix} -x\delta\beta & K_{16}(x) & 0 \\ K_{16}(x) & \Delta\beta & K_{67}(x) \\ 0 & K_{67}(x) & (x - L_{taper})\delta\beta \end{bmatrix} \quad (77)$$

One can adjust the index profile of waveguides, waveguide dimensions, spacing between waveguides, the slope of slash waveguides or the size of waveguide taper to optimize the $\beta_i(x)$, $K_{ij}(x)$ or $\delta\beta$ to seek the best configuration for pursuing the maximum splitting efficiency and the broadest band of specific required working wavelength under selected temperature environment. We then introduced the computer-aided design (CAD) via Beam Propagation Method simulation [43] to carefully optimize the parameters of AAC. The value of parameters used in optimized 1550-nm band single-mode design AAC are $L_{AAC} = 51\text{mm}$, $L_{slash} = 30\text{mm}$, $L_{taper} = 15\text{mm}$, $S_1 = 4\mu\text{m}$, $S_2 = 8\mu\text{m}$, $S_3 = 9\mu\text{m}$, $S_4 = 9\mu\text{m}$, $S_6 = 13\mu\text{m}$, $G_1 = 18\mu\text{m}$, $G_2 = 62\mu\text{m}$, $W_1 = 7\mu\text{m}$ and $W_2 = 5\mu\text{m}$.

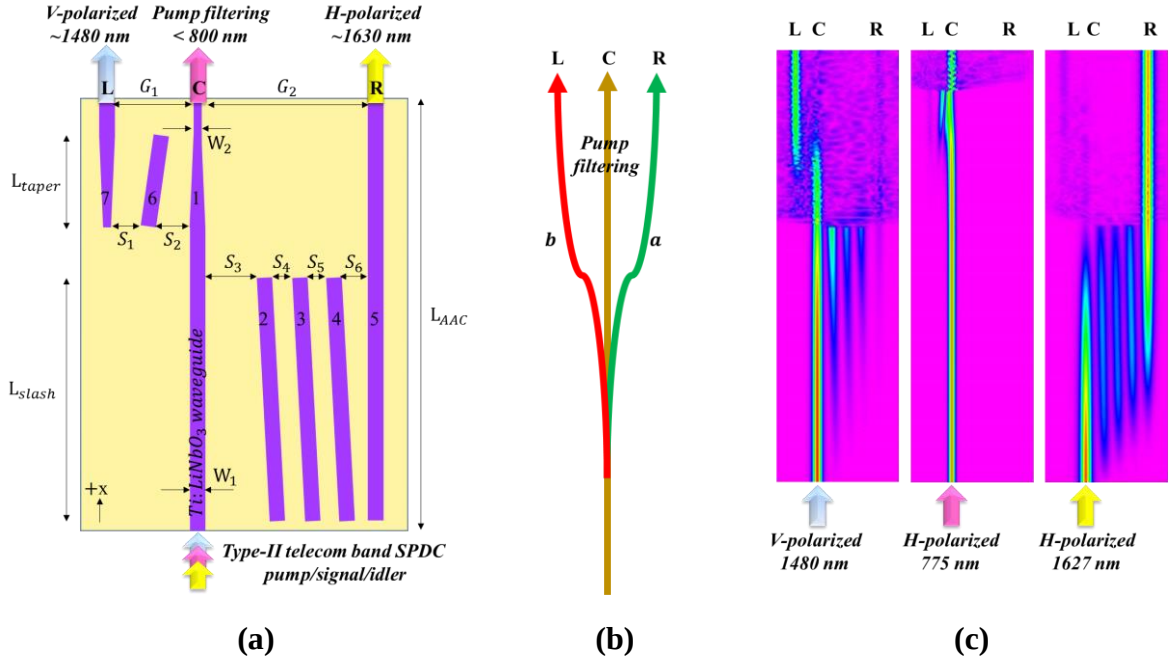


Figure 32. (a) The schematic diagram of asymmetric adiabatic coupler (AAC) under type-II SPDC wavelengths, (b) the functional diagram of exits of L, C, R, quantum path a and b for SPDC pump filtering and signal/idler splitting, (c) the simulated power adiabatic transfer during the propagation under V-polarized 1480 nm, H-polarized 775 nm pump and H-polarized 1627 nm, respectively.

We employed the titanium thermal diffusion (TTD) method [88] to fabricate the channel waveguide array on the $-z$ surface of a 51-mm long, 10-mm in width and 0.5-mm in thickness Z-cut LiNbO_3 crystal. The titanium diffused processes is

accomplish by a high-temperature furnace tube with 1035 °C operation. After end-face polishing, we used a vacuum suction fiber coupling setup [88] to examine the performance of spatial mode segmentation capability and it's bandwidth by using an external cavity laser (linearly polarized, tunable between 1495-1640 nm). The simulated power distribution for each path way and exits are shown as **Fig. 32 (c)**.

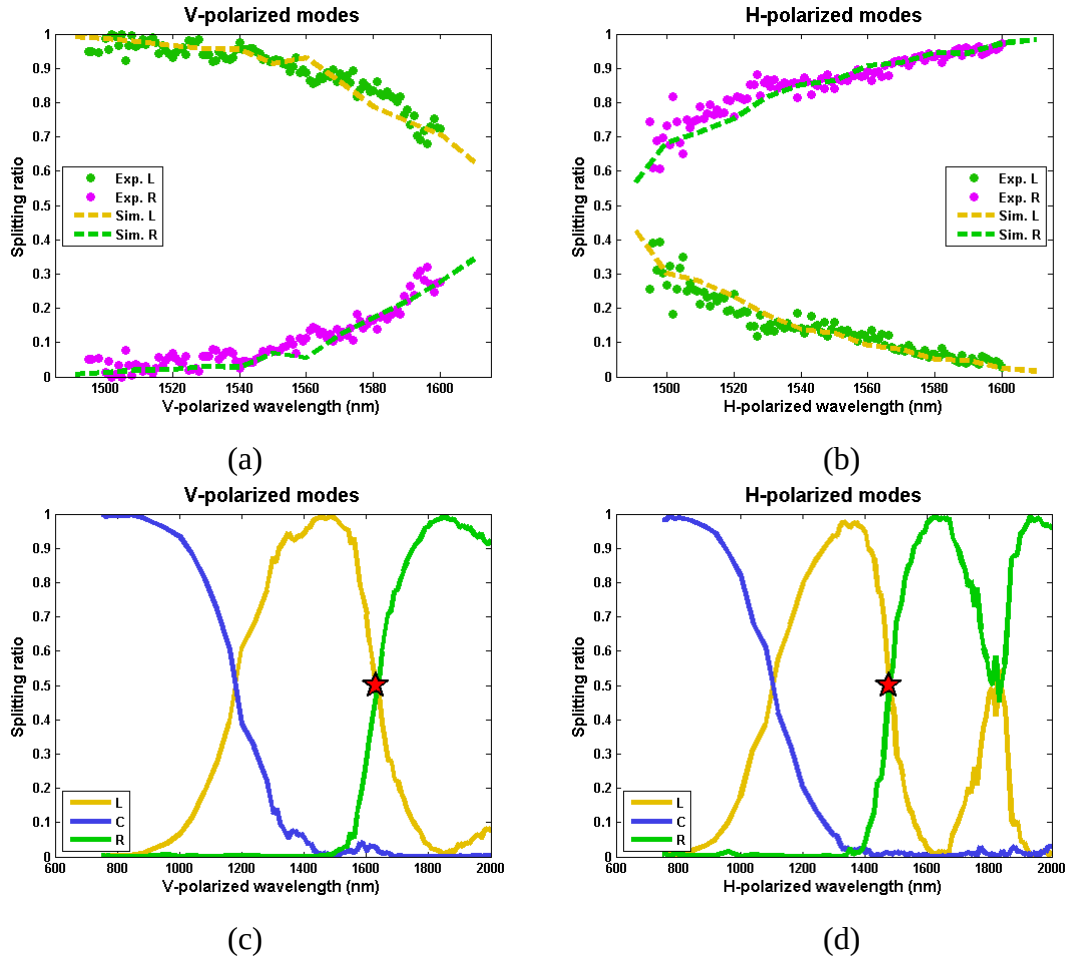


Figure 33. Measured power splitting ratio of (a) V-polarized or (b) H-polarized light in fundamental modes with a wavelength from 1495 nm to 1600 nm, where L or R exits are shown as green or purple circles and the theoretical fitting are denoted by gold or green dash curves. The theoretical splitting ratio of fundamental modes from 750 nm to 2000 nm for (c) V-polarized or (d) H-polarized modes, respectively, where the red star marks singularity wavelengths of quantum-polarization state of Type-II SPDC with V-polarized 1631 nm signal or H-polarized 1477 nm idler, pumped by H-polarized 775 nm wavelength.

The experiment and simulation fitted classical polarization-dependent splitting efficiencies of the AAC are tested under the wavelength from 1495 nm to 1600

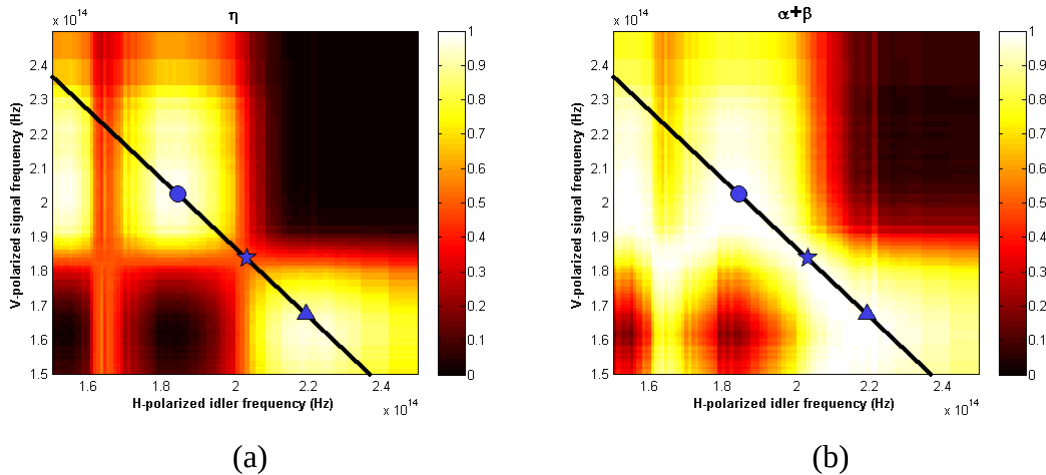
nm, which are plotted as **Fig. 33 (a)-(b)** with V-polarized and H-polarized fundamental mode, and the simulated wide range splitting spectrum of AAC with V-polarized and H-polarized fundamental mode in the wavelength from 750 to 2000 nm are shown as **Fig. 33 (c)-(d)**, respectively.

Manifestation of the AAC in quantum-polarization state preparation

For photons generated via type-II phase-matched waveguide, signal and idler photons are in pairs with orthogonal polarizations under energy conservation $\omega_{pump,H} = \omega_{signal,V} + \omega_{idler,H}$. If photons are launched into an arbitrary beam splitter, one can end up with a state described by

$$|\Psi\rangle = \alpha(|H\rangle_a|V\rangle_b) + \beta(|V\rangle_a|H\rangle_b) + \gamma(|H\rangle_a|V\rangle_a) + \xi(|V\rangle_b|H\rangle_b), \quad (78)$$

where $|\alpha|^2 + |\beta|^2 + |\gamma|^2 + |\xi|^2 = 1$. The last two terms in **Eq. (78)** describes the case when both signal and idler photons are going to the same port, which is not recognizable by on-off detectors. For our polarization- and wavelength-dependent AAC, here we focus on the investigation on how the signal and idler wavelengths influence the ratio between α and β , where we also define a preparation efficiency as $\eta = |\alpha|^2 + |\beta|^2$.



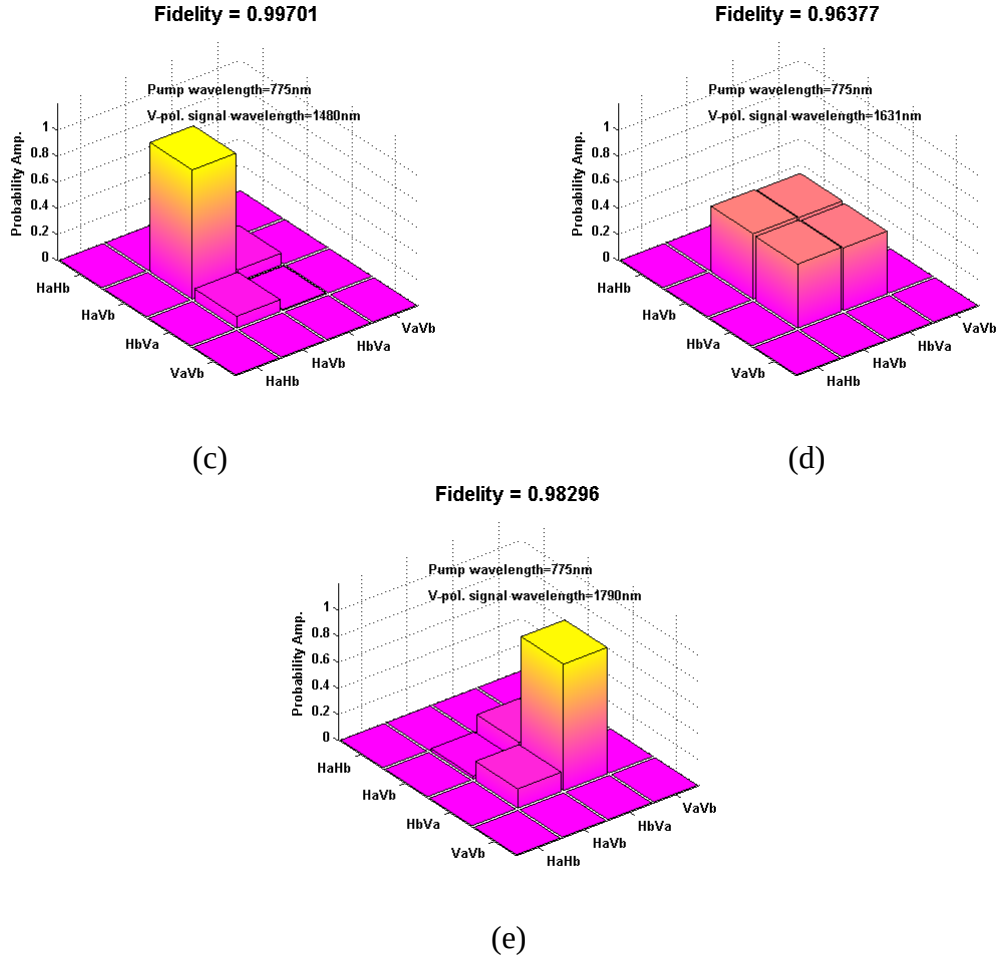


Figure 34. (a) The simulated efficiency and (b) the coefficients of $\alpha + \beta$ of quantum polarization-path states of $\Psi = \alpha(|H\rangle_a|V\rangle_b) + \beta(|V\rangle_a|H\rangle_b)$ in frequency domain with blue star markers show the singularity wavelengths of quantum polarization-path state, and the simulated density matrix after AAC splitting under QPM Type-II SPDC photon-pairs pumped by 775 nm with V-polarized signal mode wavelength equals (c) 1480 nm, (d) 1631 nm or (e) 1790 nm respectively.

The simulated preparation efficiency and coefficients of $\alpha + \beta$ are plotted in the V-polarized and H-polarized modes in frequency domain as **Fig. 34 (a-b)** shown with the QPM tuning line of 775-nm pump SPDC light source. The tuning curve of QPM-based SPDCs are the function of pump wavelength and the working temperature, which needs to satisfy to the QPM condition as $k_{pump,H}(\lambda_p, T) = k_{signal,V}(\lambda_s, T) + k_{idler,H}(\lambda_i, T) + \mathbf{K}$, with $\mathbf{K} = 2\pi m/\Lambda$ with the m -order QPM SPDC period of Λ . Due to the crosstalk of Ti-diffused LiNbO₃ waveguide is less than -20 to -25 dB in typically [86], the polarization maintaining of waveguide is relatively very well, which is very suitable to be the quantum

polarization preparation devices. On the other hand, we can use another electrode at one arm of AAC part to apply electric-field along crystal z-axis via the EO effect of LiNbO₃ waveguides with the phase modulation of $\Delta\phi = \Delta\phi(VL_{ED})$ to compensate the time-delay of cross-polarized signal and idler modes, where L_{ED} is active length of electrode. For the near path-entangled situation, the pursuing for ideal indistinguishable case can be achieved around singularity with coefficient of each states are almost the same with $\alpha \approx \beta \approx \gamma \approx \xi = 0.5$, for this case, the output splitting probability of the path **A** or path **B** are almost at the same value although the polarizations and wavelengths of signals and idlers are different, which implies that in this specific range, the quantum polarization states of these photon-pairs are indistinguishable.

In summary, we demonstrate an on-chip ultra-broadband polarization- and wavelength-dependent splitter based on a z-cut titanium-diffused LiNbO₃ waveguiding asymmetric adiabatic coupler (AAC), the simulated bandwidths of power splitting ratio > 17dB are > 60 nm or > 70 nm for H-polarized or V-polarized modes from 1610 nm to 1670 nm or from 1430 nm to 1500 nm, respectively. The AAC element is suitable for combining the existed fibre-optic telecommunication systems and developing on-chip quantum photon-pair source systems, the polarization-wavelength-dependent splitting characteristic of AAC can be applied to combine a LiNbO₃-based Type-II spontaneous parametric down conversion (SPDC) quantum light source pumped by 775 nm wave for quantum polarization-path state preparation. The simulated density matrix based on Type-II SPDC after AAC splitting shows the tunability of quantum polarization-path states can be controlled by changing the QPM conditions of SPDC, which can easily be tuned by pump wavelength or working temperature within the same QPM structure.

5-2 Topic V: Experiments of quantum integrated optical circuits

In this section, the SPDC quantum light source of Titanium-diffused PPLN waveguides (Ti:PPLN/W) play the important roles of several quantum experiments, the simulation, experiment, characteristic and analysis results will be introduced in the following paragraphs. The principle concept of SPDC generated by Ti:PPLN/W quantum integrated optical circuits (QIOCs) is described in (5-2-1), according to the polarizations of pump, signal or idler in SPDC processes, the SPDC types can be divided into three types as Type-0, Type-I, or Type-II corresponding to the polarizations of pump-signal/idler for e-ee (o-oo), e-oo (o-ee), or o-eo (e-eo), respectively. For LiNbO₃ crystal, due to the existing nonlinear coefficients non-zero terms, the most efficient SPDC processes is Type-0 for e-ee configuration for using the nonlinear coefficient $d_{33} \cong 27 \text{ pm/V}$, and second efficient processes are Type-I e-oo or Type-II o-eo configurations for using the nonlinear coefficient $d_{15} = d_{24} = d_{31} = d_{32} \cong 5 \text{ pm/V}$. For the cross-polarized entangled quantum light source generation, we designed and fabricated a Type-II Ti:PPLN/W SPDC of the pump wavelength at 785nm operated in $\sim 130^\circ\text{C}$ for generating the wavelength degenerated signal and idler in the 1570 nm in TM (e-wave, V-polarized) and TE (o-wave, H-polarized) polarizations. The design schematic diagram shows the QPM condition for Ti:PPLN waveguide SPDC as the **Fig. 35**. The measured classical SFG characteristics are shown in (5-2-2), and the SFG-SPDC analogy method is described in (5-2-3), Quantum entanglement and quantum tomography experiments are summarized in (5-2-4), and the final part of this chapter is the quantum polarization state tomography via the metasurface with SPDC source of single-photon sources and entangled photon-pair sources, the theory, experiment,

results and analysis are concluded in (5-2-5), this idea original from PhD student Kai Wang, Professor Dragomir Neshev and A/Professor Andrey A. Sukhorukov in RSPE Australian National University (ANU), and we cooperation completed this study in ANU.

5-2-1 Ti-diffused PPLN waveguide (Ti:PPLN/W) QIOC

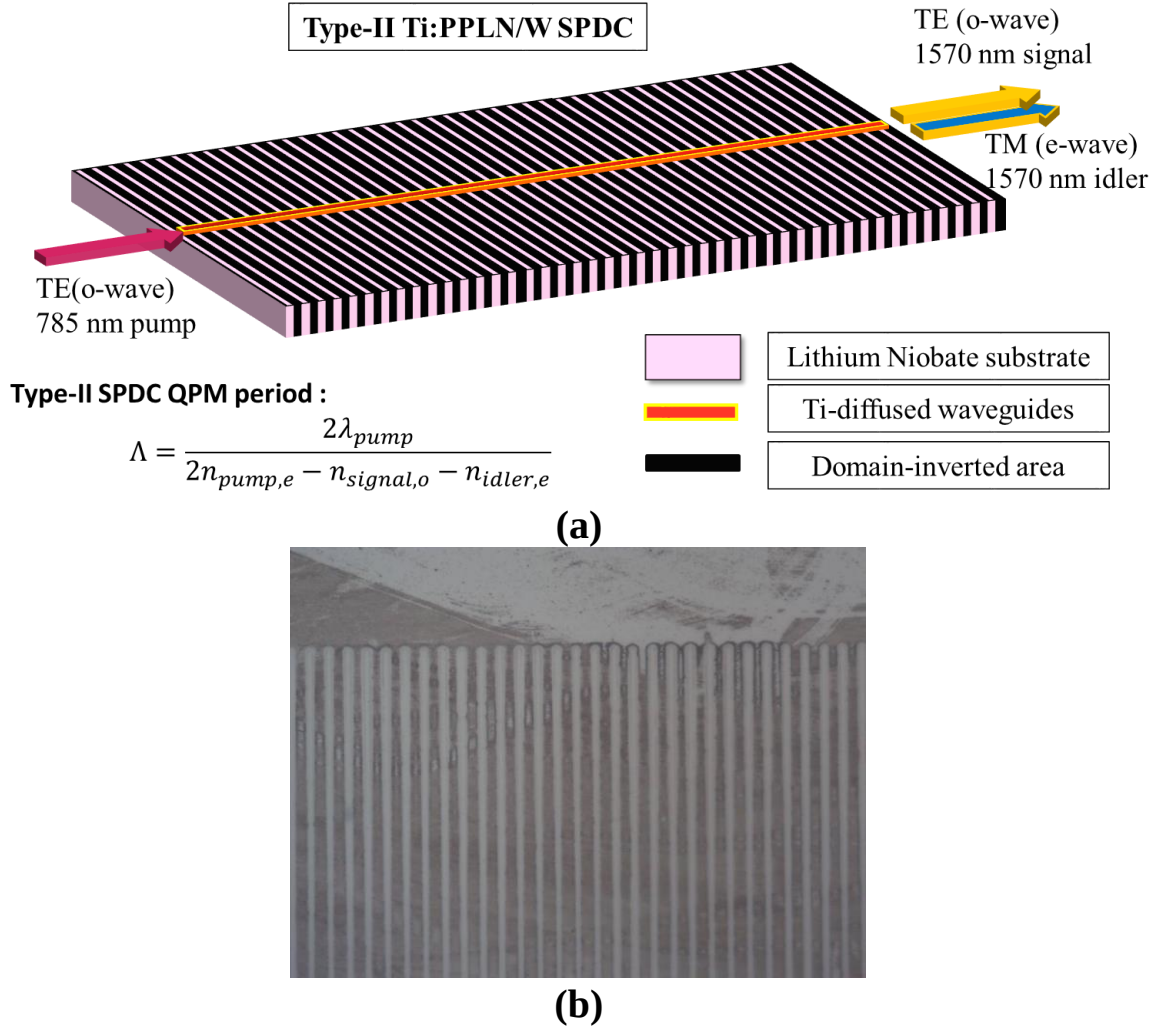


Figure 35. (a) The schematic diagram of Ti-diffused PPLN waveguide as type-II SPDC quantum polarized-entangled light source. The insert equation is the domain-inverted period for satisfying the QPM condition of type-II SPDC. (b) The real micrographs of our PPLN test sample with the periods around 9.5 μm .

The design schematic diagram of Type-II SPDC quantum entangled photon-pair generation Ti-diffused PPLN waveguide chip is shown as **Fig. 35 (a)**, the designed QPM condition is suitable for 785 nm pump of type-II SPDC

wavelength degeneracy cross-polarization 1570 nm signal and idler photons under 100 to 200 degree Celsius for poling periods from 8.7 to 9.7 μm . The chip substrate is congruent-LiNbO₃ (CLN), which is 0.5 mm in thickness, 30 mm in long and 15 mm in width. The diffused waveguide made by standard high temperature Titanium thermal diffusion (TTD) method, the end-faces are polished up to optical-grade for minimizing the coupling loss. The measured propagation loss of Ti-diffused LiNbO₃ waveguides are ~ 0.3 dB/cm and ~ 0.6 dB/cm for TM-polarization or TE-polarization modes, and the poling uniformity can be tested by HF-etch as **Fig. 35 (b)** shown.

5-2-2 Type-II SFG Ti:PPLN/W characteristic

To quantize the efficiency of quantum SPDC processes, one can use its reverse process in classical nonlinear process, SFG, to find the phase-matching wavelength under different temperature and to investigate the mode fields of different type conversion. In our experiment, we expect the working temperature of SPDC source needs to be above 100 °C to minimize the photorefractive effect [90]. The type-II SFG process of QIOC is pumped by two orthogonal polarization mode with the same wavelength at around 1570 nm, and the characteristic of SFG signals are measured by visible CCD and power detector under different pump wavelengths. We found two types of modes, the type-0 SHG mode and type-II SFG mode at the same pumping condition. The type-0 SHG signal is the higher-order modes, which is possible be generated via higher-order QPM conditions, so the relative power is 2 order less than 1st-order QPM type-II SFG process signal with fundamental mode conversion, which is 14 pW and 1546 pW under the same pump power with around 0.3 mW, respectively. The mode fields of this two type conversion processes are shown in the **Fig. 36**.

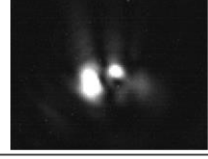
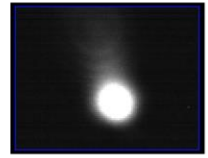
Pump wavelength : 1589.70 nm, 45° polarization, poling period: 9.5 μm, T=23.8 °C			
Type	Analyzer polarization	Measured power (after analyzer)	Mode field
Type-0 SHG (Higher-order mode)	TM	14 pW	
Type-II SFG (Fundamental mode)	TE	1546 pW	

Figure 36. Measured power and the mode field of type-0 SHG or type-II SFG signals with 45 degree polarized pump via a Ti-diffused PPLN waveguide with poling period is 9.5 μm.

The classical SFG intensity can be derived as the **Eq. 79**, which is the function of interaction length (L), pump intensity (I_p), signal intensity (I_s), effective nonlinear coefficient (d_{eff}), mode overlapping (Θ) and phase-mismatching ($\Delta\beta$) [91].

$$I_{SFG} = \frac{8d_{eff}^2\omega_{SFG}^2I_pI_s}{n_p n_s n_{SFG} \epsilon_0 c^2} \Theta L^2 \text{sinc}\left(\frac{\Delta\beta L}{2}\right)^2 \quad (79)$$

According to **Eq.79**, one can calculate the ideal first-order QPM SFG spectrum with $d_{eff} = \frac{2}{\pi} \sin(\pi D)$, D is the duty-cycle. If there are existed poling errors on the sample [92], one can slightly modified **Eq.79** to fit the experimental results as the **Fig. 37 (a)** shown. The QPM condition of SFG processes can be obtained by tuning the pump wavelength of SFG, the output power spectrum with different pump wavelength is shown as **Fig. 37 (a)**, the FWHM of spectrum is ~0.36 nm, and the SFG power under different pump power are also been measured, the results is shown as **Fig. 37 (b)**. To fully investigate the phase-matching condition of our sample, we measured 7 different periodic gratings with periods from 9.1 μm to 9.7 μm, step by 0.1 μm under different pump wavelength with temperature is

23.8, 100.0, 131.4, 150, 180°C, respectively. The QPM results show that the temperature tuning rate of Type-II phase-matched SFG pump wavelength are ~ -0.18 to -0.22 nm/°C as **Fig. 37 (c)-(d)**.

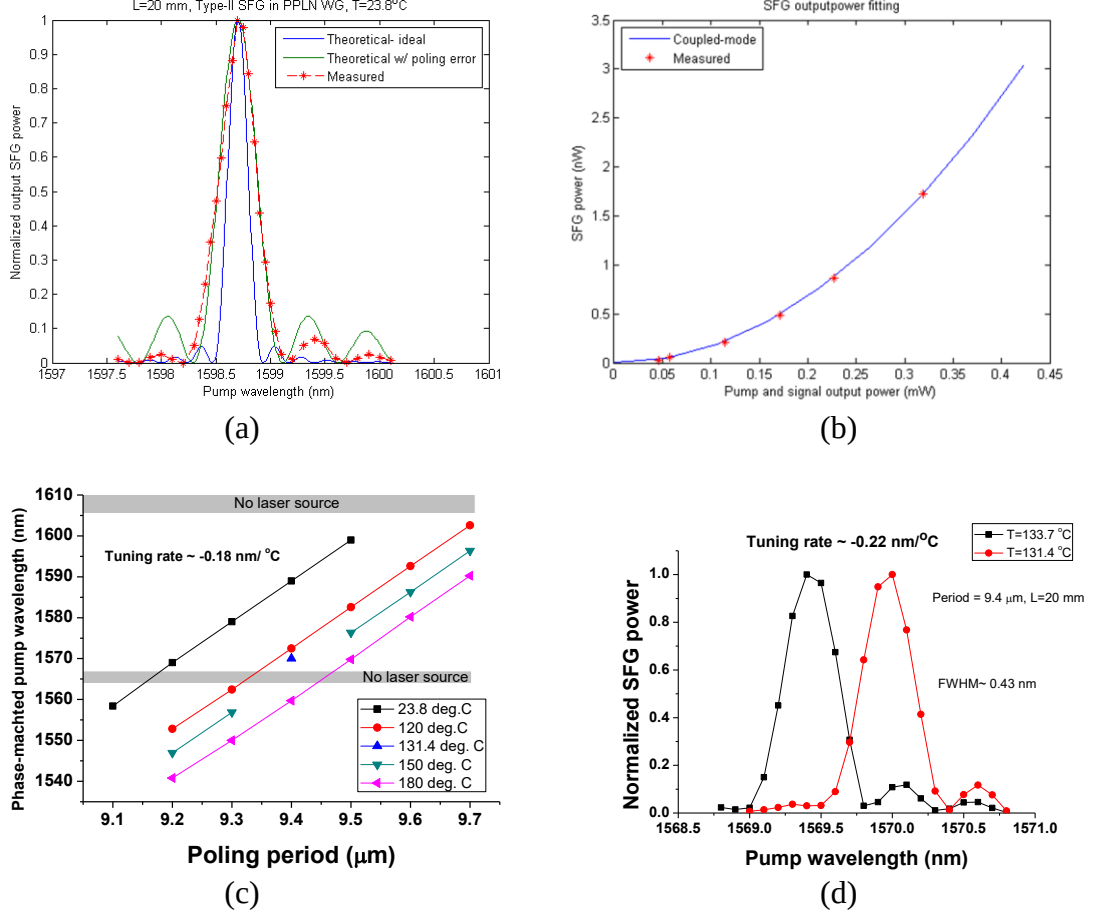


Figure 37. (a) the measured type-II SFG QPM conditions with theoretical fitting with or without poling error, (b) the SFG signal output power under gradually increasing pump power, (c) the QPM pump wavelengths of different poling under different working temperature and (d) the SFG signal spectrum under temperature tuning.

5-2-3 SFG-SPDC analogy

We can start to model the SPDC photon-pairs generation rate from the generated photon number per mode within the phase-mismatching conditions from the parametric fluorescence. [93]. The generated photon number per mode is function of the coupling coefficient (Γ), the phase-mismatching parameter (Δ) and the interaction length (L) as **Eq. (80)**.

$$D(\Gamma, \Delta) = \Gamma^2 L^2 \left| \frac{\sinh(\sqrt{\Gamma^2 - \Delta^2} L)}{\sqrt{\Gamma^2 - \Delta^2} L} \right|^2 \quad (80)$$

where, $\Gamma = \kappa^2 P_p$, $2\Delta = \beta_p - (\beta_s + \beta_i + K)$, $K = m \frac{2\pi}{\Lambda}$ in the QPM process, with coupling coefficient :

$$\kappa = \varepsilon_0 d_{eff} \sqrt{\frac{2\omega_s \omega_i}{n_p n_s n_i S_{eff}}} \left(\frac{\mu_0}{\varepsilon_0} \right)^{3/2} \quad (81)$$

and waveguide mode overlapping parameter:

$$S_{eff} = \frac{\pi}{32} \left(\frac{w_{x,s}^2 + w_{x,i}^2 + w_{x,p}^2}{w_{x,p}} \right) \left(\frac{w_{y,s}^2 + w_{y,i}^2 + w_{y,p}^2}{w_{y,p}} \right) \exp \left(\frac{16d_x^2}{w_{x,s}^2 + w_{x,i}^2 + w_{x,p}^2} \right) \quad (82)$$

with the $w_{x,s}, w_{x,i}, w_{x,p}$ is the mode width of signal, idler and pump in vertical, the $w_{y,s}, w_{y,i}, w_{y,p}$ is the mode width of signal, idler and pump in transverse and the d_x is the mode center shifting parameter of pump, signal and idler modes, respectively. According to the **Eq. (80)**, the SPDC spectrum FWHM bandwidth can be approximated as **Eq. (83)**.

$$\Delta\lambda \approx \frac{\lambda^2}{L|n_s - n_i|} \quad (83)$$

and the conversion efficiency is shown as **Eq. (84)**, which is the function of interaction length and coupling coefficients with the refractive index mismatching of signal and idler modes.

$$\eta = \frac{P_{s,i}}{P_p} \approx \frac{h\omega_{s,i}}{2\pi|n_s - n_i|} |\kappa^2| L \quad (84)$$

On the other hand, for more direct way, The SPDC efficiency can be estimated by classical SFG efficiency as **Eq. (85)**. [94].

$$\frac{1}{\Phi_P} \frac{dN_{pair}}{dt} = \sum_j \Xi_j^{SFG} \frac{\lambda_P^2}{(\lambda_s)_j (\lambda_i)_j} \frac{c \Delta \lambda}{[(\lambda_s)_j]^2} \quad (85)$$

$\frac{dN_{pair}}{dt}$, Φ_P , $\Delta \lambda$, are the photon-pair generation rate (1/sec), SPDC pump power (Watt), SPDC signal/idler spectrum FWHM, respectively, and the SFG efficiency defined by $\Xi^{SFG} \equiv \frac{P_P^{SFG}}{P_s P_i}$.

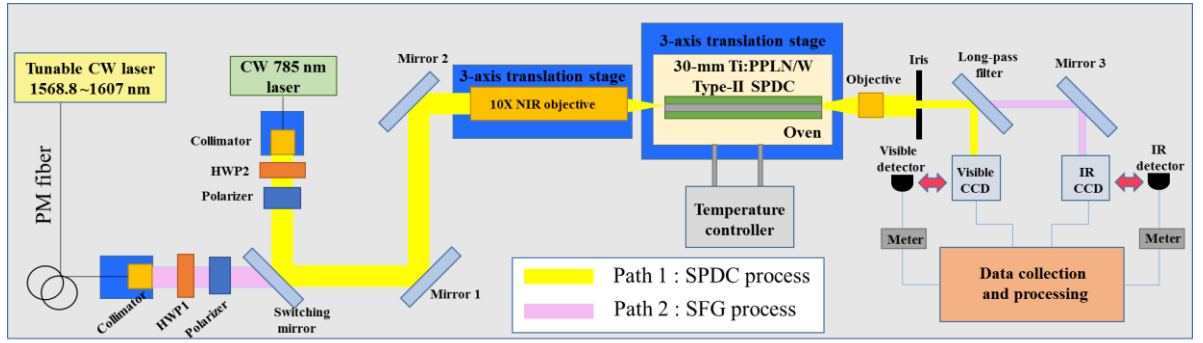


Figure 38. The dual function experimental setup schematic drawing for SPDC generation and wavelength-degenerate SFG processes with the same Type-II Ti-diffused PPLN waveguide devices.

Three filters after Ti:PPLN/W SPDC are used to filter out the signal and idler side band around 1570-nm and SPDC 785-nm pump, we observed SPDC mode by IR camera as **Fig. 39 (a)** and the bandpass filter spectrum is shown as solid curve in **Fig. 39 (b)**, the dots with error bars in **Fig. 39 (b)** are the SPDC mode intensities after filtering, which are extracted from camera data. It shows the temperature-dependent output intensities, which can be used to examine the phase-matched temperature of quantum source.

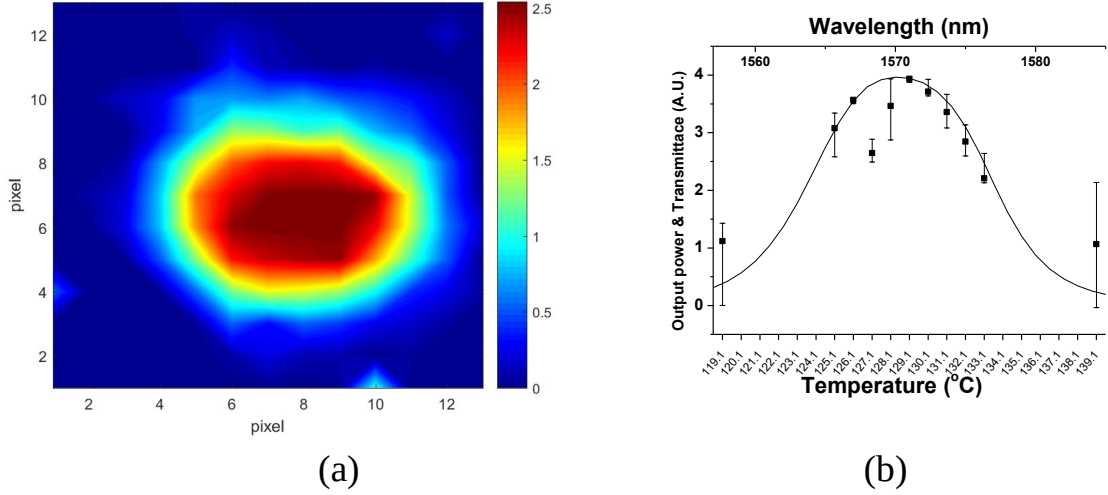


Figure 39. Measured Type-II SPDC mode field (a) and the output SPDC mode intensities under different operation temperature.

We can take more direct way to analog the SDPC quantum characteristic from classical SFG features [94], we slightly modified the launch setup before coupling objective lens in **Fig. 38** by combing two tunable C- and L-band CW lasers as SFG wavelength non-degenercized inputs with orthogonal polarization in H-polarized and V-polarized. As a result, the type-II SFG-SPDC analogy can be measured by examining the SFG output power density matrix under different wavelength of H-polarized and V-polarized inputs, which can be corresponed to the SPDC coincidence density matrix between signal and idler in spectrum domains.

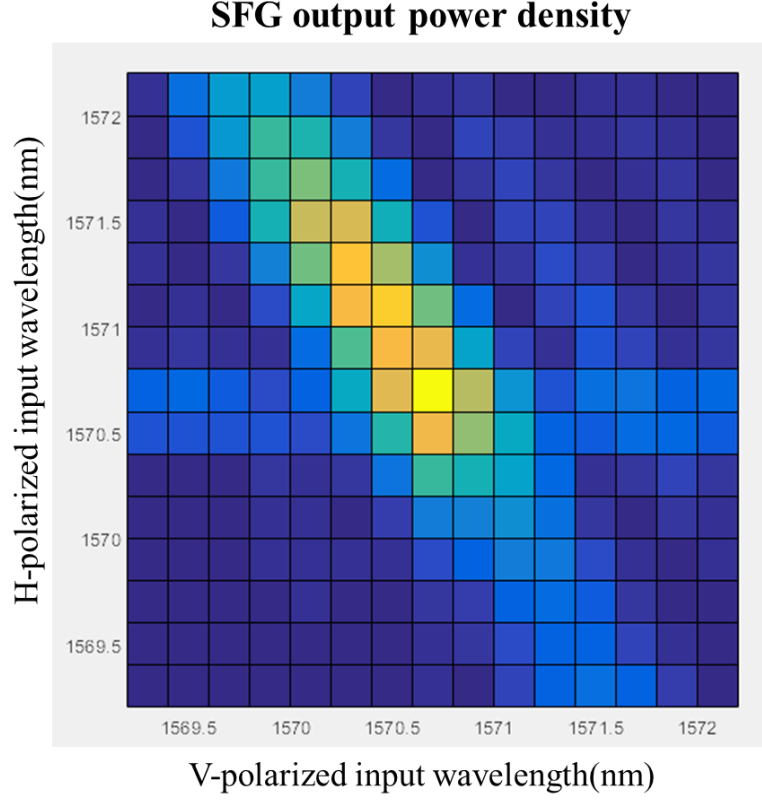


Figure 40. Measured the Type-II quasi-phase matched classical sum-frequency generation output power via H and V polarization input at 1570.6 ± 1.5 nm wavelength range.

According to our measured SFG output power density matrix result shown as **Fig. 40**, despite the density spectrum with some central offset around 1570.6 nm range, we found the output power density matrix is quite concentrated, which means that our Type-II Ti:PPLN/W SPDC has higher brightness in comparison with Type-0 SPDCs [94]. With its strict QPM condition under fixed operation temperature, we can use this feature to generate more pure entangled photon pairs, the permissible SFG pump wavelength within FWHM range is around 0.3 nm only, which is close 100 times smaller than Type-0 PPLN/W SPDCs.

5-2-4 Quantum entanglement and quantum tomography

The quantum entanglement experiment can use quantum tomography [95] to exam the quantum states, the maximum entangled states are called the Bell's

states as **Eq. (86)**, there exists different entangled Bell's states types, the Polarization-entangled Bell's states, or the Path-entangled Bell's states as the **Eq. (87)** or **Eq. (88)**. The polarization-entangled Bell's states can be generated from Type-0, Type-I or Type-II SPDC sources, the Type-0 and Type-I SPDC photon-pairs are the same polarized signal and idler as the states of $|\Phi^\pm\rangle$, but the Type-II SPDC photon-pairs are generated in the cross-polarized signal and idler as the state of $|\Psi^\pm\rangle$ in the **Eq. (87)**. In this study, our quantum light source is selected as Type-II SPDC for generating the high-brightness within rigorous QPM conditions to maximum the purity of quantum polarization states.

Bell's states:

$$\begin{aligned} |\Phi^\pm\rangle &= \sqrt{1/2}(|0\rangle_a|0\rangle_b \pm |1\rangle_a|1\rangle_b) \\ |\Psi^\pm\rangle &= \sqrt{1/2}(|0\rangle_a|1\rangle_b \pm |1\rangle_a|0\rangle_b) \end{aligned} \quad (86)$$

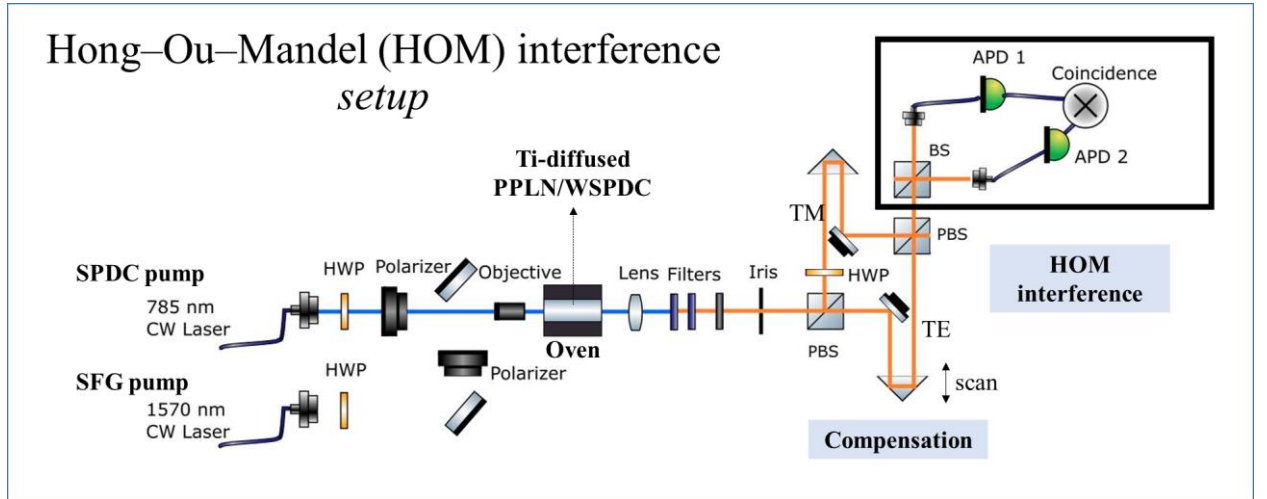
Polarization-entangled Bell's states:

$$\begin{aligned} |\Phi^\pm\rangle &= \sqrt{1/2}(|H\rangle_s|H\rangle_i \pm |V\rangle_s|V\rangle_i) \\ |\Psi^\pm\rangle &= \sqrt{1/2}(|H\rangle_s|V\rangle_i \pm |V\rangle_s|H\rangle_i) \end{aligned} \quad (87)$$

Path-entangled Bell's states:

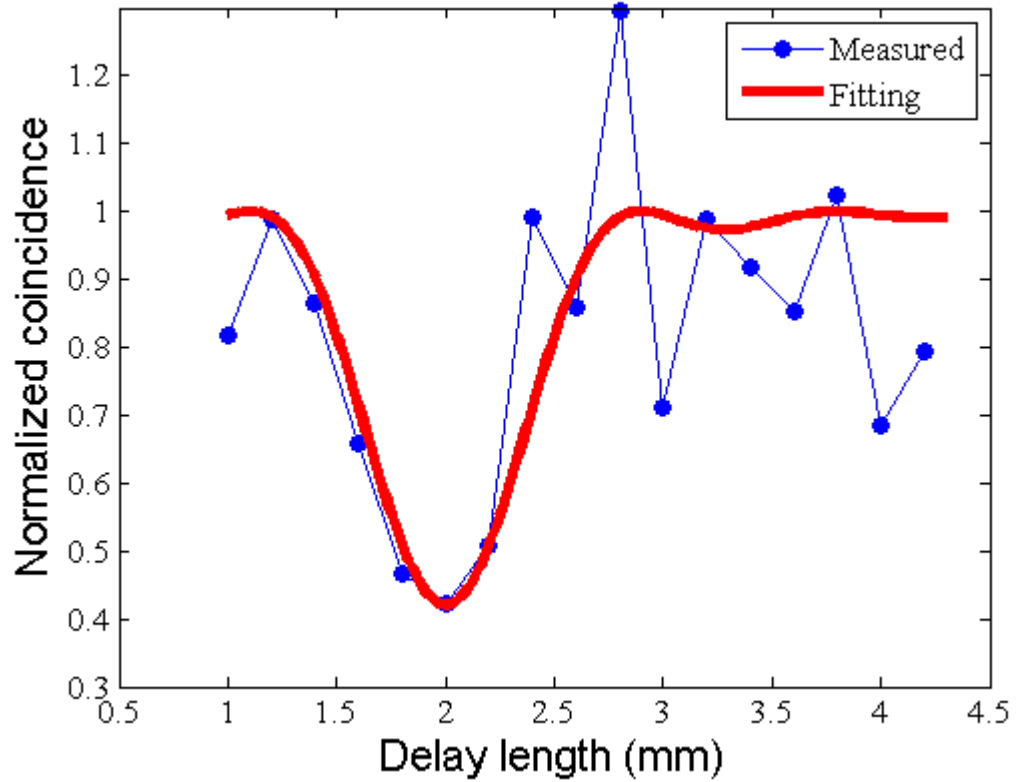
$$\begin{aligned} |\Phi^\pm\rangle &= \sqrt{1/2}(|H\rangle_a|H\rangle_b \pm |V\rangle_a|V\rangle_b) \\ |\Psi^\pm\rangle &= \sqrt{1/2}(|H\rangle_a|V\rangle_b \pm |V\rangle_a|H\rangle_b) \end{aligned} \quad (88)$$

We constructed the Hong-Ou-Mandel (HOM) interferometer, the experiment setup is shown **Fig. 41 (a)**. [96] for time-delay compensation between H-polarized and V-polarized modes, the delay time is ~ 3.3 ps, with the OPD ~ 1 mm between H-polarized and V-polarized modes, and the HOM dip preliminary measurement result shows as **Fig. 41 (b)** that the visibility of HOM interference is only 50~60 %, which limited by the experiment setup as temperature controller stability and the translation stage for time-delay compensation, the ideal visibility of HOM experiment needs to be optimized as higher as possible in the future experiment.



(a)

HOM dip, counting time : 300 sec.



(b)

Figure 41. (a) The setup of the Hong–Ou–Mandel (HOM) interferometer with time delay compensation for analyzing the type-II Ti:PPLN waveguide SPDC source. (b) Preliminary measured and fitted HOM-dip results with the visibility are around 60 % under ~3.3 ps time delay, the most ideal visibility is possible to achieve to approach to ~100% when signal and idler reaches indistinguishable condition.

According to this result, we found there are some critical and harsh experimental issues needs to be overcome to approach to the maximum visibility

for generating pure Bell's state for Type-II entangled photon-pairs. The HOM-dip fitting function is based on the simplest visibility function (V) with sinc-square of delay length (x) and the fitting constant (Φ) as **Eq. (89)** shown.

$$F(x) = 1 - V * (\text{sinc}(\Phi x))^2 \quad (89)$$

The following experiment of our SPDC source is quantum tomography [95] with the setup as **Fig. 42**, which can be used to test the quantum polarization state of photon-pairs with any unknown polarization states. The SPDC photon-pairs have been separated in two ways with H-polarized or V-polarized modes by a PBS, and after time delay compensation, and combined in the beam splitter (BS), and we used two set of quart-wave-plate (QWP), half-wave-plate (HWP) and polarization analyzer (PA) after beam splitter to prepare any polarization states before the single-photon detectors. The prepare pass polarization states combination are {'HH', 'HV', 'VV', 'VH', 'RH', 'RV', 'DV', 'DH', 'DR', 'DD', 'RD', 'HD', 'VD', 'VL', 'HL', 'RL'} for 16 different selected tomography conditions. For example, 'VH' means the selected pass polarization state is V-polarized photons for APD-1 and H-polarized photons for APD-2 and so on. (Symbol Description: H, V, R, L and D mean the polarization state is horizontal, vertical, right-handed, left-handed and $\pm 45^\circ$, respectively.)

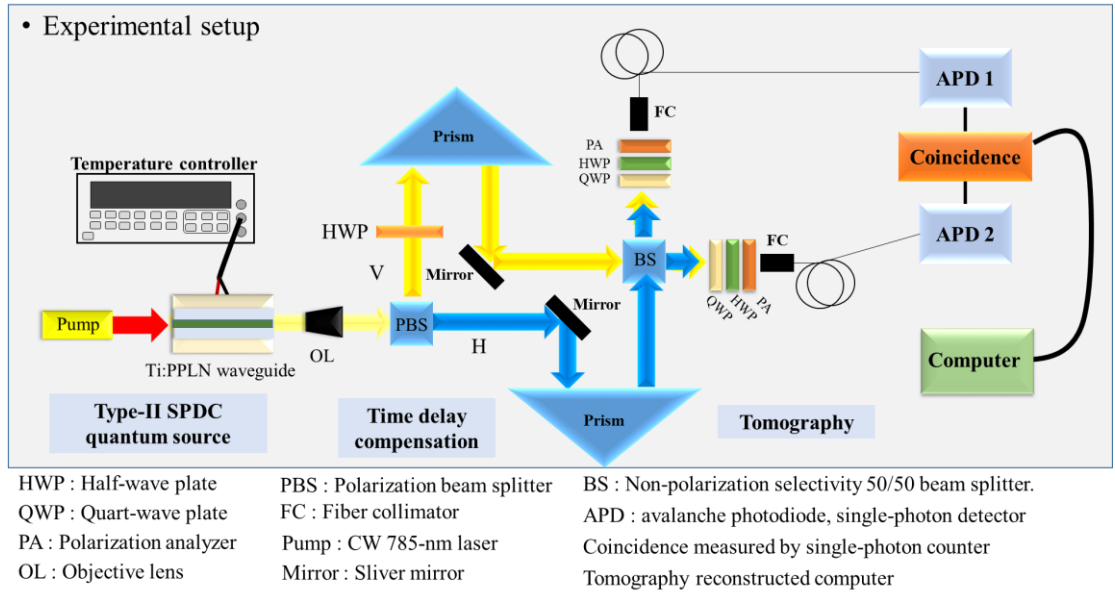
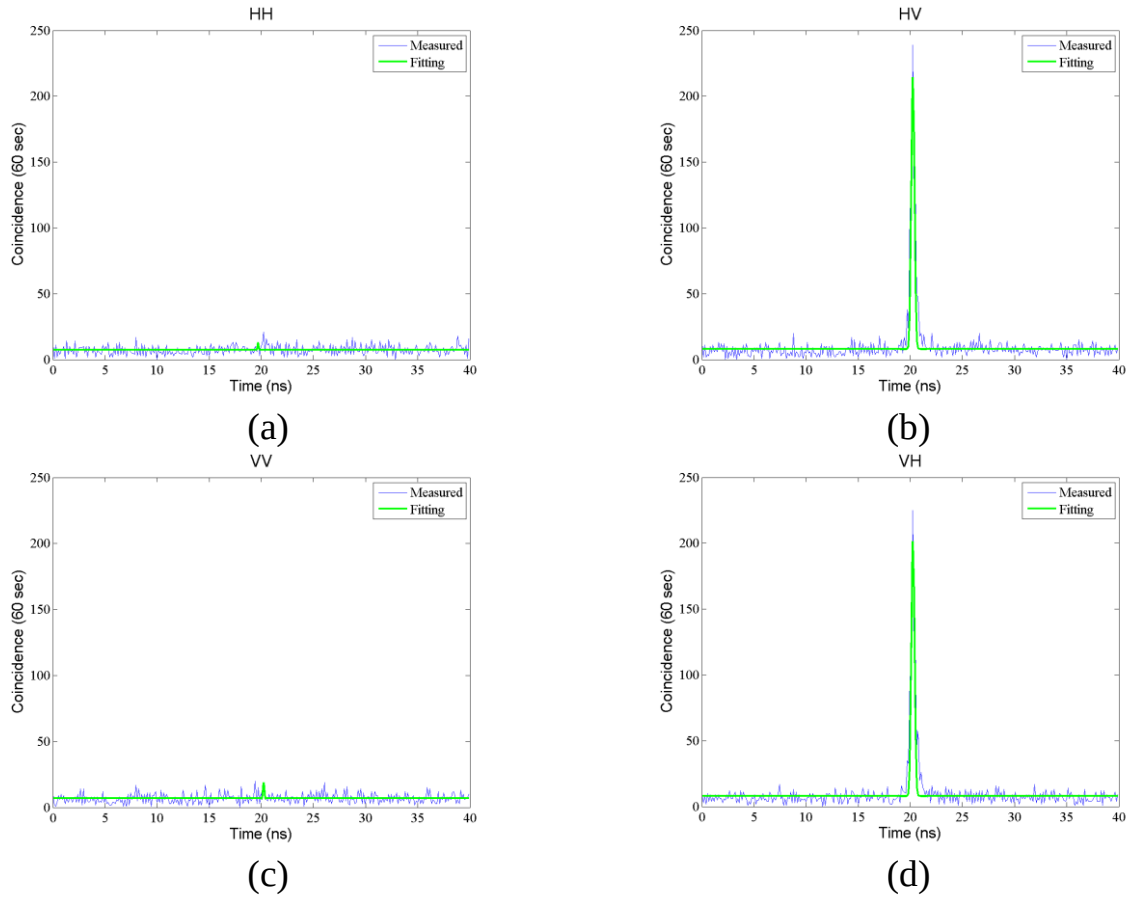
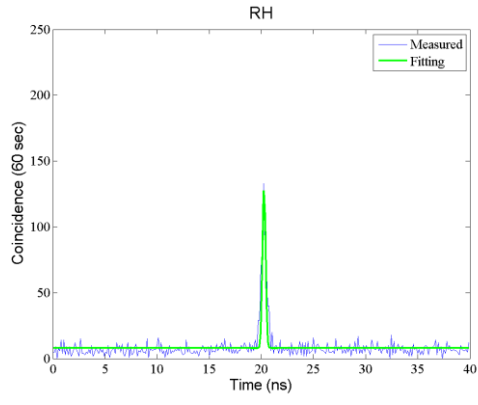
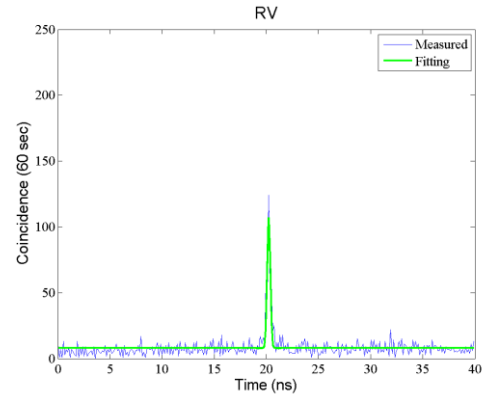


Figure 42. The experimental setup of type-II SPDC time delay compensated quantum tomography.

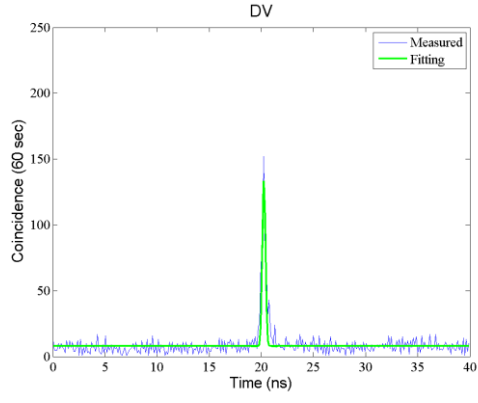




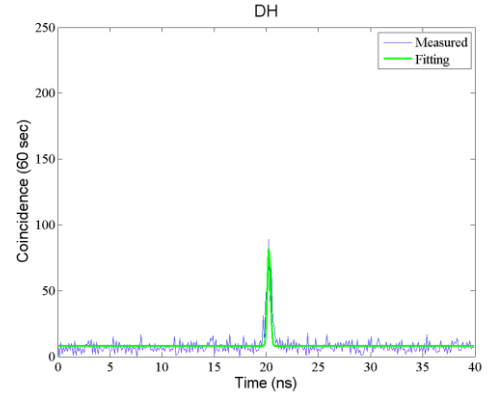
(e)



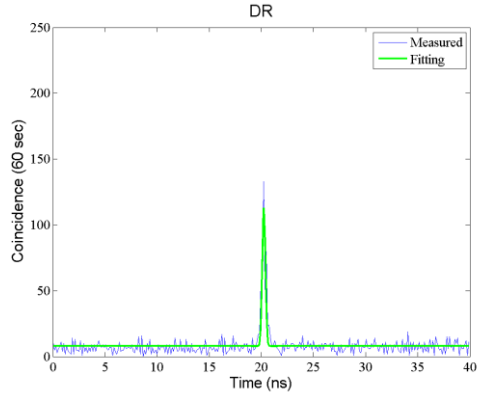
(f)



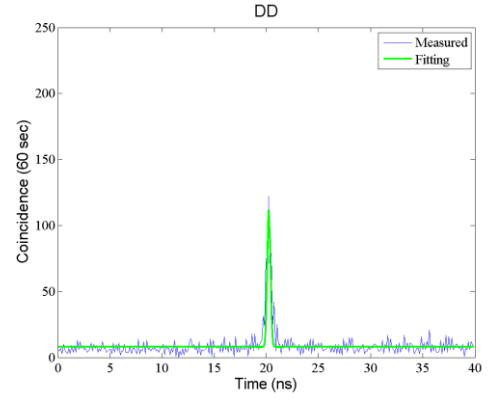
(g)



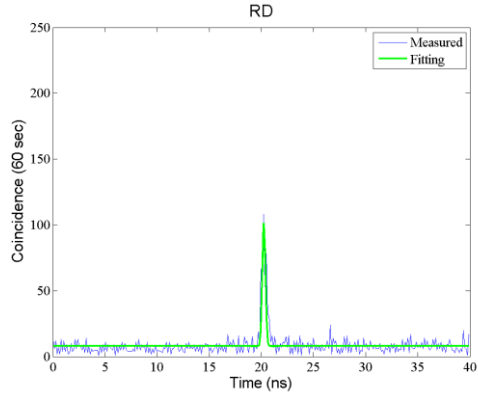
(h)



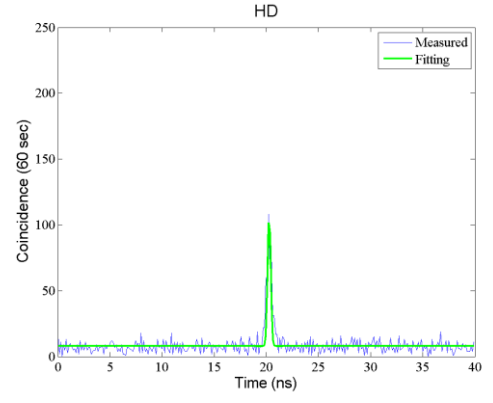
(i)



(j)



(k)



(l)

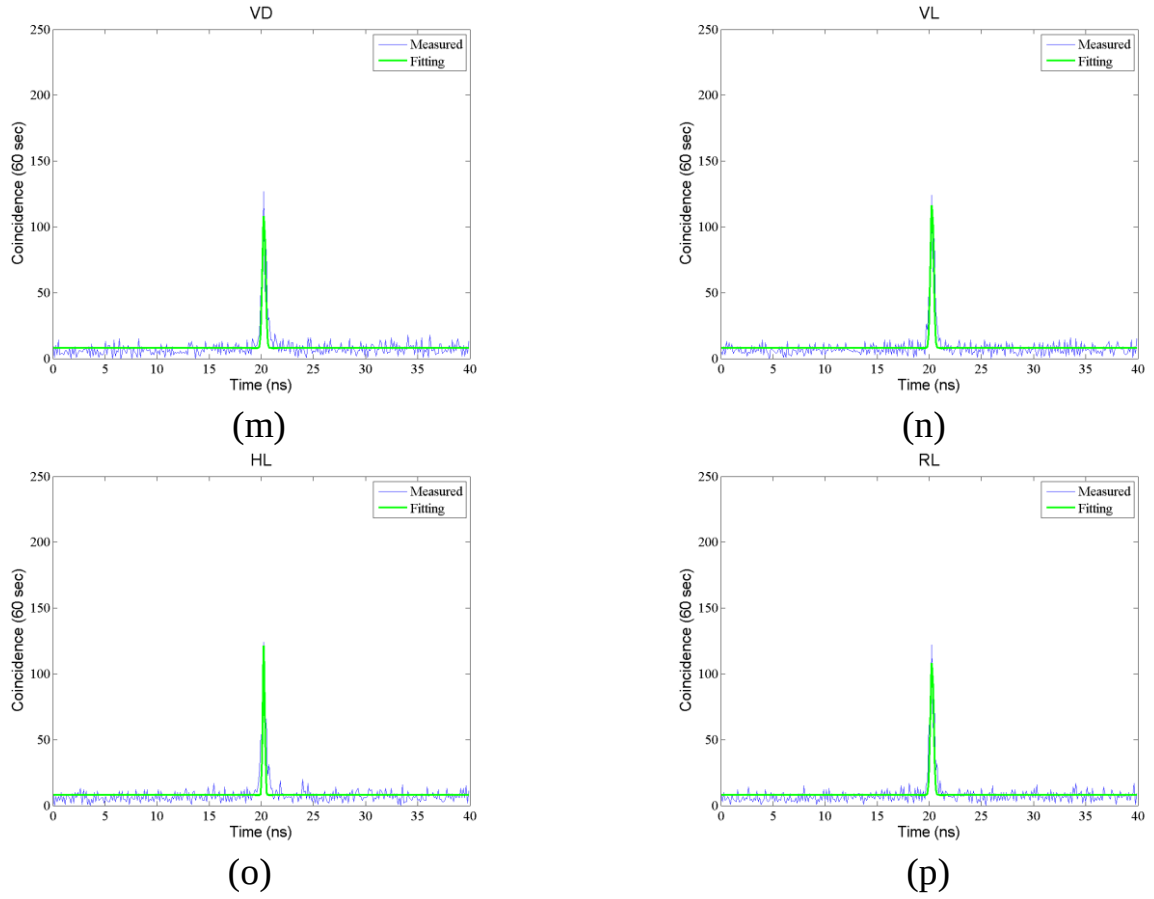


Figure 43. The experimental measured and fitted 60 second coincidence counting results of NILP type-II SPDC quantum tomography under deliberately chosen polarization setting of 'HH', 'HV', 'VV', 'VH', 'RH', 'RV', 'DV', 'DH', 'DR', 'DD', 'RD', 'HD', 'VD', 'VL', 'HL', 'RL' from (a) to (p), respectively.

We reconstructed the quantum polarization state density matrix by quantum tomography setup with or without time-delay compensation through the coincidences as the **Fig. 44** shown. The reconstructed density matrix of quantum polarization state of measured Type-II SPDC without time-delay compensation is shown as **Fig 44 (a-c)**, which is corresponding to the mixed state $|M\rangle$ with the trace of diagonal value is ~ 0.54 . The simulated Bell's states for density matrix of pure states of type-0 $|\Phi\rangle$ or type-II $|\Psi\rangle$ state, or the mixed state $|M\rangle$ are shown as **Fig.44 (d), (e) and (f)**, respectively.

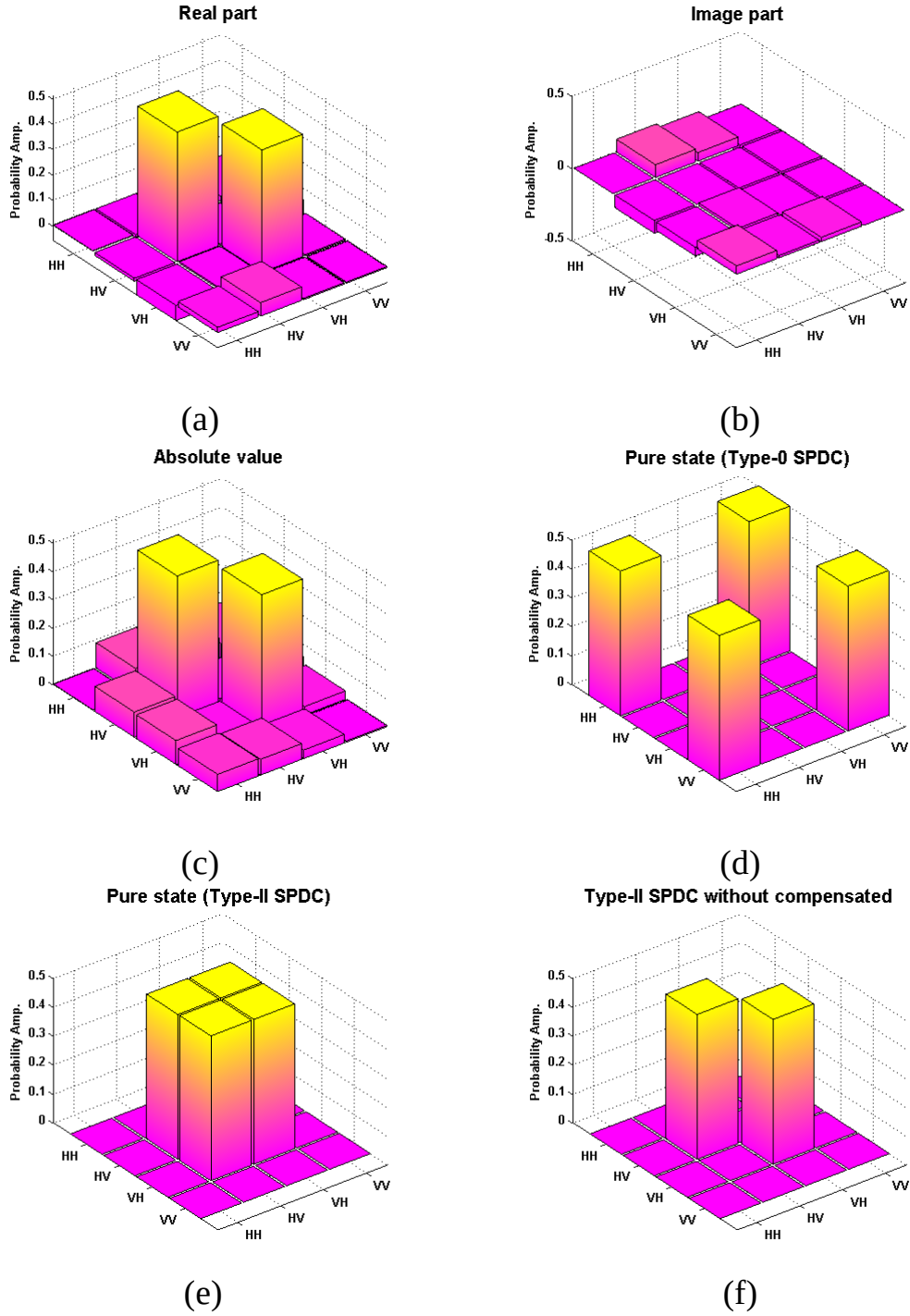


Figure 44. The reconstructed (a) real part, (b) image part and (c) absolute value of density matrix of measured NILP type-II SPDC without time delay compensated by quantum tomography, which belongs to the mixed quantum state. The simulated Bell's states for density matrix of pure states of (d) type-0 $|\Phi\rangle$ or (e) type-II $|\Psi\rangle$ state, or (f) the mixed state $|M\rangle$ of type-II SPDC without time-delay compensation.

$$|\Phi^\pm\rangle = \sqrt{1/2}(|H\rangle_a|H\rangle_b \pm |V\rangle_a|V\rangle_b) \quad (90)$$

$$|\Psi^\pm\rangle = \sqrt{1/2}(|H\rangle_a|V\rangle_b \pm |V\rangle_a|H\rangle_b) \quad (91)$$

$$|M^\pm\rangle = \alpha|H\rangle_a|V\rangle_b \pm \beta e^{i\Delta\phi}|V\rangle_a|H\rangle_b \quad (92)$$

Due to the birefringence of LiNbO₃ waveguide, the type-II SPDC photon-pairs with the cross-polarization modes have different phase velocity, which cause the phase-delay between signal and idler modes in time domain [98], the generation solution is using another half-length LiNbO₃ crystal with the crystal direction in orthogonal, (i.e. using a half-length x-cut LiNbO₃ after a z-cut type-II PPLN waveguide SPDC) [30], however, the time-delay compensation setup are usually involved the complexity from fiber/waveguide to free-space coupling alignments [99]. The pure quantum polarization state (Bell's state) needs to overcome the following issues in basically, first, for PPLN waveguides-based type-II SPDC telecom band source, the bandwidth of telecom band signals and idlers in spectrum domain are quite narrow with FWHM < 0.1~1 nm in typically [100], which means that the temperature control of SPDC source needs to be very stable and accurate to make sure the signals and idlers are overlapped in spectral domain in maximally. Second, as involved the fiber/waveguide to free-space coupling in the free-space time-delay compensation setup as the Fig. 42 shown, the overlapping of H-polarized and V-polarized modes in spatially are the key issue for distinguishability. Third, the last and most important, in the time domain, the mode overlapping of cross-polarization modes are suffered by the birefringence, which leads to different arrival time in two single-photon detectors. These issues can be minimized by a fully integrated quantum optical source [101] via the EO phase tunability of LiNbO₃ substrate.

Discussion and improvement of type-II entangled photon-pair source

The prior results of density matrix from quantum tomography shows the indistinguishability of generated photon-pairs are not completed, we discussed the possible of distinguishability of type-II SPDC sources, the distinguishability may come from the spatial, spectral temporal or polarization are not exactly the same

of generated photon-pairs. The spatial distinguishable can be resulted from mode size difference and imperfect spatial overlapping, the spectrum distinguishable are resulted from spectrum shape and temperature stability, the temporal distinguishable are resulted from imperfect time-delay compensation and the polarization distinguishable may come from the imperfect polarization-dependent elements on the setup.

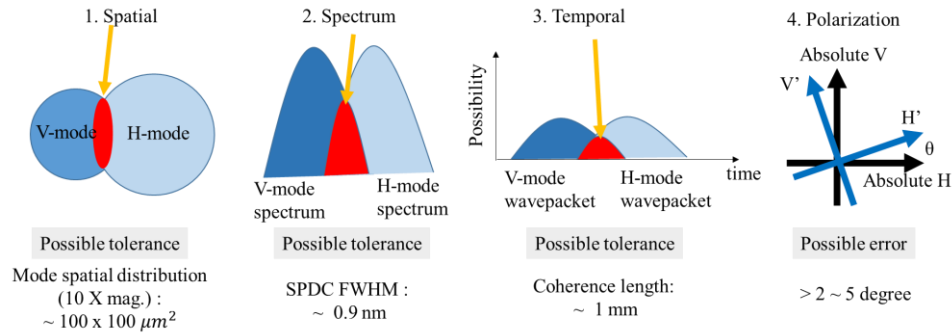


Figure 45. The possible sources of photon-pair distinguishability and possible tolerance of type-II SPDC source.

In order to optimize the entangled photon-pairs, we used the spatial filter after mode magnification, spectrum filters, highly precision temperature control system and checking the pump laser stability. To improve the coincidence visibility, we used the electric control compensation stage with longer integration time of coincidence counting. **Fig.46** shows quantum tomography density matrix from optimized type-II SPDC photon-pairs generation and delay compensation setup, which indicates the photon-pairs are closer to indistinguishable, in other words, they are more *entangled*.

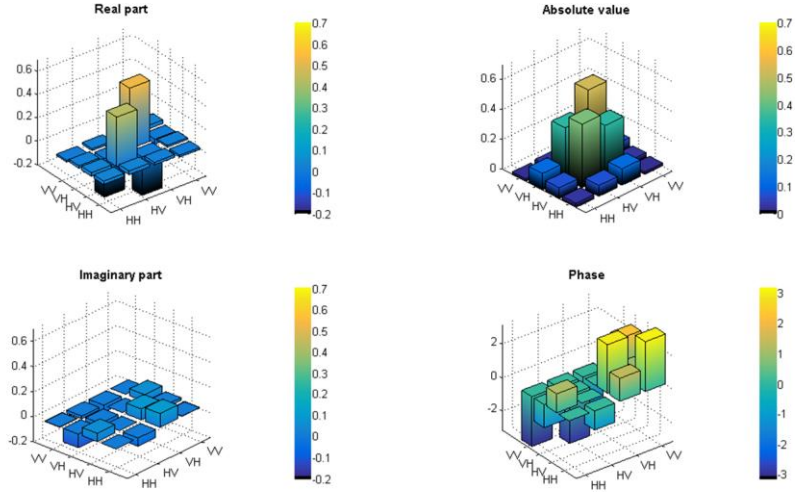


Figure 46. The density matrix of optimized type-II SPDC photon-pairs with delay compensation.

5-2-5 Quantum source experiment of metasurface (MS)

We combine the capacity of quantum entanglement simulation and measurement technique of the Nonlinear Physics Centre (NPC) in the Australian National University and the electric-field poling technique, the diffusion-type waveguide design and fabrication capacity of the Nonlinear Integrated Laser Photonic lab (NILP) in the National Central University, we want to simulate and experiment the characteristic of the spontaneous parametric down-conversion (SPDC) entangled photon pair through a special designed integrated Ti-diffused periodic poled Lithium Niobate waveguide (Ti:PPLN WGs) devices.

After established and tested the non-time delay compensated Type-II SPDC base-on Ti-diffused PPLN waveguide, we start to exam the quantum polarization states of generated photon-pairs via a specially designed metasurface (MS) polarization-dependent spatial mode splitter by measuring the coincidence of two single-photon detectors from any two output ports of spatial splitted H-polarized and V-polarized signal and idler modes at degeneracy wavelength.

Figure 47. (a) Experiment setup of SPDC photon-pair generation and metasurface quantum polarization state preparation (b) Measured photon-pair coincidence counting via metasurface polarization-dependent spatial mode splitting. The reconstructed density matrix of measured NILP type-II SPDC without delay compensated by metasurface quantum polarization state tomography without compensation in real part (c), image part (d) and absolute value (e)

The schematic diagram for experimental setup of quantum polarization state preparations of collinear Ti-diffused PPLN waveguide Type-II SPDC with metasurface system are shown as **Fig. 47 (a)**, including the quantum photon-pair generation, metasurface polarization spatial splitting via polarization-depended diffraction, multi-channel fiber coupling, coincidence, and the quantum polarization state reconstruction, respectively. One of representative measured coincidence counting in time domain via metasurface is plotted as **Fig. 47 (b)** with the sampling time is 25 minutes. The reconstructed 3x3 density matrix with axis of VV, HV, HH of measured results are plotted as **Fig. 47 (c)-(e)** with the real part, imaginary part and absolute values, respectively.

We used the heralded quantum entanglement experiment setup [97], with the metasurface to exam the single-photons quantum polarization states. The heralded quantum entangled states measurement setup is based on high splitting ratio polarization beam splitter (PBS) with splitting ratio > 60 dB, for separating the SPDC signal/idler H-polarized and V-polarized modes in space as **Fig. 48** shown. For the single-photon detector 1 (APD₁), the SPDC V-polarized mode is *Slave* port and the single-photon detector 2 (APD₂) is the *Master* port, which received the SPDC H-polarized photons after metasurface polarization-depended separation in space, we can prepare any polarization state of *Master* by adjusting the quart-wave plate and half-wave plate with appropriate angles. The coincidences are measured by changing the fibers for six ports of *Master* with the one fixed *slave* port, so there are 6 *Slave-Master* combinations. For each combination, we prepared 6 x 6 QWP and HWP setting for generating 36 different

polarization of single-photon sources for *Master* port, and calculating the basis of quantum polarization states tomography of this metasurface by numerical fitting the response curve for reconstructing the coincidence basis curves of each *Slave* ports with *Master* port.

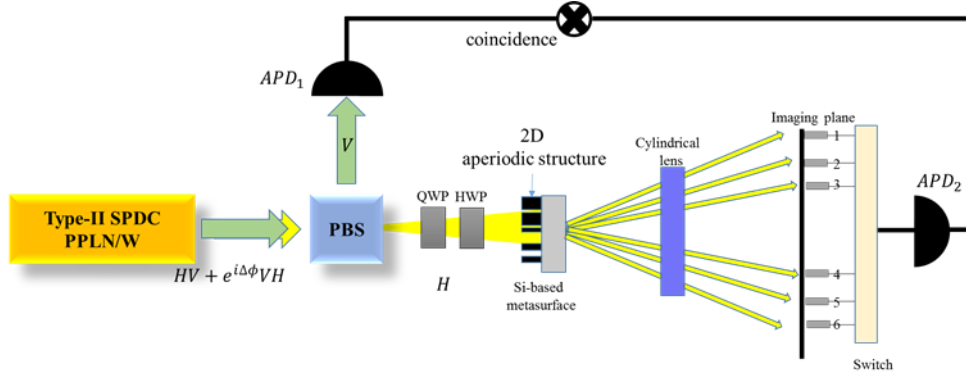


Figure 48. The single-photon quantum polarization state reconstruction experiment setup via a type-II SPDC PPLN waveguide within a metasurface.

For the single-photon detector 1 (APD₁), the SPDC V-polarized mode is *Slave* port and the single-photon detector 2 (APD₂) is the *Master* port, which received the SPDC H-polarized photons after metasurface polarization-depended separation in space, we can prepare any polarization state of *Master* by adjusting the quart-wave plate and half-wave plate with appropriate angles. The coincidences are measured by changing the fibers for six ports of *Master* with the one fixed *slave* port, so there are 6 *Slave-Master* combinations. For each combination, we prepared 6 x 6 QWP and HWP setting for generating 36 different polarization of single-photon sources for *Master* port, and calculating the basis of quantum polarization states tomography of this metasurface by numerical fitting the response curve for reconstructing the coincidence basis curves of each *Slave* ports with *Master* port. One of representative measured and curve fitting results shows the coincidence basis of the metasurface, shown as **Fig. 49 (a)-(b)** with discrete or dense correlation matrix.

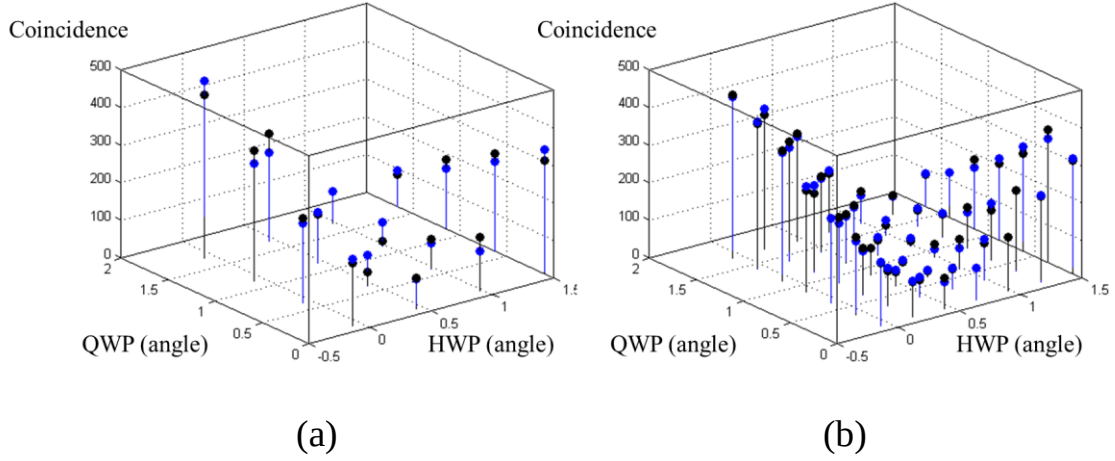


Figure 49. One of the measured (black) and fitting (blue) response points of coincidence between the *Slave* and *Master* under different polarization preparation of single photon source after metasurface quantum polarization state tomography for discrete (a) or dense (b) correlation matrix.

After number of times of 6x6x6 measurements with prepared discrete or dense quantum polarization states of *Master* single-photon, according to the measured coincidence results, we can use the fitted curve as the basis to reconstruction the density matrix of heralded single-photon quantum polarization states as **Fig. 50**. We can use the measured metasurface quantum polarization basis curves to predict any prepared quantum polarization states, the fidelity of the most representative tested density matrix of measured quantum polarization states is approach to 99.99% in comparison with ideal (prepared) states as the **Fig. 50** shown,.

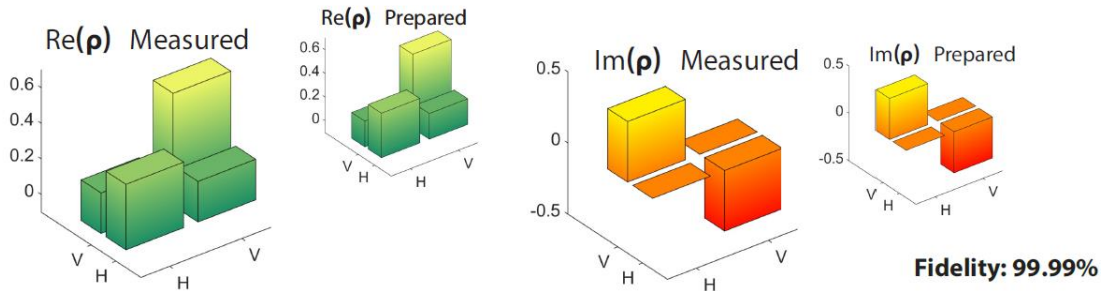


Figure 50. The reconstructed density matrix of measured and prepared single photon quantum polarization state tomography in real and image parts of polarization state density matrix.

5-3 Summary

For adiabatic light transfer and wavelength and polarization-dependent mode spatial splitting asymmetric adiabatic coupler, We demonstrate an on-chip ultra-broadband polarization- and wavelength-dependent splitter based on a z-cut titanium-diffused LiNbO₃ waveguiding asymmetric adiabatic coupler (AAC), the simulated bandwidths of power splitting ratio >17dB are > 60 nm or > 70 nm for H-polarized or V-polarized modes from 1610 nm to 1670 nm or from 1430 nm to 1500 nm, respectively. The AAC element is suitable for combining the existed fibre-optic telecommunication systems and developing on-chip quantum photon-pair source systems, the polarization-wavelength-dependent splitting characteristic of AAC can be applied to combine a LiNbO₃-based Type-II spontaneous parametric down conversion (SPDC) quantum light source pumped by 775 nm wave for quantum polarization-path state preparation. The simulated density matrix based on Type-II SPDC after AAC splitting shows the tunability of quantum polarization-path states can be controlled by changing the quasi-phase-matching (QPM) conditions of SPDC, which can easily be tuned by pump wavelength or working temperature within the same QPM structure. Further, with specially designed splitting spectrum of cross-polarized modes under an appropriate temperature, we can integrate the AAC structure with the QPM SPDC source and the electro-optical polarization-mode converter (EOPMC) in a single LiNbO₃ substrate to demonstrate a fully integrated time-delay compensated EO tunable Type-II quantum polarization-path state preparation optical circuit.

In the Type-II SPDC quantum light source, we design, fabrication, simulation, and verification the quantum light SPDC source made by an integrated optical circuit based on Ti-diffused periodic-poled LiNbO₃ chip, in addition to the

substrate purchased by foreign countries, which is 100% made in Taiwan from NILP Laboratory, National Central University. This international research cooperation project is also greatly assisted by PhD student Kai Wang, Prof. Dragomir N. Neshev and A/Prof. Andrey A. Sukhorukov from Nonlinear Physics Centre, RSPE in Australian National University. The quantum light source of SPDC characteristics are verified by SFG-SPDC analogy from experiment of measured the Type-II quasi-phase matched classical sum-frequency generation output power via H and V polarization input at 1570.6 ± 1.5 nm wavelength range, the measured results shows our SPDC source belongs high-brightness quantum light source due to its FWHM of SFG spectrum is only ~ 0.4 nm, which represents the quantum entangled state of this light source is very rigorous for generating the pure Bell's states in indistinguishable conditions. On the extended experiment, combining our SPDC source with ANU's metasurface, we developed a new technology for quantum source polarization characteristics verification, which is known as the quantum polarization state tomography as **Fig. 51** shown. Despite there are some issues needed to be overcome, we have completed a preliminary experimental test for single-photon quantum polarization state tomography, the measured maximum reconstruction accuracy of quantum polarization state via this quantum polarization state tomography is already approached to 99.99% in comparison with ideal setting polarization state.

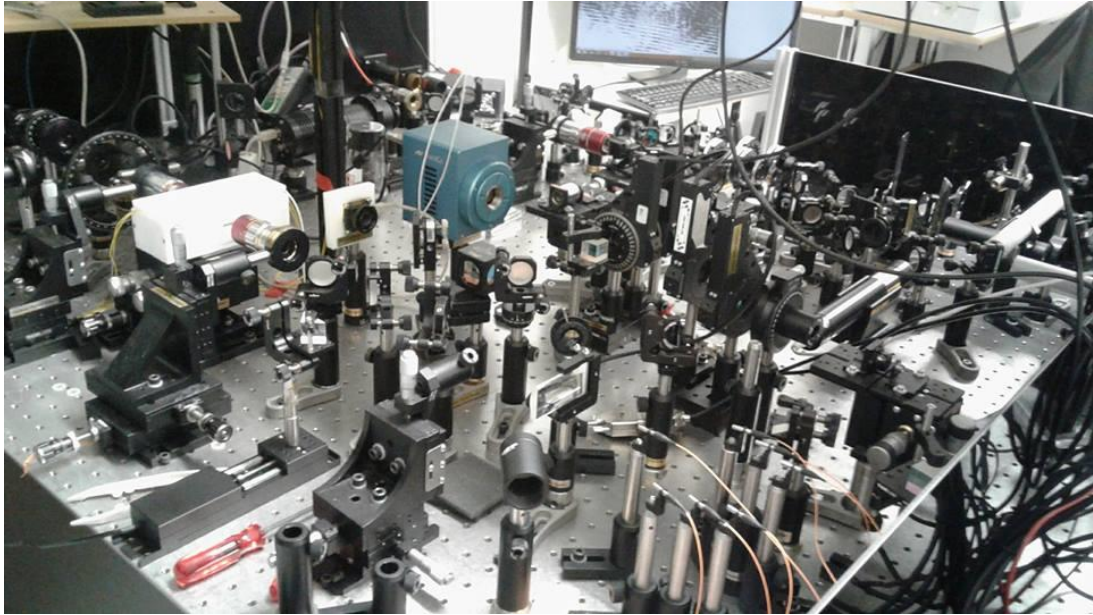


Figure 51. The quantum entangled SPDC source with the metasurface quantum polarization state tomography experimental setup.

CUDOS:2017

Best poster in nonlinear quantum photonics (EO APPLN directional couplers)

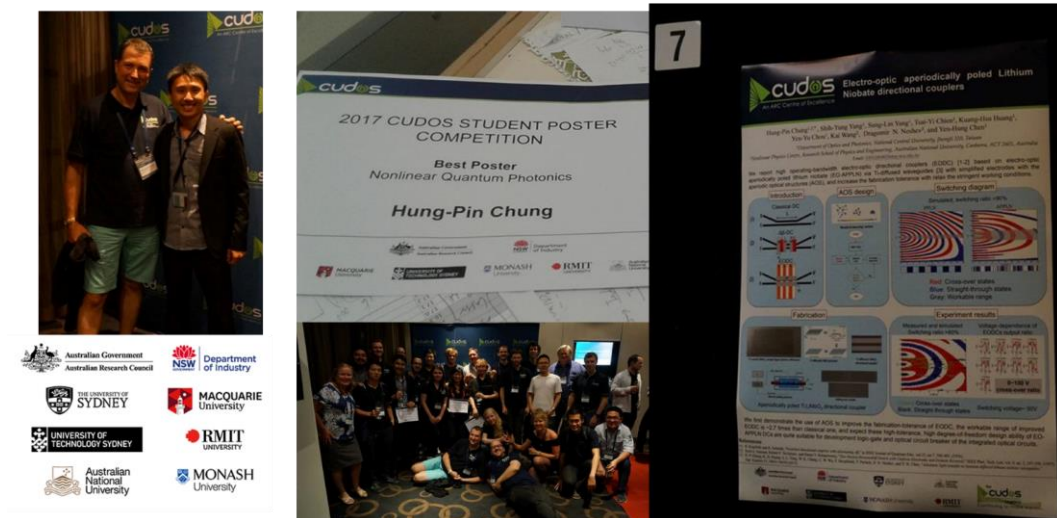


Figure 52. The photo with my project-associated advisor: Prof. Dragomir N. Neshev. I got the Best poster award in CUDOS:2017 annual workshop in nonlinear quantum photonics session.

Chapter 6 、 Conclusion and future outlook

6-1 Conclusion

This thesis included my major studies and research results, with the cooperation of professor Chen and his laboratory members, we invented and developed several devices and systems for providing the new vision in many applications in industry and academia, the main contribution are EO spectrum tunable or tailorable OPG/IPO systems for nonlinear laser generation, the 100 GHz high-frequency EO optical intensity modulator array for classical fiber-telecommunication network systems and the on-chip integrated quantum light generation and polarization state preparation optical circuits for next generation quantum communication networks. The related principle, design, modification, fabrication, quantization and analysis methods of EO modulation, waveguide modeling, and optical nonlinear characteristic are included in the chapter 2. My thesis composed of five themes with six research results, which are included in the chapter 3 to 5, respectively. The main research results are described as the following.

1. Topic I: Electro-optically tunable, multi-wavelength optical parametric generators in APPLN.

We demonstrated an EO spectrum tunable, multi-wavelength OPGs via a specially designed aperiodically poled LiNbO₃ device through the EO effect of LiNbO₃ crystal with unbalanced poled/unpoled cumulative lengths by two adjustment methods of ΔL or ΔN enhanced methods, the spectrum tuning rate of two methods are 0.38 and 0.49 nm/kV/mm, and the signal conversion efficiencies of ~8.6% and ~13% were obtained under a 10-Hz, bandwidth <

0.2 nm, 9-ns pulse duration, Q-switched Nd:YAG 1064 nm laser with pump intensity $\sim 92\text{MW/cm}^2$, for minimizing the photorefraction effect, the crystal temperature was stabilized at 140 degree Celsius.

2. Topic II: Electro-optically spectrum tailorable intracavity optical parametric oscillator.

With the integration of electro-optical polarization mode filtering and optical parametric generation medium in a single APPLN substrate via AOS technique, we invented a novel, sample configuration electro-optical spectrum tailorable IOPO systems under 809-nm CW fiber-coupled pump within the EO Bragg PPLN Q-switching devices. The spectrum of tailored IOPO signals are tunable from single-peak ,dual-peak to multi-peak by only adjusting the applied Ey-field equals 200, 400 and 500 V/mm via different polarization mode filtering spectrum of IOPO signals, the spectral width of single-peak signal is ~ 0.067 nm, which is narrowed by the factor of ~ 13.7 in comparison with original untailored-signals and the peak intensity of single-peak signals is ~ 6.1 times than non-tailored signal with the peak-to-peak intensity fluctuation of single peak signals is $\sim 9\%$.

3. Topic III: High-speed LiNbO₃ electro-optical modulators array.

We achieve effective 100 GHz bandwidth with separately control ability with a 12-modulator-integrated EO LiNbO₃ intensity modulator array, measurement result shows the each modulator -3dB bandwidth of EO response is $> 8\sim 10$ GHz, and the $V\pi < 10$ V, from DC-10GHz, for single modulator, all specification meets the commercial standards of 10 GHz EO LiNbO₃ intensity modulator.

4. Topic IV-1: Ultra-broadband adiabatic light transfer in titanium diffused Lithium Niobate waveguides.

Ultra-broadband adiabatic light transfer was demonstrated for the first time in lithium niobate. Coupling bandwidths and efficiencies of ~ 540 nm and >0.96 , respectively, were obtained with 5-cm long Ti:LiNbO₃ adiabatic couplers for both polarization modes., in the same LiNbO₃ chip of 50-mm in length, 5-mm in width, and 0.5-mm in thickness using the titanium thermal diffusion method.

5. Topic IV-2: Ultra-broadband asymmetric adiabatic coupler for two-photon quantum-polarization state preparation.

We demonstrate an on-chip ultra-broadband polarization- and wavelength-dependent splitter based on a z-cut titanium-diffused LiNbO₃ waveguiding asymmetric adiabatic coupler (AAC), the polarization-wavelength-dependent splitting characteristic of AAC can be applied to combine a LiNbO₃-based Type-II spontaneous parametric down conversion (SPDC) quantum light source pumped by 775 nm wave for quantum polarization-path state preparation. the simulated bandwidths of power splitting ratio > 17 dB are > 60 nm or > 70 nm for H-polarized or V-polarized modes from 1610 nm to 1670 nm or from 1430 nm to 1500 nm, respectively. The concept of fully integrated type-II SPDC EO polarization state tunable quantum polarization preparation optical circuit are also be presented on a single Ti-diffused multi-section PPLN waveguide device, the design detail and device schematic are included in the future work of thesis section 6-2.

6. Topic V: Experiments of quantum integrated optical circuits

We design, fabrication, simulation, and verification the quantum light Type-II SPDC source made by an integrated optical circuit based on Ti-

diffused periodic-poled LiNbO₃ chip, the quantum light source of SPDC characteristics are verified by SFG-SPDC analogy from experiment of measured the Type-II quasi-phase matched classical sum-frequency generation output power via H and V polarization input at 1570.6 ± 1.5 nm wavelength range in the Nonlinear Physics Centre, RSPE in Australian National University under the greatly help of PhD student Kai Wang, Prof. Dragomir N. Neshev and A/Prof. Andrey A. Sukhorukov. The measured results shows our SPDC source belongs high-brightness quantum light source due to its FWHM of SFG spectrum is only ~ 0.4 nm. We also developed a new technology of the quantum polarization state tomography, and have completed a preliminary experimental test for single-photon quantum polarization state tomography, the measured maximum reconstruction accuracy of quantum polarization state is already approached to 99.99% in comparison with ideal setting polarization state.

6-2 Future outlook

In the future work, we first propose a concept of the fully integrated quantum-polarization state source with three sections for photon-pair generation, polarization-mode conversion and the pump filtering and quantum-polarization state preparations, respectively. The PPLN EOPMC section is used to modulate the polarization of photon-pairs modes generated via the first section PPLN Type-II SPDC, one can control the polarization of the photon-pairs by adjusting the applied electric-fields along crystal y-axis via the EO effect in the QPM PPLN waveguide through electrode **PA**.

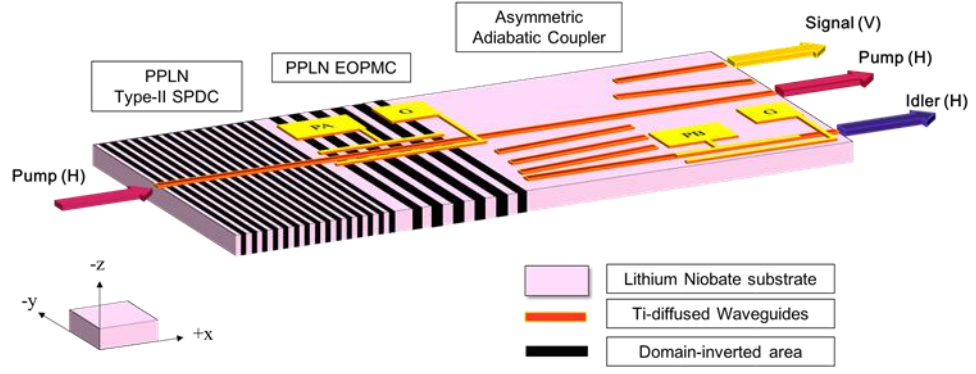


Figure 53. Concept of a fully integrated quantum-polarization state source, where the type-II Ti-diffused PPLN waveguide, EOPMC with the electrode PA and AAC are used for photon-pair generation, EO-tunable polarization state conversion and polarization-wavelength beam splitting, respectively, and the time-delay compensation of cross-polarization mode can be completed by the electrode PB within the AAC.

In the third section, for the pump filtering and quantum-polarization state preparation, we specially used the different coupling feature of two cross-polarization modes of z-cut Ti-diffused LiNbO₃ waveguides with different refractive index change and firstly proposed an integrated high efficiency ultra-broadband polarization-wavelength-dependent spatial splitting asymmetric adiabatic coupler (AAC) based on a specially designed waveguide array with two parts for filtering out unused SPDC pump and separating the V-polarized signals and the H-polarized idlers in spatially, and we can modulate two-photon quantum-polarization states by taking the advantage of the wavelength-dependent splitting spectrum of AAC. The time-delay of two cross-polarization modes can be compensated via the electrode **PB** by adjusting the applied electric-fields along crystal z-axis in the arm of H-polarization modes in AAC section.

The main goal of this research cooperation project is to design, simulate, fabricate and measure a multifunctional integrated Ti-diffused PPLN waveguide devices for adiabatic spatially pumps filtering and entangled photon pair generation. The conceptual diagram of devices in this project is shown as **Fig. 53**. This device is composed of three sections, which are type-II SPDC section, the EOPMC section and AAC section. First part of devices is used to generate the

entangled photon pairs by SPDC. Second part is used to modulate the quantum polarization states of photon-pairs and third is the polarized-dependent spatial splitter to convert the polarization entangled states to spatial entangled states. Because of effective indexes of each mode are fabrication-dependent, we need to establish the dispersion model of our Ti-diffused waveguides in the beginning of this study. The correct poling periods in Ti-diffused waveguides can be experimented by SFG measurement of 785 nm generation (pump by 1570 nm) or can be simulated by numerical analysis of waveguide simulation approaching. The correlation measurement of entangled photon pairs and the analysis of quantum entanglement of the completed devices need to rely on the equipment, experience and recommendation of Nonlinear Physics Centre for further studies. In the case of type-II QPM SPDC, pump wavelength is setting as 785 nm, the degeneracy signal and idler are 1570 nm but with cross-polarization, the period of QPM domain structure of Ti:PPLN waveguide is $\sim 9.5 \mu\text{m}$ at 135°C operating. According our prior device, the QPM Type-II SPDC degeneracy wavelength of Ti-diffused PPLN waveguides are around 1530-1600 nm with periods around $8.8\sim 9.7 \mu\text{m}$ under working temperature above 120°C with temperature tuning rate is around $d\lambda/dT = -0.18 \text{ nm}/^\circ\text{C}$. On the other hand, the spectrum of EOPMC based-on Ti-diffused LiNbO₃ waveguides depends the active length with poled area [101-103], the calculated spectrum FWHM under 1550-nm band is around 5 nm of 6-mm long Ti-diffused LiNbO₃ EOPMC with the temperature tuning rate is around $d\lambda/dT = -0.8 \text{ nm}/^\circ\text{C}$ [104]. The working temperature of each sections of fully integrated quantum circuits can be designed under specific temperature, for instance, we can adjust the poling grating periods of SPDC and EOPMC section for a selective QPM wavelengths, and the AAC part can also modify the coupling matrix by carefully designing the structures.

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